

Risk Bounds for Quantile Temporal-Spatial Analysis

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Revision key. Red text is new or materially revised relative to the supplied manuscript. The main paper and the complete supplementary material are retained in this single file. All mathematical comments that previously appeared in the appendix have now been implemented as corrected statements and proofs; no unresolved proof-comment boxes remain. Experimental designs and numerical values are not substantively changed.

Abstract

We study nonparametric conditional-quantile estimation for observations indexed by time and space. Observations collected at nearby times may be dependent, while observations at the same time share a time-specific random spatial effect. We analyze constrained empirical quantile-risk minimizers over bounded convex and nonconvex function classes. The theory allows heavy-tailed measurement errors and imposes no moment condition on the response noise. The main bound is expressed through a blockwise localized Rademacher fixed point and an additional term measuring the fluctuation induced by the random spatial effect. We specialize the framework to univariate trend filtering, high-dimensional linear quantile prediction, K -nearest-neighbor graph fused lasso, and clipped sparse ReLU networks for hierarchical composition models. For trend filtering, the usual minimax exponent is retained up to logarithmic factors and the spatial-effect term. We also provide ADMM algorithms and numerical studies under light- and heavy-tailed noise.

Keywords: total variation, nonparametric quantile regression, spatio-temporal dependence, beta mixing, localized Rademacher complexity

1 Introduction

Technological advancements have significantly improved data collection and analysis techniques, resulting in an abundance of data with both temporal and spatial dimensions across various disciplines (Wen et al. (2019), Briz-Redón et al. (2020), Liu et al. (2020), Tian et al. (2021), Jayasiri et al. (2022)). This proliferation of complex datasets has necessitated the development and adoption of advanced temporal-spatial models designed to address the unique challenges and harness the full potential of temporally and spatially diverse data.

Current leading results for temporal-spatial estimators typically assume that error distributions are sub-Gaussian. In contrast, quantile regression, known for its robustness and versatility across different sectors (Coad et al. (2006); Sanderson et al. (2006); Perlich et al. (2007); Benoit et al. (2009); Wasko et al. (2014); Pata et al. (2015); Belloni et al. (2023); Zhang et al. (2023)), often requires fewer presuppositions about error distributions. This approach diverges from

conventional models that rely heavily on such assumptions. We extend the quantile regression approach to incorporate a thorough theory of risk bounds for an extensive array of quantile temporal-spatial functions, doing so with minimal assumptions about error variables. This analytical enhancement broadens the scope of quantile regression for temporal-spatial data analysis, bolstering its theoretical underpinnings and inclusivity towards error distributions, including both normal and heavy-tailed distributions.

This paper studies nonparametric quantile regression in a spatio-temporal setting. For time points $i = 1, \dots, n$, let m_i be the number of spatial observations collected at time i . We observe

$$y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}, \quad j = 1, \dots, m_i, \quad (1)$$

where $x_{ij} \in [0, 1]^d$, $f^* : [0, 1]^d \rightarrow \mathbb{R}$ is deterministic, δ_i is a time-specific random spatial function, and e_{ij} is a measurement error. For a fixed quantile level $\tau \in (0, 1)$, we impose

$$\mathbb{P}\{\delta_i(x_{ij}) + e_{ij} \leq 0 \mid x_{ij}\} = \tau. \quad (2)$$

Thus $f^*(x)$ is the conditional τ -quantile of y_{ij} given $x_{ij} = x$. The shared function δ_i induces dependence among observations within time i , and the time-indexed clusters may be beta-mixing across i .

1.1 Notation

For an integer $q \geq 1$, write $[q] = \{1, \dots, q\}$. Let Q denote the common marginal distribution of the design points and define the weighted empirical design distribution

$$Q_{n,m} = \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \delta_{x_{ij}}.$$

For a measurable $g : [0, 1]^d \rightarrow \mathbb{R}$, let

$$\|g\|_{L_p(Q)} = \left(\int_{[0,1]^d} |g(x)|^p dQ(x) \right)^{1/p}, \quad \|g\|_\infty = \sup_{x \in [0,1]^d} |g(x)|,$$

with the usual modification for $p = \infty$. We abbreviate $\|g\|_2 = \|g\|_{L_2(Q)}$ and define

$$\langle f, g \rangle_{n,m} = \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} f(x_{ij})g(x_{ij}), \quad \|g\|_{n,m}^2 = \langle g, g \rangle_{n,m}.$$

Let $N = \sum_{i=1}^n m_i$ and let

$$m = \left(\frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \right)^{-1}$$

be the harmonic mean of the group sizes. When a function is evaluated on all observed design points, the same symbol denotes the vector of evaluations in $\mathbb{R}^N$. For positive sequences a_n, b_n , write $a_n \lesssim b_n$ if $a_n \leq Cb_n$ for a constant independent of n , and $a_n \asymp b_n$ if both inequalities hold. The symbols $O_{\mathbb{P}}$ and $o_{\mathbb{P}}$ have their standard meanings.

1.2 Summary of Results

Let $\rho_\tau(u) = u\{\tau - \mathbf{1}(u < 0)\}$ be the check loss. We estimate f^* by

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau\{y_{ij} - f(x_{ij})\}. \quad (3)$$

Each time point receives total weight $1/n$, so unequal group sizes are accommodated. The estimation criterion is the check loss, whereas the risk is reported in $L_2(Q)$. A Huber-type loss appears naturally as a lower bound for the excess population quantile risk and bridges these two quantities.

The temporal dependence is modeled through absolute regularity of the joint cluster process $W_i = (X_i, \delta_i, E_i)$, rather than through separate and potentially incompatible assumptions for the design, spatial effect, and measurement errors. For

$$L = \left\lceil \frac{2}{C_\beta} \log n \right\rceil,$$

define the interlaced index sets

$$\begin{aligned} \mathcal{J}_{e,\ell} &= \{\ell + 2qL : q \geq 0\} \cap [n], \\ \mathcal{J}_{o,\ell} &= \{\ell + (2q+1)L : q \geq 0\} \cap [n], \quad \ell \in [L]. \end{aligned} \quad (4)$$

These sets partition $[n]$, and observations in each fixed set are separated by at least $2L$. Berbee coupling is applied once to the joint clusters, preserving the contemporaneous relationship among X_i , δ_i , and E_i .

For a centered class $\mathcal{H}$ and $a \in \{e, o\}$, the relevant block complexity retains the weights of the original empirical loss:

$$\mathcal{R}_{a,\ell}(\mathcal{H}) = \mathbb{E} \sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i \in \mathcal{J}_{a,\ell}} \frac{1}{m_i} \sum_{j=1}^{m_i} \varepsilon_{ij} h(x_{ij}^*) \right|.$$

The convex theorem defines a critical radius through the sum of these localized block complexities and a remainder

$$\Gamma_n \asymp \frac{(s+1)L \log(nT_n)}{n},$$

which quantifies the random spatial effect under the low-dimensional Lipschitz condition in Assumption 3. If r_n denotes that critical radius and $b_B^2 = 1 \vee 2B$, then Theorem 6 gives

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}(b_B^2 r_n^2).$$

No generic $O\{\text{polylog}(n)/(nm)\}$ rate follows without evaluating the entropy or geometry of the particular function class.

For univariate trend filtering with $J_k(f) = \text{TV}(f^{(k-1)}) \leq V$, the localized entropy calculation yields

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ V^{2/(2k+1)} (L/N)^{2k/(2k+1)} + \Gamma_n \right\} \right],$$

up to constants and the logarithmic factors made explicit in the proof. This retains the usual exponent $2k/(2k+1)$. For high-dimensional linear quantile regression, the result controls prediction error through weighted block-column norms and does not claim coefficient recovery without an additional identifiability condition.

For $d > 1$, the K -nearest-neighbor graph fused-lasso result separates in-sample estimation from off-sample nearest-neighbor approximation. Its stochastic bound is expressed through the explicit incidence-pseudoinverse parameter $\rho(G_K)$ and is proved in Proposition 30. Under the stated graph-geometry conditions and piecewise-Lipschitz regularity, the leading nonparametric term is $N^{-1/d}$ up to logarithms, plus Γ_n . The graph total variation of a regular target scales as $N^{1-1/d}$, not $N^{-1/d}$.

For nonconvex classes, we select a near-best function $f^\circ \in \mathcal{F}$, work with the centered star hull, and add the approximation error $\|f^\circ - f^*\|_\infty^2$. The neural-network corollary applies to clipped sparse ReLU networks whose depth, width, and sparsity are chosen to balance approximation and stochastic complexity for hierarchical composition models. The paper also gives corrected ADMM updates for trend filtering and graph fused lasso and reports numerical studies under light- and heavy-tailed noise.

2 General Methods

We formulate a general quantile-regression problem over a bounded function class. Throughout the theory,

$$\|f^*\|_\infty \leq B, \quad \sup_{f \in \mathcal{F}} \|f\|_\infty \leq B. \quad (5)$$

For the convex result, $\mathcal{F}$ is closed under convex combinations:

$$\lambda f + (1 - \lambda)g \in \mathcal{F}, \quad f, g \in \mathcal{F}, \quad \lambda \in [0, 1]. \quad (6)$$

The nonconvex result removes this restriction and allows $f^* \notin \mathcal{F}$.

2.1 Assumptions

Assumption 1 (Sampling, dependence, and quantile calibration). *The following conditions hold.*

1. *The observations satisfy $y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$ and the conditional quantile calibration in (2).*
2. *Each x_{ij} has a common density v on $[0, 1]^d$, with $0 < v_{\min} \leq v(x) \leq v_{\max} < \infty$.*
3. *Let $X_i = (x_{i1}, \dots, x_{im_i})$ and $E_i = (e_{i1}, \dots, e_{im_i})$. The spatial effect δ_i is independent of X_i . Conditional on δ_i , the design points x_{ij} are independent with common law Q ; conditional on (δ_i, X_i) , the measurement errors e_{ij} are independent. Their conditional distributions may depend on δ_i and on the corresponding design point; within a fixed time cluster, the conditional kernel is the same for all j .*
4. *The joint process $W_i = (X_i, \delta_i, E_i)$ is beta-mixing with coefficients $\beta_W(k) \leq C_\beta^{(0)} e^{-C_\beta k}$ for constants $C_\beta^{(0)}, C_\beta > 0$.*
5. *The random spatial function δ_i may induce arbitrary dependence among the responses within time i through the shared evaluations $\delta_i(x_{ij})$.*
6. *No moment condition is imposed on e_{ij} or $\delta_i(x_{ij})$.*
7. *There are constants $0 < c_m \leq C_m < \infty$ such that $c_m m \leq m_i \leq C_m m$ for every i , where m is the harmonic mean defined above.*

The joint mixing condition replaces the contradictory combination of cross-time independence and beta mixing in the supplied version. Coupling the joint cluster is essential: separate couplings of X_i , δ_i , and E_i need not preserve their contemporaneous joint distribution.

Assumption 2 (Local quantile curvature). *There are constants $R > 0$ and $0 < \underline{f} \leq \bar{f} < \infty$ such that, for every i, j , every $x \in [0, 1]^d$, and every $|u| \leq R$,*

$$\underline{f}|u| \leq |F_{ij}\{f^*(x) + u \mid x\} - F_{ij}\{f^*(x) \mid x\}| \leq \bar{f}|u|, \quad (7)$$

where $F_{ij}(\cdot \mid x)$ is the conditional distribution function of y_{ij} given $x_{ij} = x$, and $F_{ij}\{f^*(x) \mid x\} = \tau$.

Assumption 3 (Low-dimensional Lipschitz spatial fluctuation). *Let $U_{ij} = \delta_i(x_{ij}) + e_{ij}$ and define*

$$D_i(x, u) = \mathbb{P}(U_{i1} \leq u \mid x_{i1} = x) - \mathbb{P}(U_{i1} \leq u \mid \delta_i, x_{i1} = x). \quad (8)$$

There is a coordinate set $S \subset [d]$, $|S| = s$, such that $D_i(x, u)$ depends on x only through x_S . Almost surely, the map $(x_S, u) \mapsto D_i(x_S, u)$ is uniformly Lipschitz on $[0, 1]^s \times [-T_n, T_n]$, with a deterministic constant that does not depend on i or n . Since $\mathbb{E}\{D_i(x, u)\} = 0$, the joint-block coupling and a finite-grid argument imply, rather than assume,

$$\max_I \sup_{x \in [0, 1]^d, |u| \leq T_n} \left| \frac{1}{|I|} \sum_{i \in I} D_i(x, u) \right| = O_{\mathbb{P}} \left\{ \sqrt{\frac{(s+1)L \log\{n(1+T_n)\}}{n}} \right\}. \quad (9)$$

The complete derivation is given in Lemma 23.

Assumption 4 (Localization range). *Set $T_n = 2B$. If the envelope $B = B_n$ is allowed to grow, assume only that $\log\{n(1+T_n)\} = O(\log n)$. Because $|f(x) - f^*(x)| \leq 2B$, every CDF increment in the proof is evaluated on $[-T_n, T_n]$. No tail probability or moment condition on e_{ij} or $\delta_i(x_{ij})$ is required.*

Example 5 (Heavy-tailed errors). The theory permits Cauchy, Student- t , and other heavy-tailed measurement errors without any moment restriction. The required conditions concern the conditional quantile calibration, the local CDF slope near the target quantile, and the Lipschitz behavior of the conditional CDF difference in (8); they do not require truncating the realized response or controlling $\max_{i,j} |e_{ij}|$.

2.2 Results for General Methods

For $a \in \{e, o\}$, $\ell \in [L]$, and a centered class $\mathcal{H}$, define

$$\mathcal{R}_{a,\ell}(\mathcal{H}) = \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i \in \mathcal{J}_{a,\ell}} \frac{1}{m_i} \sum_{j=1}^{m_i} \varepsilon_{ij} h(x_{ij}^*) \right| \right]. \quad (10)$$

The denominator and weights are those of the full empirical criterion. If one instead uses a within-block normalization, it must be multiplied by that block's weight in the full objective.

Let $b_B = (1 \vee 2B)^{1/2}$ and

$$\mathcal{H}_{\mathcal{F}}(r) = \{f - f^* : f \in \mathcal{F}, \|f - f^*\|_2 \leq r\}.$$

Set

$$\Gamma_n = \frac{(s+1)L \log\{n(1+T_n)\}}{n}, \quad (11)$$

and, for a sufficiently large universal constant C_* , define

$$\Psi_n(r) = C_* \left[\sum_{\ell=1}^L \sum_{a \in \{e, o\}} \mathcal{R}_{a, \ell} \{ \mathcal{H}_{\mathcal{F}}(c_0^{-1/2} b_B r) \} + b_B^2 \Gamma_n \right],$$

$$r_n = \inf \{ r > 0 : \Psi_n(r) \leq r^2/64 \}. \quad (12)$$

A nondecreasing sub-root majorant of the complexity term is used when needed.

2.3 For $\mathcal{F}$ as a Convex Set

Theorem 6 (Convex spatio-temporal quantile regression). *Suppose Assumptions 1, 2, 3, and 4 hold, $\mathcal{F}$ is convex, $f^* \in \mathcal{F}$, and (5) holds. Let*

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_{\tau} \{ y_{ij} - f(x_{ij}) \}. \quad (13)$$

If r_n is defined by (12), then

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}(b_B^2 r_n^2). \quad (14)$$

More explicitly, for every $\epsilon > 0$ there is $C_{\epsilon} < \infty$ such that, for all sufficiently large n ,

$$\mathbb{P}\{\|\hat{f} - f^*\|_2^2 > C_{\epsilon} b_B^2 r_n^2\} \leq \epsilon + C n \beta_W(2L).$$

The theorem is stated through a critical radius because the rate depends on the local geometry of $\mathcal{F}$. The proof uses Knight's identity, local quantile curvature, a single joint-cluster coupling, centered conditional symmetrization, contraction, and the blockwise spatial-effect bound. Since $|f - f^*| \leq 2B$, $\|f - f^*\|_2^2 \leq (1 \vee 2B) \Delta^2(f - f^*)$; this corrects the factor used in the supplied proof.

3 Application To Trend Filtering

Over the past decade, trend filtering has become a widely used nonparametric method for fitting piecewise-polynomial functions (Rudin et al., 1992; Mammen et al., 1997; Tibshirani et al., 2005; Kim et al., 2009; Tibshirani, 2014).

Definition 7 (Trend-filtering class). *For $d = 1$, an integer $k \geq 1$, and $V > 0$, let*

$$J_k(f) = \text{TV}\{f^{(k-1)}\}, \quad \mathcal{F}_k(V) = \{f \in \mathcal{F} : J_k(f) \leq V\}. \quad (15)$$

For a univariate function g ,

$$\text{TV}(g) = \sup_{0 \leq t_1 < \dots < t_M \leq 1} \sum_{q=1}^{M-1} |g(t_{q+1}) - g(t_q)|.$$

If g is absolutely continuous, then $\text{TV}(g) = \int_0^1 |g'(t)| dt$. The fitted pieces have degree $k - 1$: $k = 1$ gives a piecewise-constant fit and $k = 2$ a piecewise-linear fit.

Assumption 8 (Trend-filtering target). *The target satisfies $f^* \in \mathcal{F}_k(V)$ for a fixed integer $k \geq 1$ and a radius $V \geq J_k(f^*)$. In the canonical regime, $V = O(1)$ and $\|f^*\|_{\infty} \leq B$. No mean-zero constraint is required solely for identifiability because the loss identifies the intercept; a centering constraint may be added for a particular parameterization.*

3.1 Result for Quantile Temporal-Spatial Trend Filtering

The constrained estimator is

$$\hat{f} \in \arg \min_{f \in \mathcal{F}_k(V)} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau\{y_{ij} - f(x_{ij})\}. \quad (16)$$

Theorem 9 (Trend filtering). *Suppose the assumptions of Theorem 6 and Assumption 8 hold. Assume the empirical entropy of the localized trend-filtering class satisfies the standard bound*

$$\log N\{\epsilon, \mathcal{F}_k(V), \|\cdot\|_{n,m}\} \lesssim V^{1/k} \epsilon^{-1/k}.$$

Then

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ V^{2/(2k+1)} \left(\frac{L}{N} \right)^{2k/(2k+1)} + \Gamma_n \right\} \right]. \quad (17)$$

For bounded V and B , the leading exponent is $2k/(2k+1)$, up to the logarithmic factor from $L \asymp \log n$ and the spatial-effect term Γ_n .

The block calculation must retain each block's weight in the full empirical criterion. If $M_I \asymp N/L$ is the number of observations in a block, a normalized within-block Dudley bound is multiplied by M_I/N . Summing over the $2L$ blocks gives a stochastic term of order $V^{1/(2k)} r^{1-1/(2k)} \sqrt{L/N}$ and hence the fixed point in (17).

4 Application To High-dimensional Quantile Linear Regression

We next consider the constrained ℓ_1 quantile-regression estimator studied by Fu et al. (2000) and Belloni et al. (2011). It is a robust prediction method in high-dimensional linear models.

Assumption 10 (Linear target). *The conditional quantile is $f^*(x) = x^\top \beta^*$ for some $\beta^* \in \mathbb{R}^d$ with $\|\beta^*\|_1 \leq V$, and the functions $x \mapsto x^\top \beta$ in the constraint set satisfy the common bound (5).*

Define

$$\mathcal{H}(V) = \{x \mapsto x^\top \beta : \|\beta\|_1 \leq V\}, \quad \hat{f} \in \arg \min_{f \in \mathcal{H}(V)} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau\{y_{ij} - f(x_{ij})\}. \quad (18)$$

For block $I = \mathcal{J}_{a,\ell}$, let X_I^w be the matrix whose row corresponding to (i, j) is $x_{ij}^\top / m_i$, and set

$$\mathcal{A}_n = \sum_{a \in \{e, o\}} \sum_{\ell=1}^L \mathbb{E} \max_{q \in [d]} \|X_{a,\ell,q}^w\|_2. \quad (19)$$

4.1 Result for High-dimensional Quantile Linear Regression

Theorem 11 (High-dimensional linear prediction). *Under Theorem 6 and Assumption 10,*

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ \frac{V \sqrt{\log(2d)}}{n} \mathcal{A}_n + \Gamma_n \right\} \right]. \quad (20)$$

If the unweighted block columns have Euclidean norm $O(\sqrt{M_{a,\ell}})$ and $m_i \asymp m$, the stochastic term is of order $V \sqrt{L \log d / N}$ up to constants. This is a prediction bound. Estimation of β^ or variable-selection consistency requires an additional restricted-eigenvalue or identifiability condition.*

5 Application To Quantile K -NN fused lasso

For $d > 1$, we consider a quantile version of the K -nearest-neighbor graph fused lasso (Elmoataz et al., 2008; Ferradans et al., 2014; Madrid Padilla et al., 2020). Construct the symmetric union K -NN graph $G_K = (V_K, E_K)$ on the N observed design points: an undirected edge is included when either endpoint is among the other's K nearest neighbors.

For $\theta \in \mathbb{R}^N$, let D_{G_K} be an oriented incidence matrix and define

$$\|D_{G_K}\theta\|_1 = \sum_{(u,v) \in E_K} |\theta_u - \theta_v|. \quad (21)$$

For a piecewise-Lipschitz target, the natural graph-TV scale is

$$\|D_{G_K}\theta^*\|_1 \lesssim N^{1-1/d} \text{polylog}(N), \quad (22)$$

not $N^{-1/d}$. The latter is the scale of the prediction risk, not the unnormalized graph total variation.

Assumption 12 (Graph geometry and target regularity). *The design density is bounded above and below on $[0, 1]^d$, $d > 1$, and K is polylogarithmic in N . On an event whose probability tends to one: (i) the symmetric union K -NN graph is connected; (ii) with $D_{G_K}^\dagger$ the Moore–Penrose inverse of its incidence matrix,*

$$\rho(G_K) := \max_{e \in E_K} \|(D_{G_K}^\dagger)_\cdot e\|_2 \leq \rho_N;$$

(iii) the K -NN radius is at most $C_R(K/N)^{1/d}$; (iv) the neighbor-overlap bound $Q\{x : x_v \in N_K(x)\} \leq C_O K/N$ holds for every sampled vertex v ; and (v), writing $r_N = C_R(K/N)^{1/d}$ and $\mathcal{S}$ for the singularity set, the empirical boundary-tube count satisfies $N^{-1} \sum_{v=1}^N \mathbf{1}\{x_v \in B_{r_N}(\mathcal{S})\} \leq C_S r_N$. The target is bounded and piecewise Lipschitz with a singularity set whose r -neighborhood has Q -measure $O(r)$, and $V_n \geq \|D_{G_K}\theta^\|_1$. Lemma 31 shows that one may take $V_n \lesssim N^{1-1/d} K^{1+1/d}$ on the stated geometry event.*

5.1 Result for Quantile K -NN fused lasso

The constrained in-sample estimator is

$$\hat{\theta} \in \arg \min_{\theta \in \mathbb{R}^N} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau(y_{ij} - \theta_{ij}) \quad \text{subject to } \|D_{G_K}\theta\|_1 \leq V_n. \quad (23)$$

Set $\hat{f}(x_{ij}) = \hat{\theta}_{ij}$ at sampled points and, for a new x , define

$$\hat{f}(x) = \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{\theta}_{ij}. \quad (24)$$

Theorem 13 (Quantile K -NN fused lasso). *Suppose Theorem 6 and Assumption 12 hold. Then*

$$\|\hat{\theta} - \theta^*\|_{n,m}^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ \frac{L}{N} + \frac{LV_n \rho_N \sqrt{\log(2|E_K|)}}{N} + \Gamma_n \right\} \right], \quad (25)$$

and the nearest-neighbor extension satisfies

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ \frac{L}{N} + \frac{LV_n \rho_N \sqrt{\log(2|E_K|)}}{N} + \Gamma_n + \left(\frac{K}{N} \right)^{1/d} \right\} \right]. \quad (26)$$

If ρ_N and K are polylogarithmic and $V_n \lesssim N^{1-1/d} K^{1+1/d}$, the leading nonparametric term is $N^{-1/d}$ up to logarithmic factors and Γ_n . Proposition 30 proves the localized graph-complexity inequality used here.

6 For $\mathcal{F}$ as a Non-convex Set and $f^* \notin \mathcal{F}$

For a possibly nonconvex class $\mathcal{F}$, define the centered star hull

$$\text{Star}(\mathcal{F} - f^*) = \{\mu(f - f^*) : f \in \mathcal{F}, \mu \in [0, 1]\}. \quad (27)$$

Choose a function $f^\circ \in \mathcal{F}$ satisfying

$$\|f^\circ - f^*\|_\infty \leq \inf_{f \in \mathcal{F}} \|f - f^*\|_\infty + n^{-1}, \quad \phi_n = \|f^\circ - f^*\|_\infty. \quad (28)$$

This is a function-valued near-best approximation; the scalar infimum used in the supplied manuscript was not a projection.

Theorem 14 (Nonconvex temporal-spatial quantile regression). *Suppose Assumptions 1, 2, 3, and 4 hold, and f^* and all $f \in \mathcal{F}$ satisfy (5). Let*

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau\{y_{ij} - f(x_{ij})\}. \quad (29)$$

Define the critical radius as in (12), with $\mathcal{H}_{\mathcal{F}}(r)$ replaced by the localized centered star hull. Then

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}\{b_B^2(r_n^2 + \phi_n^2)\}. \quad (30)$$

The proof controls the absolute value of one centered localized empirical process, rather than duplicating twelve one-sided decompositions. The approximation-adjusted radius is used consistently throughout.

7 Application to Neural Networks

Deep ReLU networks provide flexible nonparametric estimators and can exploit hierarchical composition structure to mitigate the ambient-dimensional curse (Schmidhuber, 2015; Schmidt-Hieber, 2020; Kohler et al., 2019).

For depth L , width r , sparsity S , and clipping level B , let

$$\mathcal{F}_{\text{NN}}(L, r, S, B) = \{\text{clip}_B(f) : f \text{ is a ReLU network with depth } L, \text{ width at most } r, \text{ and at most } S \text{ nonzero parameters}\}. \quad (31)$$

The clipping enforces (5); the sparsity restriction makes the entropy finite and explicit.

Definition 15 (Hierarchical composition model). *Let $\mathcal{P} \subset (0, \infty) \times \mathbb{N}$ be finite. A level-zero component is a coordinate projection. Recursively, a level- $(\ell + 1)$ component is $g(f_1, \dots, f_K)$, where $(p, K) \in \mathcal{P}$, $g : \mathbb{R}^K \rightarrow \mathbb{R}$ is (p, C) -smooth on the relevant bounded domain, and each f_k is a level- ℓ component. The target f^* belongs to a fixed-level hierarchical composition class of this form.*

Assumption 16 (Network approximation, entropy, and balance). *There are architecture sequences (L_N, r_N, S_N) and $f_N^\circ \in \mathcal{F}_{\text{NN}}(L_N, r_N, S_N, B)$ such that, with*

$$a_N^2 = \max_{(p, K) \in \mathcal{P}} N^{-2p/(2p+K)} \text{polylog}(N),$$

we have $\|f_N^\circ - f^\|_\infty^2 \lesssim a_N^2$ and, uniformly over empirical measures,*

$$\log N \{\epsilon, \mathcal{F}_{\text{NN}}(L_N, r_N, S_N, B), \|\cdot\|_{n,m}\} \lesssim S_N L_N \log(2S_N) \log(CB/\epsilon).$$

The architecture is selected so that

$$\frac{S_N L_N \log(2S_N) L \log N}{N} \lesssim a_N^2.$$

These three displayed conditions isolate the approximation theorem, the entropy theorem, and their required balance; the proof uses no unspecified logarithmic exponent beyond the one encoded in a_N .

The estimator is

$$\hat{f} \in \arg \min_{f \in \mathcal{F}_{\text{NN}}(L_N, r_N, S_N, B)} \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau\{y_{ij} - f(x_{ij})\}. \quad (32)$$

7.1 Results For Neural Networks

Theorem 17 (Neural-network quantile regression). *Under Theorem 14, Assumption 16, and the hierarchical composition condition in Definition 15,*

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ \max_{(p, K) \in \mathcal{P}} N^{-2p/(2p+K)} \text{polylog}(N) + \Gamma_n \right\} \right]. \quad (33)$$

The exponent depends on the local input dimension K of the composition components rather than on the ambient dimension d .

8 Algorithms

We use scaled-form ADMM. The supplied dual “argmax” was unbounded; the correct scaled-dual step is an additive update.

8.1 Quantile Trend Filtering

Let $M = N$, let $\tilde{x}_1 \leq \dots \leq \tilde{x}_M$ be the ordered univariate design points, and let $D_{\tilde{x}}^{(k)}$ denote the standard order- k trend-filtering difference operator on unevenly spaced inputs. The penalized problem is

$$\min_{\theta \in \mathbb{R}^M} \sum_{i,j} w_{ij} \rho_\tau(y_{ij} - \theta_{ij}) + \lambda \|D_{\tilde{x}}^{(k)} \theta\|_1, \quad w_{ij} = \frac{1}{nm_i}. \quad (34)$$

With the split $\theta = z$, the scaled updates are

$$\theta^{t+1} = \arg \min_{\theta} \sum_{i,j} w_{ij} \rho_{\tau}(y_{ij} - \theta_{ij}) + \frac{\varrho}{2} \|\theta - z^t + u^t\|_2^2, \quad (35)$$

$$z^{t+1} = \arg \min_z \frac{1}{2} \|\theta^{t+1} + u^t - z\|_2^2 + \frac{\lambda}{\varrho} \|D_{\tilde{x}}^{(k)} z\|_1, \quad (36)$$

$$u^{t+1} = u^t + \theta^{t+1} - z^{t+1}. \quad (37)$$

The first update is the weighted check-loss proximal map. For $a = z_{ij}^t - u_{ij}^t$,

$$\theta_{ij}^{t+1} = \begin{cases} a + w_{ij}\tau/\varrho, & y_{ij} > a + w_{ij}\tau/\varrho, \\ a - w_{ij}(1-\tau)/\varrho, & y_{ij} < a - w_{ij}(1-\tau)/\varrho, \\ y_{ij}, & \text{otherwise.} \end{cases} \quad (38)$$

Algorithm 1: ADMM for quantile trend filtering

Input: y , ordered design points, λ , τ , ϱ , tolerance κ , maximum iteration count.

Initialize $\theta^0 = z^0 = y$ and $u^0 = 0$;

for $t = 0, 1, 2, \dots$ *until the primal and dual residual criteria are met* **do**

Update θ^{t+1} coordinatewise using (38);
 Update z^{t+1} by the trend-filtering proximal problem (36);
 Update $u^{t+1} = u^t + \theta^{t+1} - z^{t+1}$;

Output: $\hat{\theta} = \theta^{t+1}$.

8.2 Quantile K-NN Fused Lasso

For the graph problem, replace $D_{\tilde{x}}^{(k)}$ by the graph incidence matrix D_{G_K} :

$$\theta^{t+1} = \arg \min_{\theta} \sum_{i,j} w_{ij} \rho_{\tau}(y_{ij} - \theta_{ij}) + \frac{\varrho}{2} \|\theta - z^t + u^t\|_2^2, \quad (39)$$

$$z^{t+1} = \arg \min_z \frac{1}{2} \|\theta^{t+1} + u^t - z\|_2^2 + \frac{\lambda}{\varrho} \|D_{G_K} z\|_1, \quad (40)$$

$$u^{t+1} = u^t + \theta^{t+1} - z^{t+1}. \quad (41)$$

The z update is a graph total-variation proximal problem.

Algorithm 2: ADMM for quantile K -NN graph fused lasso

Input: y , design points, K , λ , τ , ϱ , tolerance κ , maximum iteration count.

Construct the symmetric union K -NN graph and its incidence matrix D_{G_K} ;

Initialize $\theta^0 = z^0 = y$ and $u^0 = 0$;

for $t = 0, 1, 2, \dots$ *until the primal and dual residual criteria are met* **do**

Update θ^{t+1} coordinatewise using the weighted check-loss proximal map;
 Update z^{t+1} by (40);
 Update $u^{t+1} = u^t + \theta^{t+1} - z^{t+1}$;

Output: $\hat{\theta} = \theta^{t+1}$.

9 Experiments

The experimental designs, numerical values, tuning grids, and substantive interpretations are retained. Edits in this section are limited to grammar, notation, and reproducibility statements; no new experimental result is inserted.

This section provides numerical evidence to support our theoretical results. We start by conducting empirical experiments to evaluate the performance of the penalized estimators for Trend Filtering, High Dimensional Quantile Regression and K -NN-FL, respectively, for both settings $d = 1$ and $d > 1$.

To establish a comparison, we include various competing methods as benchmarks.

9.1 Simulated data

To conduct our simulations, we consider various generative models, referred to as scenarios, for both settings $d = 1$ and $d > 1$. We describe these scenarios in the following subsections. Next, we outline the common aspects of our simulations for both the univariate and multivariate settings.

For the $\tau = 0.5$ setting, the performance of each method is evaluated by the MSE obtained when using the true quantile function. We then report the average MSE over 50 Monte Carlo simulations. For other τ , we evaluate the performance using the test set coverage, averaged over 5 train-test splits. In both cases, we then report the average metric over 50 Monte Carlo simulations.

For each setting below, we vary n in $\{100, 200, 400\}$. For each fix value n we then vary a parameter m_{mult} in $\{0.5, 1, 2\}$, to obtain different sets of values for $\{m_i\}_{i=1}^n$.

Specifically, we consider:

- $m_1 = \dots = m_{n/4} = 16m_{mult}$,
- $m_{n/4+1} = \dots = m_{n/2} = 24m_{mult}$,
- $m_{n/2+1} = \dots = m_{3n/4} = 20m_{mult}$,
- $m_{3n/4+1} = \dots = m_n = 10m_{mult}$.

The spatial effect is generated as

$$\delta_i(x) = 0.5\delta_{i-1}(x) + \sum_{t=1}^{25} t^{-1}b_{i,t}h_t(x), \quad (42)$$

where

$$h_t(x) = \left(\frac{1}{\sqrt{2}} \pi \sin(tx(j)) \right)_{j=1}^{25}, \quad (43)$$

are basis functions and $b_{i,t} \sim G_i$ for some distributions G_i . The measurement error is generated as follows: For each $i \in [n]$, the vector $\tilde{e}_i \in \mathbb{R}^M$ is generated recursively as,

$$\tilde{e}_i = 0.3\tilde{e}_{i-1} + \xi_i, \quad (44)$$

where $\{\xi_i\}_{i=1}^n$ are independent errors with $\xi_i \sim F_i$ for some distributions F_i . To form the measurement errors, for each $i \in [n]$, we take the first m_i entries of the vector $\tilde{e}_i$ and denote the resulting vector as e_i , since m_i are varied across i . We note, also, that when $\tau \neq 0.5$, a normalization factor is included for both the spatial noise and measurement error to ensure the variance remains constant across i .

We observe the noisy spatio-temporal data $\{y_{ij}\}_{i=1, j=1}^{n, m_i}$ at design points $\{x_{ij}\}_{j=1}^{m_1}$ evenly sampled from $\text{Unif}([0, 1]^d)$ as follows. First we generate $\{x_{1,j}\}_{j=1}^{m_1} \sim \text{Unif}([0, 1]^d)$, then for any $1 < i \leq n$,

$$x_{ij} = \begin{cases} x_{i-1,j}, & \text{with probability } \phi = 0.1, \\ x_{ij} \sim \text{Unif}([0, 1]^d), & \text{otherwise,} \end{cases} \quad (45)$$

for all $j = 1, \dots, m_i$.

9.2 Univariate Study

For quantile trend filtering, we consider tuning parameters λ such that $\log_{10}(\lambda)$ is in a grid of 50 evenly spaced points between -7 and 1 , where $a, b \in \mathbb{R}$. We then perform Algorithm 1 on λ/M .

To select λ , we use Bayesian Information Criteria (BIC; citeschwarz1978estimating). The BIC for quantile regression citeyu2001bayesian can be computed as

$$\text{BIC}(\tau) = \frac{2}{\sigma} \sum_{i=1}^n \rho_{\tau}(y_i - \hat{\theta}_i) + \nu \log n,$$

where ν denotes the degrees of freedom of the estimator, and $\sigma > 0$ can be empirically chosen as $\sigma = 1 - |1 - 2\tau|$. We estimate ν by the $d + k + 1$, where d is the number of elements of the k -th order difference operator larger than 0.05 in absolute value.

As benchmark methods, we consider the usual (mean regression) penalized trend filtering estimator of order $k = 1$ and $k = 2$ denoted as TF1 and TF2 respectively, and quantile cubic smoothing splines, denoted as QSS. Note that TF1 and TF2 only provide estimates for $\tau = 0.5$.

Mean regression trend filtering is implemented using the `trendfilter` function in the `glmgen` package, and the tuning parameter selected via BIC.

Quantile smoothing splines are implemented with a similar method as Algorithm 1, with step (ii) replaced by the `smooth.spline` function in the `fields` package. A full derivation of this Algorithm is presented in the appendix. The tuning parameter is selected via K -fold cross validation.

For all of the following scenarios, we consider $f^* = f - \int_0^1 f(x)dx$ for the following sinusoidal function, which is constructed to have with nonhomogeneous smoothness:

$$f(x) = 2x^{\frac{1}{4}} \sin\left(\frac{2\pi}{(x + 0.1)^{\frac{1}{2}}}\right)$$

1. **Scenario 1:** We take $G_i \sim N(0, 1)$ and $F_i \sim N(0, 0.5)$ This scenario may serve as a baseline.
2. **Scenario 2:** We take $G_i \sim \text{laplace}(0, 2)$ and $F_i \sim t(3)$. Here $t(3)$ denotes the t -distribution with 3 degrees of freedom.
3. **Scenario 3:** We take $G_i \sim N(0, 1)$, and $F_i \sim N(0, 1)$. However, here we consider $\tau = 0.2$ and $\tau = 0.8$.

Other scenarios, with other underlying functions, are demonstrated in the Appendix.

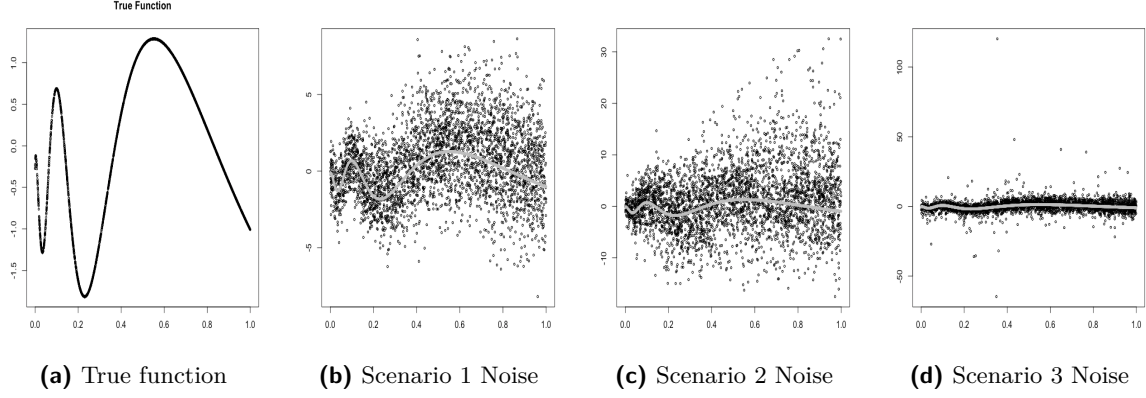


Figure 1: Panel a is the true function f^* . Panels b, c, and d are example plots of the noisy data with $n = 200$ and $m_mult = 1$.

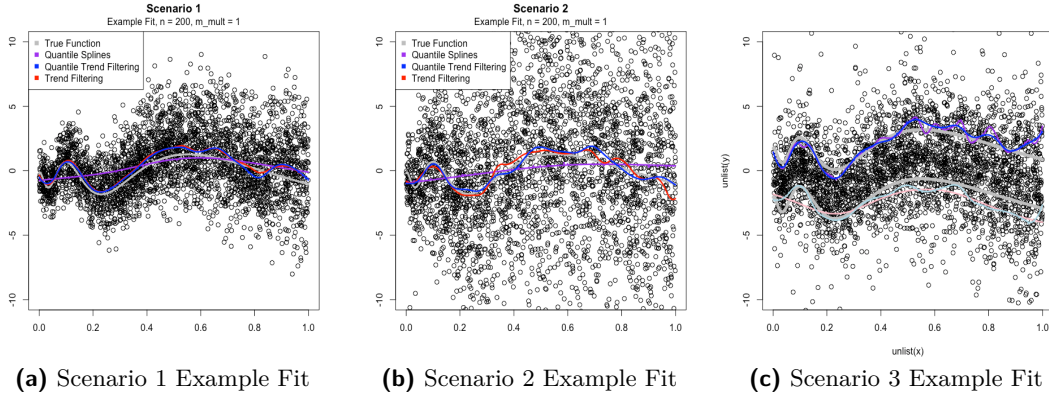


Figure 2: Panels a, b, and c are examples of the fitted curves of TF, QTF, and QSS with $n = 200$ and $m_mult = 1$.

9.3 Multivariate Study

For all of the following scenarios, we consider $f^* = f - \int_0^1 f(x)dx$ for the following piecewise constant function:

$$f_0(x) = \begin{cases} 1 & \text{if } \|x - \frac{1}{4}\mathbf{1}_d\|_2 < \|x - \frac{3}{4}\mathbf{1}_d\|_2, \\ -1 & \text{otherwise,} \end{cases}$$

1. **Scenario 1:** We take $G_i \sim N(0, 1)$, $F_i \sim N(0, 0.5)$, and $d = 2$. This scenario may serve as a baseline.
2. **Scenario 2:** We take $G_i \sim t(2)$, $F_i \sim \text{cauchy}(0, 1)$, and $d = 2$. $t(2)$ denotes the t-distribution with 2 degrees of freedom.
3. **Scenario 3:** We take $G_i \sim N(0, 1)$, $F_i \sim N(0, 1)$, and consider $\tau = 0.2$ and $\tau = 0.8$ $d = 2$.
4. **Scenario 4:** We take $G_i \sim N(0, 2)$, and $F_i \sim t(3)$. However, here we consider $d = 5$ and $d = 10$.

9.4 Non-Convex Study

We also study a **nonconvex setting** by replacing the convex estimators in the multivariate study with neural networks. We consider two neural-network baselines with identical architectures, differing only in the training loss: (i) a squared-error network (SqErr Network) trained by minimizing the mean squared error loss, and (ii) a quantile network (Quantile Network) trained by minimizing the pinball loss for a set of target quantile levels $\mathcal{Q} = \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

We consider three spatial-temporal scenarios (Scenarios 6–8) in the multivariate setting with $d = 2$. All three scenarios follow the common sampling scheme described in Section 9.1 (Simulated data), including the heterogeneous per-time sample sizes $\{m_i\}_{i=1}^n$, the temporally dependent design points $\{x_{ij}\}$, and the spatio-temporal noise components $\delta_i(\cdot)$ and e_i . Specifically, for each $i \in [n]$ and $j \in [m_i]$, the response is generated as

$$y_{ij} = \beta(x_{ij}) + \delta_i(x_{ij}) + e_{ij}, \quad (46)$$

where the signal $\beta(\cdot)$ is the piecewise-constant function

$$\beta(x) = \begin{cases} 1, & \text{if } \|x - \frac{1}{4}\mathbf{1}_d\|_2 < \|x - \frac{3}{4}\mathbf{1}_d\|_2, \\ -1, & \text{otherwise.} \end{cases} \quad (47)$$

Design points and flattening. For a fixed temporal length n , we first generate a padded array $\{x_{i\ell}\}_{\ell=1}^M$ with $M = \max_i m_i$ by sampling $x_{1\ell} \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([0, 1]^d)$ and then, for $i \geq 2$, copying each previous location with probability $\phi = 0.1$ (as in (45) in Section 9.1); we only keep the first m_i locations at time i , i.e., $\{x_{ij}\}_{j=1}^{m_i}$. The training set used by neural networks is obtained by flattening the panel data $\{(x_{ij}, y_{ij})\}_{i=1, j=1}^{n, m_i}$ into i.i.d.-style pairs:

$$X_{\text{train}} = [x_{11}; \dots; x_{1m_1}; x_{21}; \dots; x_{nm_n}] \in \mathbb{R}^{(\sum_i m_i) \times d}, \quad y_{\text{train}} = [y_{11}, \dots, y_{nm_n}]^\top \in \mathbb{R}^{\sum_i m_i}.$$

The test set is formed in the same way using an independently generated panel of design points.

Spatial noise. The spatio-temporal component $\delta_i(\cdot)$ follows the recursive construction in (42):

$$\delta_i(x) = 0.5 \delta_{i-1}(x) + \sum_{t=1}^{25} t^{-1} b_{i,t} h_t(x), \quad b_{i,t} \sim G_i, \quad (48)$$

where $\{h_t\}_{t=1}^{25}$ are sine-basis functions as in (43). In implementation, for each location $x = (x(1), \dots, x(d)) \in [0, 1]^d$, we use

$$h_t(x) = \prod_{r=1}^d \left(\frac{\pi}{\sqrt{2}} \sin(tx(r)) \right), \quad t = 1, \dots, 25. \quad (49)$$

Measurement error. The measurement error is generated via the AR-type recursion in (44):

$$\tilde{e}_i = 0.3 \tilde{e}_{i-1} + \xi_i, \quad \xi_i \sim F_i, \quad (50)$$

where $\tilde{e}_i \in \mathbb{R}^M$, and we set e_{ij} to be the j th entry of $\tilde{e}_i$ for $j \leq m_i$.

Scenario specifications. The scenarios differ only through the innovation distributions G_i and F_i and the quantile level τ :

- **Scenario 6 (light-tailed):** $b_{i,t} \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$ and $\xi_i(\ell) \stackrel{\text{i.i.d.}}{\sim} N(0, 0.5^2)$.
- **Scenario 7 (heavy-tailed):** $b_{i,t} \stackrel{\text{i.i.d.}}{\sim} t(5)$ and $\xi_i(\ell) \stackrel{\text{i.i.d.}}{\sim} t(5)$, leading to heavier tails in both $\delta_i(\cdot)$ and e_i .

- **Scenario 8 :** we consider $\tau = 0.5$ and define the target quantile signal as $\beta_\tau(x) = f_0(x) + \Phi^{-1}(\tau)$. The spatial-noise coefficients follow an AR-type recursion: for $t = 1, \dots, 25$, let $z_{i,t} \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$ and set

$$b_{i,t} = \frac{0.5 b_{i-1,t} + t^{-1} z_{i,t}}{s_\delta}, \quad s_\delta = \sqrt{0.5^2 + 1^2},$$

(with $s_\delta = 1$ at $i = 1$), and define $\delta_i(x) = \sum_{t=1}^{25} b_{i,t} h_t(x)$. The measurement error is generated by $u_{i,\ell} \stackrel{\text{i.i.d.}}{\sim} t(3)$ and

$$\tilde{e}_i(\ell) = \frac{0.3 \tilde{e}_{i-1}(\ell) + u_{i,\ell}}{s_e}, \quad s_e = \sqrt{0.3^2 + 1^2},$$

(with $s_e = 1$ at $i = 1$), and we take $e_{ij} = \tilde{e}_i(j)$ for $j \leq m_i$. Finally, $y_{ij} = \beta_\tau(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$.

Evaluation metric. For each quantile level $q \in \mathcal{Q}$, we evaluate the squared error between the learned quantile function and the oracle quantile function on a large test set, and report the average over Monte Carlo trials. We vary the temporal length $n \in \{100, 500, 1000\}$ (with the corresponding $\{m_i\}$ pattern described above).

Summary of findings. Table 1 shows that the Quantile Network consistently outperforms the SqErr Network across all three scenarios and all quantile levels. The improvements are particularly pronounced for tail quantiles ($q = 0.1$ and $q = 0.9$), where minimizing squared error is misaligned with the quantile target and heavy-tailed errors exacerbate the discrepancy. In contrast, the pinball loss directly targets the conditional quantiles and yields substantially smaller errors, especially in Scenarios 6 and 8.

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Table 1: Non-convex study: test-set MSE between predicted and oracle quantile functions for five quantile levels $\mathcal{Q} = \{0.1, 0.3, 0.5, 0.7, 0.9\}$. We compare a squared-error network (SqErr) and a quantile network (Quantile). The best performance is **bolded**.

Scenario	Method (n)	$q = 0.1$	$q = 0.3$	$q = 0.5$	$q = 0.7$	$q = 0.9$
Scenario 6	SqErr (100)	21.717	6.522	3.060	5.422	19.034
	Quantile (100)	3.702	2.486	2.457	3.097	7.141
	SqErr (500)	23.058	7.071	3.059	4.870	17.690
	Quantile (500)	2.625	2.243	2.205	2.333	2.799
	SqErr (1000)	23.409	7.220	3.067	4.737	17.355
	Quantile (1000)	2.516	2.251	2.150	2.136	2.308
Scenario 7	SqErr (100)	37.480	9.810	4.067	7.505	30.190
	Quantile (100)	4.913	3.982	4.513	5.058	7.696
	SqErr (500)	37.040	9.618	4.031	7.630	30.553
	Quantile (500)	3.864	3.730	3.810	4.186	5.119
	SqErr (1000)	32.095	7.612	3.893	9.374	35.126
	Quantile (1000)	4.742	3.965	3.687	3.792	4.796
Scenario 8	SqErr (100)	19.419	5.226	2.750	7.078	26.661
	Quantile (100)	2.409	2.079	1.854	1.883	2.237
	SqErr (500)	22.925	6.557	2.557	5.264	22.440
	Quantile (500)	1.803	1.631	1.644	1.667	1.805
	SqErr (1000)	22.870	6.533	2.556	5.287	22.498
	Quantile (1000)	2.018	1.671	1.643	1.637	1.696

A Proof of Theorem 6

This appendix gives a complete proof of the convex theorem using the same weights as the empirical criterion. In particular, all dependence is handled at the level of the joint time cluster

$$W_i = (X_i, \delta_i, E_i), \quad X_i = (x_{i1}, \dots, x_{im_i}), \quad E_i = (e_{i1}, \dots, e_{im_i}),$$

and every empirical process is centered at f^* . No separate coupling of X_i , δ_i , and E_i is used.

A.1 Losses, weights, and local classes

Write

$$N = \sum_{i=1}^n m_i, \quad w_{ij} = \frac{1}{nm_i}, \quad h_f = f - f^*.$$

Assumption 1.7 implies that there are constants $0 < c_w < C_w < \infty$ such that

$$\frac{c_w}{N} \leq w_{ij} \leq \frac{C_w}{N} \quad \text{for every } (i, j). \quad (51)$$

Let

$$\ell_{ij,f} = \rho_\tau\{y_{ij} - f(x_{ij})\} - \rho_\tau\{y_{ij} - f^*(x_{ij})\}, \quad (52)$$

$$\hat{Z}(f) = \sum_{i=1}^n \sum_{j=1}^{m_i} w_{ij} \ell_{ij,f}, \quad \bar{Z}(f) = \mathbb{E} \hat{Z}(f). \quad (53)$$

For $t > 0$, define the Huber-localized and $L_2(Q)$ -localized classes

$$\mathcal{F}_\Delta(t) = \{f \in \mathcal{F} : c_0 \Delta^2(h_f) \leq t^2\}, \quad (54)$$

$$\mathcal{H}_2(r) = \{h_f : f \in \mathcal{F}, \|h_f\|_2 \leq r\}. \quad (55)$$

The classes used below are centered because they contain $h_{f^*} = 0$.

Lemma 18 (Knight identity and curvature). *Under Assumption 2, for every measurable f satisfying $\|f\|_\infty \leq B$,*

$$\bar{Z}(f) \geq C_0 \Delta^2(h_f), \quad C_0 = \underline{f} \min \left\{ \frac{1}{2}, \frac{R}{4} \right\}. \quad (56)$$

Moreover, if $\|h_f\|_\infty \leq R$, then

$$\bar{Z}(f) \leq \frac{\bar{f}}{2} \|h_f\|_2^2. \quad (57)$$

Proof. Let $U_{ij} = y_{ij} - f^*(x_{ij})$ and $h = h_f(x_{ij})$. Knight's identity gives

$$\rho_\tau(U_{ij} - h) - \rho_\tau(U_{ij}) = -h\{\tau - \mathbb{1}(U_{ij} \leq 0)\} + \int_0^h \{\mathbb{1}(U_{ij} \leq u) - \mathbb{1}(U_{ij} \leq 0)\} du.$$

After conditioning on $x_{ij} = x$, the linear term vanishes because $\mathbb{P}(U_{ij} \leq 0 \mid x_{ij} = x) = \tau$. Thus

$$\mathbb{E}(\ell_{ij,f} \mid x_{ij} = x) = \int_0^{h_f(x)} \{F_{U,ij}(u \mid x) - F_{U,ij}(0 \mid x)\} du. \quad (58)$$

If $|h_f(x)| \leq R$, Assumption 2 bounds this integral below by $\underline{f} h_f(x)^2/2$. If $h_f(x) > R$, monotonicity of the conditional CDF and the same assumption at $R/2$ imply

$$\int_0^{h_f(x)} \{F_{U,ij}(u \mid x) - F_{U,ij}(0 \mid x)\} du \geq \frac{\underline{f}R}{2} \{h_f(x) - R/2\} \geq \frac{\underline{f}R}{4} |h_f(x)|.$$

The case $h_f(x) < -R$ follows by reversing the direction of integration. Since $|z| \geq z^2$ when $|z| \leq 1$, these bounds imply

$$\mathbb{E}(\ell_{ij,f} \mid x_{ij} = x) \geq C_0 \min\{|h_f(x)|, h_f(x)^2\}.$$

Integrating over Q and averaging with the weights w_{ij} proves (56). The upper bound follows from the upper inequality in Assumption 2 applied to (58). $\square$

Lemma 19 (Comparison with squared prediction loss). *If $\|f\|_\infty, \|f^*\|_\infty \leq B$, then*

$$\|h_f\|_2^2 \leq b_B^2 \Delta^2(h_f), \quad b_B^2 = 1 \vee 2B. \quad (59)$$

Consequently,

$$\{h_f : f \in \mathcal{F}_\Delta(t)\} \subseteq \mathcal{H}_2(c_0^{-1/2} b_B t). \quad (60)$$

Proof. Pointwise, $|h_f| \leq 2B$ and $h_f^2 \leq (1 \vee 2B) \min\{|h_f|, h_f^2\}$. Integrating proves (59); (60) follows immediately. $\square$

A.2 Interlaced blocks and joint-cluster coupling

For an integer $L \geq 1$ and $\ell \in [L]$, define

$$\mathcal{J}_{e,\ell} = \{\ell + 2qL : q \geq 0\} \cap [n], \quad (61)$$

$$\mathcal{J}_{o,\ell} = \{\ell + (2q+1)L : q \geq 0\} \cap [n]. \quad (62)$$

These $2L$ sets partition $[n]$, and consecutive indices within a fixed set are separated by $2L$. Under Assumption 1.4, we take

$$L = L_n = \left\lceil \frac{2}{C_\beta} \log n \right\rceil. \quad (63)$$

Lemma 20 (Joint Berbee coupling). *On an enlarged probability space, for every block $I \in \{\mathcal{J}_{e,\ell}, \mathcal{J}_{o,\ell} : \ell \in [L]\}$ there are random elements $\{W_i^* : i \in I\}$ such that:*

- (i) W_i^* has the same marginal law as W_i ;
- (ii) the variables $\{W_i^* : i \in I\}$ are mutually independent;
- (iii)

$$\mathbb{P}\{W_i \neq W_i^* \text{ for some } i \in I\} \leq (|I| - 1)\beta_W(2L). \quad (64)$$

If Ω_β denotes the event that all coupled blocks agree with their original blocks, then

$$\mathbb{P}(\Omega_\beta^c) \leq n\beta_W(2L) = O(n^{-3}). \quad (65)$$

The coupled copies need not be independent across different blocks; the proof only uses independence within one block at a time.

Proof. Apply Berbee's coupling lemma successively to the subsequence $\{W_i : i \in I\}$. Its consecutive elements are separated by $2L$, which gives (64). Summing the block errors yields at most $n\beta_W(2L)$. The exponential mixing bound and (63) give the last order statement. $\square$

A.3 Conditional decomposition of the empirical process

For each (i, j) and f , define

$$\tilde{Z}_{ij}(f) = \mathbb{E}(\ell_{ij,f} \mid X_i, \delta_i), \quad (66)$$

$$Z_{ij}(f) = \mathbb{E}(\ell_{ij,f} \mid x_{ij}), \quad (67)$$

$$\bar{Z}_{ij}(f) = \mathbb{E}\ell_{ij,f}. \quad (68)$$

The distinction between conditioning on (X_i, δ_i) and conditioning on x_{ij} is essential.

Lemma 21 (Exact spatial-effect identity). *Let*

$$D_i(x, u) = \mathbb{P}(U_{i1} \leq u \mid x_{i1} = x) - \mathbb{P}(U_{i1} \leq u \mid \delta_i, x_{i1} = x), \quad (69)$$

where $U_{ij} = \delta_i(x_{ij}) + e_{ij}$. Then

$$Z_{ij}(f) - \tilde{Z}_{ij}(f) = r_{ij}(f) := \int_0^{h_f(x_{ij})} D_i(x_{ij}, u) du. \quad (70)$$

For every fixed (x, δ_i) , the map $z \mapsto \int_0^z D_i(x, u) du$ is one-Lipschitz and vanishes at zero.

Proof. Conditional Knight identities give

$$\begin{aligned} Z_{ij}(f) &= \int_0^{h_f(x_{ij})} \mathbb{P}(U_{ij} \leq u \mid x_{ij}) du - \tau h_f(x_{ij}), \\ \tilde{Z}_{ij}(f) &= \int_0^{h_f(x_{ij})} \mathbb{P}(U_{ij} \leq u \mid \delta_i, x_{ij}) du - \tau h_f(x_{ij}). \end{aligned}$$

The linear terms therefore cancel exactly, proving (70). Since $|D_i(x, u)| \leq 1$, the Lipschitz assertion follows. $\square$

For a block I , define the weighted centered process

$$\mathbb{G}_I(f) = \sum_{i \in I} \sum_{j=1}^{m_i} w_{ij} \{\bar{Z}_{ij}(f) - \ell_{ij,f}\}. \quad (71)$$

The identity

$$\bar{Z}_{ij}(f) - \ell_{ij,f} = \{\bar{Z}_{ij}(f) - Z_{ij}(f)\} + \{Z_{ij}(f) - \tilde{Z}_{ij}(f)\} + \{\tilde{Z}_{ij}(f) - \ell_{ij,f}\} \quad (72)$$

is used below.

Lemma 22 (Pointwise symmetrization terms). *Let I be a coupled block and let ε_{ij} be independent Rademacher variables, independent of the coupled data. For every centered class $\mathcal{H}$,*

$$\mathbb{E} \sup_{h_f \in \mathcal{H}} \left| \sum_{i \in I} \sum_j w_{ij} \{\tilde{Z}_{ij}(f) - \ell_{ij,f}\} \right| \leq 4\mathcal{R}_I(\mathcal{H}), \quad (73)$$

$$\mathbb{E} \sup_{h_f \in \mathcal{H}} \left| \sum_{i \in I} \sum_j w_{ij} \{\bar{Z}_{ij}(f) - Z_{ij}(f)\} \right| \leq 4\mathcal{R}_I(\mathcal{H}), \quad (74)$$

where

$$\mathcal{R}_I(\mathcal{H}) = \mathbb{E} \sup_{h \in \mathcal{H}} \left| \sum_{i \in I} \sum_{j=1}^{m_i} w_{ij} \varepsilon_{ij} h(x_{ij}^*) \right|. \quad (75)$$

Furthermore, with

$$R_i(f) = \mathbb{E}\{r_{ij}(f) \mid \delta_i\}, \quad (76)$$

we have

$$\mathbb{E} \sup_{h_f \in \mathcal{H}} \left| \sum_{i \in I} \sum_j w_{ij} \{r_{ij}(f) - R_i(f)\} \right| \leq 4\mathcal{R}_I(\mathcal{H}). \quad (77)$$

Proof. For (73), condition on $(X_i^*, \delta_i^*)_{i \in I}$. The measurement errors are independent across all (i, j) in the coupled block by Assumption 1.3. Conditional symmetrization gives a factor two, and the check-loss difference is one-Lipschitz in $h_f(x_{ij}^*)$ and equals zero for $h_f = 0$. The scalar contraction inequality gives the second factor two.

For (74), $Z_{ij}(f)$ is a deterministic one-Lipschitz function of $h_f(x_{ij})$. The design points are independent within a coupled block, so ordinary symmetrization and contraction apply. Finally, conditional on δ_i^* , the variables x_{ij}^* are independent and the map in Lemma 21 is one-Lipschitz. The same argument proves (77). $\square$

A.4 Uniform control of the blockwise spatial mean

For a block I , set

$$\Pi_I(x, u) = \frac{1}{|I|} \sum_{i \in I} D_i(x, u). \quad (78)$$

By the tower property, $\mathbb{E}D_i(x, u) = 0$ for every (x, u) .

Lemma 23 (Grid bound for Π_I). *Suppose Assumptions 3 and 4 hold. There is a constant C_Π depending only on the uniform Lipschitz constant such that, for every coupled block I and every $z \geq 1$,*

$$\mathbb{P} \left[\sup_{x \in [0,1]^d, |u| \leq T_n} |\Pi_I(x, u)| > C_\Pi \sqrt{\frac{(s+1) \log\{n(1+T_n)\} + z}{|I|}} \right] \leq 2e^{-z}. \quad (79)$$

Consequently, simultaneously over all $2L$ blocks,

$$\max_I \sup_{x, |u| \leq T_n} |\Pi_I(x, u)| = O_{\mathbb{P}}(\sqrt{\Gamma_n}), \quad \Gamma_n = \frac{(s+1)L \log\{n(1+T_n)\}}{n}. \quad (80)$$

Proof. Only the coordinates x_S matter. Cover $[0, 1]^s \times [-T_n, T_n]$ by a rectangular grid with mesh q^{-1} , where $q = \lceil n(1+T_n) \rceil$. Its cardinality is at most $Cq^s\{q(1+T_n)\}$. At each grid point, the summands D_i are independent, centered, and bounded by one, so Hoeffding's inequality and a union bound give

$$\max_{\text{grid}} |\Pi_I| \lesssim \sqrt{\frac{(s+1) \log\{n(1+T_n)\} + z}{|I|}}$$

with probability at least $1 - 2e^{-z}$. The Lipschitz condition adds at most C/q , which is smaller than the displayed stochastic term. Since $|I| \asymp n/L$, a union bound over the $2L$ blocks proves (80). $\square$

Lemma 24 (Spatial-mean contribution). *On the event in (80), for every $t > 0$,*

$$\sup_{f \in \mathcal{F}_\Delta(t)} \left| \sum_I \sum_{i \in I} \sum_j w_{ij} R_i(f) \right| \leq Cc_0^{-1/2} b_B t \sqrt{\Gamma_n}. \quad (81)$$

Hence, for every $\eta > 0$,

$$Cc_0^{-1/2} b_B t \sqrt{\Gamma_n} \leq \eta t^2 + C_\eta b_B^2 \Gamma_n. \quad (82)$$

Proof. Because $T_n = 2B$ and $|h_f| \leq 2B$, Fubini's theorem and (78) give

$$\sum_{i \in I} \sum_j w_{ij} R_i(f) = \frac{|I|}{n} \int_{[0,1]^d} \int_0^{h_f(x)} \Pi_I(x, u) du dQ(x).$$

Its absolute value is at most

$$\frac{|I|}{n} \sup_{x, u} |\Pi_I(x, u)| \|h_f\|_1 \leq \frac{|I|}{n} \sup_{x, u} |\Pi_I(x, u)| \|h_f\|_2.$$

The blocks partition $[n]$, and Lemma 19 gives $\|h_f\|_2 \leq c_0^{-1/2} b_B t$. Summing over the blocks proves (81); Young's inequality gives (82). $\square$

A.5 Localized empirical-process inequality

Proposition 25 (Localized process bound). *There are constants $C_1, C_2 < \infty$ such that, for every $t > 0$,*

$$\begin{aligned} & \mathbb{E} \left[\sup_{f \in \mathcal{F}_\Delta(t)} |\bar{Z}(f) - \hat{Z}(f)| \mathbb{1}_{\Omega_\beta} \right] \\ & \leq C_1 \sum_{\ell=1}^L \sum_{a \in \{e, o\}} \mathcal{R}_{a, \ell} \{ \mathcal{H}_2(c_0^{-1/2} b_B t) \} + C_2 c_0^{-1/2} b_B t \sqrt{\Gamma_n}. \end{aligned} \quad (83)$$

Proof. Apply (72) block by block. The measurement error term and the design term are bounded by (73) and (74). For the middle term, use Lemma 21, split $r_{ij}(f)$ into $r_{ij}(f) - R_i(f)$ and $R_i(f)$, apply (77) to the first part and Lemma 24 to the second part. Finally use the local inclusion (60). All suprema are nonnegative because the centered class contains zero, so the coupling indicator can be removed after passing to the coupled block. $\square$

A.6 Convex localization and completion of the proof

Lemma 26 (Convex boundary reduction). *Suppose $\mathcal{F}$ is convex, $f^* \in \mathcal{F}$, and $\hat{f}$ minimizes (13). If $c_0 \Delta^2(\hat{f} - f^*) > t^2$, there exists $f_t \in \mathcal{F}_\Delta(t)$ satisfying $c_0 \Delta^2(f_t - f^*) = t^2$ and $\hat{Z}(f_t) \leq 0$. Consequently,*

$$\{c_0 \Delta^2(\hat{f} - f^*) > t^2\} \subseteq \left\{ \sup_{f \in \mathcal{F}_\Delta(t)} |\bar{Z}(f) - \hat{Z}(f)| \geq t^2 \right\}. \quad (84)$$

Proof. The map $\mu \mapsto c_0 \Delta^2\{\mu(\hat{f} - f^*)\}$ is continuous, vanishes at zero, and exceeds t^2 at one. Choose $\mu_t \in (0, 1)$ at the first crossing and let $f_t = f^* + \mu_t(\hat{f} - f^*)$. Convexity of $\mathcal{F}$ gives $f_t \in \mathcal{F}$, while convexity of the empirical check loss gives

$$\hat{Z}(f_t) \leq (1 - \mu_t) \hat{Z}(f^*) + \mu_t \hat{Z}(\hat{f}) \leq 0.$$

By Lemma 18, $\bar{Z}(f_t) \geq C_0 \Delta^2(f_t - f^*) \geq t^2$. Therefore $|\bar{Z}(f_t) - \hat{Z}(f_t)| \geq t^2$, proving (84). $\square$

Lemma 27 (Critical-radius implication). *Let*

$$\Psi_n(r) = C_* \left[\sum_{\ell=1}^L \sum_{a \in \{e, o\}} \mathcal{R}_{a, \ell} \{ \mathcal{H}_2(c_0^{-1/2} b_B r) \} + b_B^2 \Gamma_n \right], \quad (85)$$

where C_* is a sufficiently large universal constant, and let $r_n = \inf\{r > 0 : \Psi_n(r) \leq r^2/64\}$. If a sub-root majorant of the first term is used, then for every $M \geq 1$,

$$\mathbb{P}\{c_0 \Delta^2(\hat{f} - f^*) > M^2 r_n^2\} \leq \frac{C}{M} + \mathbb{P}(\Omega_\beta^c) + o(1). \quad (86)$$

Proof. Apply Lemma 26 with $t = M r_n$, split according to Ω_β , and use Markov's inequality together with Proposition 25. Apply (82) with a sufficiently small η and absorb the resulting multiple of t^2 into the left-hand side. The sub-root property gives $\Psi_n(M r_n) \leq M \Psi_n(r_n) \leq M r_n^2/64$, whereas the threshold is $M^2 r_n^2$. This yields the C/M term. The grid event from Lemma 23 has probability tending to one, and the coupling error is handled separately. $\square$

Proof of Theorem 6. Lemma 27 implies $\Delta^2(\hat{f} - f^*) = O_{\mathbb{P}}(r_n^2)$. The norm comparison (59) then gives

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}(b_B^2 r_n^2).$$

The explicit coupling term follows from (65). This proves (14). $\square$

B Trend Filtering

B.1 Localized entropy calculation

For a block I , let $M_I = \sum_{i \in I} m_i$ and define

$$\|g\|_I^2 = M_I^{-1} \sum_{i \in I} \sum_{j=1}^{m_i} g(x_{ij})^2.$$

By (51), the full-objective block process is bounded by a constant times M_I/N multiplied by the usual within-block empirical Rademacher process.

Lemma 28 (Trend-filtering block complexity). *Assume the entropy bound in Theorem 9. Uniformly over the interlaced blocks,*

$$\mathcal{R}_I\{\mathcal{H}_2(r) \cap (\mathcal{F}_k(V) - f^*)\} \lesssim \frac{\sqrt{M_I}}{N} V^{1/(2k)} r^{1-1/(2k)} + \frac{B}{N}. \quad (87)$$

Consequently,

$$\sum_I \mathcal{R}_I\{\mathcal{H}_2(r) \cap (\mathcal{F}_k(V) - f^*)\} \lesssim V^{1/(2k)} r^{1-1/(2k)} \sqrt{\frac{L}{N}} + \frac{BL}{N}. \quad (88)$$

Proof. Condition on the design. Dudley's entropy integral, truncated below at a constant multiple of B/M_I , yields

$$\begin{aligned} & \mathbb{E}_\varepsilon \sup_{\substack{f \in \mathcal{F}_k(V) \\ \|f - f^*\|_I \leq Cr}} \left| \frac{1}{M_I} \sum_{i \in I} \sum_j \varepsilon_{ij} \{f(x_{ij}) - f^*(x_{ij})\} \right| \\ & \lesssim \frac{1}{\sqrt{M_I}} \int_0^{Cr} \sqrt{V^{1/k} \epsilon^{-1/k}} d\epsilon + \frac{B}{M_I} \\ & \lesssim \frac{V^{1/(2k)} r^{1-1/(2k)}}{\sqrt{M_I}} + \frac{B}{M_I}. \end{aligned}$$

Multiplication by M_I/N proves (87). Since $M_I \asymp N/L$ and there are $2L$ blocks, summing proves (88). $\square$

Proof of Theorem 9. Insert (88) into the critical-radius inequality. Ignoring the lower-order term BL/N , the stochastic fixed point is

$$r^2 \asymp V^{1/(2k)} r^{1-1/(2k)} \sqrt{L/N}.$$

Equivalently,

$$r^2 \asymp V^{2/(2k+1)} (L/N)^{2k/(2k+1)}.$$

Adding Γ_n and applying Theorem 6 proves (17). $\square$

C High-dimensional Quantile Linear Regression

Lemma 29 (Weighted linear-class complexity). *Let I be an interlaced block and let X_I^w be the weighted matrix from (19). Then*

$$\mathcal{R}_I\{x \mapsto x^\top (\beta - \beta^*) : \|\beta\|_1, \|\beta^*\|_1 \leq V\} \leq \frac{2V \sqrt{2 \log(2d)}}{n} \mathbb{E} \max_{q \in [d]} \|X_{I,q}^w\|_2. \quad (89)$$

Proof. For $b = \beta - \beta^*$, $\|b\|_1 \leq 2V$, and therefore

$$\begin{aligned} \mathcal{R}_I &\leq \mathbb{E} \sup_{\|b\|_1 \leq 2V} \left| \frac{1}{n} \sum_{i \in I} \sum_{j=1}^{m_i} \frac{\varepsilon_{ij}}{m_i} x_{ij}^\top b \right| \\ &= \frac{2V}{n} \mathbb{E} \left\| \sum_{i \in I} \sum_j \frac{\varepsilon_{ij}}{m_i} x_{ij} \right\|_\infty. \end{aligned}$$

Conditional on the design, the q th coordinate is sub-Gaussian with variance proxy $\|X_{I,q}^w\|_2^2$. The standard maximal inequality for d sub-Gaussian variables proves (89). $\square$

Proof of Theorem 11. Sum (89) over the $2L$ blocks and insert the result into Theorem 6. This gives (20). If $|x_{ij,q}| \leq C_x$ and $m_i \asymp m$, then

$$\|X_{I,\cdot,q}^w\|_2^2 = \sum_{i \in I} \sum_j \frac{x_{ij,q}^2}{m_i^2} \lesssim \frac{|I|}{m} \asymp \frac{n}{Lm}.$$

Thus $n^{-1} \mathcal{A}_n \lesssim \sqrt{L/(nm)} = \sqrt{L/N}$. The theorem concerns prediction error. A bound for $\|\hat{\beta} - \beta^*\|$ would require a separate identifiability condition on the design. $\square$

D Quantile K -NN Fused Lasso

D.1 A graph-complexity inequality

Let $G = (V, E)$ be a connected graph on N vertices, let D_G be an oriented incidence matrix, and let $D_G^\dagger$ be its Moore–Penrose inverse. Define

$$\rho(G) = \max_{e \in E} \|(D_G^\dagger)_{\cdot,e}\|_2. \quad (90)$$

This is a deterministic graph parameter. The revised theorem states its required bound explicitly rather than hiding it inside an unspecified “polylogarithmic graph event.”

Proposition 30 (Weighted graph-TV complexity). *Suppose (51) holds and every interlaced block contains $M_I \asymp N/L$ vertices. For*

$$\mathcal{U}(r, V) = \{u \in \mathbb{R}^N : \|u\|_{n,m} \leq r, \|D_G u\|_1 \leq 2V\},$$

we have

$$\sum_I \mathcal{R}_I\{\mathcal{U}(r, V)\} \lesssim r \sqrt{\frac{L}{N}} + \frac{LV \rho(G) \sqrt{\log(2|E|)}}{N}. \quad (91)$$

Proof. For a block I , let $a_I \in \mathbb{R}^N$ have coordinates $(a_I)_v = w_v \varepsilon_v \mathbb{1}(v \in I)$. Since G is connected,

$$u = \bar{u} \mathbf{1}_N + D_G^\dagger D_G u, \quad \bar{u} = N^{-1} \mathbf{1}_N^\top u.$$

Hence

$$|a_I^\top u| \leq |\bar{u}| |\mathbf{1}_N^\top a_I| + \|(D_G^\dagger)^\top a_I\|_\infty \|D_G u\|_1. \quad (92)$$

The balance condition gives $\|u\|_2 \lesssim \sqrt{Nr}$, so $|\bar{u}| \lesssim r$. Also,

$$\mathbb{E}|\mathbf{1}_N^\top a_I| \leq \left(\sum_{v \in I} w_v^2 \right)^{1/2} \lesssim (NL)^{-1/2}.$$

For each edge e , the random variable $\{(D_G^\dagger)^\top a_I\}_e$ is sub-Gaussian with variance proxy at most $C\rho(G)^2/N^2$. Therefore

$$\mathbb{E}\|(D_G^\dagger)^\top a_I\|_\infty \lesssim \frac{\rho(G)\sqrt{\log(2|E|)}}{N}.$$

Taking the supremum in (92) and summing over the $2L$ blocks proves (91). $\square$

D.2 Graph geometry and target total variation

The following deterministic conditions are the graph events invoked in Assumption 12. They are stated explicitly so that every rate in the theorem is tied to a checkable quantity:

$$G_K \text{ is connected and } \rho(G_K) \leq \rho_N, \quad (93)$$

$$\sup_{x \in [0,1]^d} R_K(x) \leq C_R(K/N)^{1/d}, \quad (94)$$

$$Q\{x : x_v \in N_K(x)\} \leq C_O K/N \quad \text{for every vertex } v. \quad (95)$$

Here $R_K(x)$ is the distance from x to its K th nearest sample point.

Lemma 31 (Graph-TV scale for a piecewise-Lipschitz target). *Suppose the maximum degree of G_K is at most $C_D K$, the radius event (94) holds, the empirical fraction of vertices in the r_N -neighborhood of the singularity set is at most $C_S r_N$, and f^* is bounded and piecewise Lipschitz with a discontinuity set whose r -neighborhood has Q -measure at most $C_S r$. Then*

$$\|D_{G_K} \theta^*\|_1 \lesssim N^{1-1/d} K^{1+1/d}, \quad \theta_v^* = f^*(x_v). \quad (96)$$

In particular, if K is polylogarithmic in N , this is $N^{1-1/d} \text{polylog}(N)$.

Proof. Let $r_N = C_R(K/N)^{1/d}$. Vertices outside the r_N -neighborhood of the singularity set have all graph neighbors in the same Lipschitz region, so each incident edge contributes at most $C r_N$. There are at most $C N K$ such edges, giving $C N K r_N$. The number of vertices in the boundary tube is at most $C N r_N$ on the design regularity event; each has degree at most $C_D K$ and each edge difference is at most $2B$. This contributes the same order $C N K r_N$. Substituting $r_N = (K/N)^{1/d}$ proves (96). $\square$

Lemma 32 (Off-sample transfer). *On the events (94)–(95), for every $u \in \mathbb{R}^N$,*

$$\int \left\{ \frac{1}{K} \sum_{x_v \in N_K(x)} u_v \right\}^2 dQ(x) \leq C \|u\|_{n,m}^2. \quad (97)$$

Moreover, for a bounded piecewise-Lipschitz f^* ,

$$\int \left\{ f^*(x) - \frac{1}{K} \sum_{x_v \in N_K(x)} f^*(x_v) \right\}^2 dQ(x) \lesssim (K/N)^{1/d}. \quad (98)$$

Proof. Jensen's inequality and (95) give

$$\begin{aligned} \int \left\{ K^{-1} \sum_{v \in N_K(x)} u_v \right\}^2 dQ(x) &\leq K^{-1} \sum_v u_v^2 Q\{x : v \in N_K(x)\} \\ &\leq CN^{-1} \sum_v u_v^2 \lesssim \|u\|_{n,m}^2. \end{aligned}$$

For the approximation term, set $r_N = C_R(K/N)^{1/d}$. Outside the r_N -tube around the discontinuity set, all neighbors lie in the same Lipschitz component and the squared error is $O(r_N^2)$. Inside the tube, the squared error is at most $4B^2$, while the tube has probability $O(r_N)$. Therefore the integrated error is $O(r_N^2 + r_N) = O(r_N)$. $\square$

Proof of Theorem 13. For $u = \theta - \theta^*$, feasibility gives $\|D_{G_K} u\|_1 \leq 2V_n$. Proposition 30 and the critical-radius theorem yield

$$\|\hat{\theta} - \theta^*\|_{n,m}^2 = O_{\mathbb{P}} \left[b_B^2 \left\{ \frac{L}{N} + \frac{LV_n \rho_N \sqrt{\log(2|E_K|)}}{N} + \Gamma_n \right\} \right]. \quad (99)$$

Apply Lemma 32 to $u = \hat{\theta} - \theta^*$ and add the approximation bound (98). This proves the off-sample statement in the revised Theorem 13. Finally, Lemma 31 gives the displayed specialization when $V_n \asymp N^{1-1/d} K^{1+1/d}$ and ρ_N is polylogarithmic. $\square$

E For $f^* \notin \mathcal{F}$ and a Nonconvex Class

Let $f^\circ \in \mathcal{F}$ be the near-best approximant in (28), write $h^\circ = f^\circ - f^*$, and suppose $\phi_n = \|h^\circ\|_\infty \leq R$. Define the centered star hull

$$\mathcal{H}^* = \{\mu(f - f^*) : f \in \mathcal{F}, \mu \in [0, 1]\}.$$

Lemma 33 (Approximation-adjusted localization). *Let $\hat{f}$ minimize (29). There is a constant C_A depending only on the curvature constants such that, if*

$$c_0 \Delta^2(\hat{f} - f^*) > t^2 + C_A \phi_n^2, \quad (100)$$

then there is $\tilde{h} \in \mathcal{H}^$ satisfying $c_0 \Delta^2(\tilde{h}) = t^2 + C_A \phi_n^2$ and*

$$t^2 \leq 2 \sup_{\substack{h \in \mathcal{H}^* \\ c_0 \Delta^2(h) \leq t^2 + C_A \phi_n^2}} |(\mathbb{P} - \mathbb{P}_n)\ell_h|. \quad (101)$$

Proof. Choose $\mu \in (0, 1)$ so that $\tilde{h} = \mu(\hat{f} - f^*)$ lies on the boundary in (100). Convexity of the check loss and optimality of $\hat{f}$ over $\mathcal{F}$ imply

$$\mathbb{P}_n \ell_{\tilde{h}} \leq \mu \mathbb{P}_n \ell_{\hat{f} - f^*} \leq \mu \mathbb{P}_n \ell_{h^\circ}.$$

By curvature, $\mathbb{P} \ell_{\tilde{h}} \geq C_0 \Delta^2(\tilde{h})$, while (57) gives $\mathbb{P} \ell_{h^\circ} \leq (\bar{f}/2) \phi_n^2$. Add and subtract the population losses of $\tilde{h}$ and h° . Since both $\tilde{h}$ and h° belong to the displayed localized star hull once C_A is large enough, the two stochastic differences are bounded by the same absolute empirical-process supremum. Choosing C_A to absorb the deterministic approximation terms proves (101). $\square$

Proof of Theorem 14. Apply Proposition 25 to the localized centered star hull. The proof of that proposition uses only centering, conditional independence, and Lipschitz contraction, so it remains valid without convexity of $\mathcal{F}$. Lemma 33 then replaces the convex boundary lemma. The critical-radius argument gives

$$\Delta^2(\hat{f} - f^*) = O_{\mathbb{P}}(r_n^2 + \phi_n^2).$$

Finally use Lemma 19 to obtain (30). $\square$

F Proof of Theorem 17

Set

$$A_N = S_N L_N \log(2S_N). \quad (102)$$

Assumption 16 supplies both an approximating network and the uniform empirical entropy bound

$$\log N(\epsilon, \mathcal{F}_{\text{NN}}, \|\cdot\|_{n,m}) \lesssim A_N \log(CB/\epsilon).$$

Lemma 34 (Localized network complexity). *For $0 < r \leq B$,*

$$\sum_I \mathcal{R}_I\{\mathcal{H}_2(r) \cap (\mathcal{F}_{\text{NN}} - f^*)\} \lesssim r \sqrt{\frac{A_N L}{N} \log \frac{CB}{r}} + \frac{BA_N L \log N}{N}. \quad (103)$$

Proof. For a block with $M_I \asymp N/L$ observations, the full-objective process is M_I/N times the within-block normalized process. Dudley's inequality and the entropy assumption give

$$\begin{aligned} \mathcal{R}_I &\lesssim \frac{\sqrt{M_I}}{N} \int_0^{Cr} \sqrt{A_N \log(CB/\epsilon)} d\epsilon + \frac{BA_N \log N}{N} \\ &\lesssim \frac{r}{\sqrt{NL}} \sqrt{A_N \log(CB/r)} + \frac{BA_N \log N}{N}. \end{aligned}$$

Summing over the $2L$ blocks proves (103); constants are absorbed into $\lesssim$. $\square$

Proof of Theorem 17. Let

$$a_N^2 = \max_{(p,K) \in \mathcal{P}} N^{-2p/(2p+K)} \text{polylog}(N).$$

Assumption 16 gives a near-best network with $\phi_N^2 \lesssim a_N^2$ and chooses the architecture so that

$$\frac{A_N L \log N}{N} \lesssim a_N^2. \quad (104)$$

The fixed point generated by (103) is therefore $r_N^2 \lesssim a_N^2 + \Gamma_n$. Apply Theorem 14 to obtain

$$\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}\{b_B^2(a_N^2 + \Gamma_n)\},$$

which is (33). The logarithmic factor is the one explicitly encoded in the approximation, entropy, and architecture-balance conditions; no additional unverified exponent is used. $\square$

G ADMM Derivations

G.1 Weighted check-loss proximal map

For $w > 0$, $\varrho > 0$, and $a, y \in \mathbb{R}$, consider

$$\text{prox}(a) = \arg \min_{\theta} \left\{ w \rho_{\tau}(y - \theta) + \frac{\varrho}{2}(\theta - a)^2 \right\}. \quad (105)$$

The subgradient condition is

$$0 \in \varrho(\theta - a) - w\{\tau - \mathbb{1}(y - \theta < 0)\},$$

with the usual interval-valued subgradient at $\theta = y$. Solving the three cases gives

$$\text{prox}(a) = \begin{cases} a + w\tau/\varrho, & y > a + w\tau/\varrho, \\ a - w(1 - \tau)/\varrho, & y < a - w(1 - \tau)/\varrho, \\ y, & \text{otherwise.} \end{cases} \quad (106)$$

Taking $w = w_{ij}$ yields the coordinate update in the main paper.

G.2 Scaled dual update

For a split problem $\min_{\theta, z} g(\theta) + h(z)$ subject to $\theta = z$, the scaled augmented Lagrangian is

$$g(\theta) + h(z) + \frac{\varrho}{2}\|\theta - z + u\|_2^2 - \frac{\varrho}{2}\|u\|_2^2.$$

Alternating minimization in θ and z is followed by

$$u^{t+1} = u^t + \theta^{t+1} - z^{t+1}. \quad (107)$$

There is no maximization of $\|\theta + u - z\|_2^2$ over u ; such a maximization is unbounded. Equations (106) and (107) establish the corrected Algorithms 1 and 2.

H Auxiliary Empirical-process Lemmas

Lemma 35 (Conditional symmetrization). *Let $Z_1, \dots, Z_M$ be conditionally independent given a sigma-field $\mathcal{A}$, and let $\mathcal{G}$ be a class containing the zero function. Then*

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} \left\| \sum_{q=1}^M \{g(Z_q) - \mathbb{E}(g(Z_q) \mid \mathcal{A})\} \right\| \middle| \mathcal{A} \right] \leq 2 \mathbb{E} \left[\sup_{g \in \mathcal{G}} \left\| \sum_{q=1}^M \varepsilon_q g(Z_q) \right\| \middle| \mathcal{A} \right].$$

Proof. Introduce a conditionally independent ghost sample $Z'_1, \dots, Z'_M$, apply Jensen's inequality, and use the conditional exchangeability of (Z_q, Z'_q) to insert Rademacher signs. The triangle inequality gives the factor two. $\square$

Lemma 36 (Contraction). *If $\phi_q(0) = 0$ and every ϕ_q is one-Lipschitz, then for every centered class $\mathcal{V} \subset \mathbb{R}^M$,*

$$\mathbb{E}_{\varepsilon} \sup_{v \in \mathcal{V}} \left| \sum_{q=1}^M \varepsilon_q \phi_q(v_q) \right| \leq 2 \mathbb{E}_{\varepsilon} \sup_{v \in \mathcal{V}} \left| \sum_{q=1}^M \varepsilon_q v_q \right|.$$

Proof. This is the Ledoux–Talagrand contraction inequality in its absolute-value form. It is applied in this paper only after centering at f^* , which ensures that the index set contains zero. $\square$

Lemma 37 (Dudley integral with full-objective weighting). *Let a block contain M observations and let its total weight in the full criterion be $\omega_I = \sum_{v \in I} w_v$. Conditional on the design,*

$$\mathbb{E}_\varepsilon \sup_{g \in \mathcal{G}} \left| \sum_{v \in I} w_v \varepsilon_v g_v \right| \lesssim \omega_I \inf_{\alpha > 0} \left[\alpha + \frac{1}{\sqrt{M}} \int_\alpha^{\text{diam}(\mathcal{G})} \sqrt{\log N(\epsilon, \mathcal{G}, \|\cdot\|_I)} d\epsilon \right].$$

Proof. By balanced weights, the left-hand side is bounded by a constant times ω_I multiplied by the normalized empirical Rademacher process on the block. Apply the usual chaining proof of Dudley’s inequality to that normalized process. $\square$

Lemma 38 (Sub-root scaling). *If ψ is nondecreasing and $r \mapsto \psi(r)/r$ is nonincreasing, then $\psi(Mr) \leq M\psi(r)$ for $M \geq 1$. Consequently, if $\psi(r_*) \leq r_*^2/c$, then $\psi(Mr_*)/(M^2r_*^2) \leq 1/(cM)$.*

Proof. The first claim follows directly from monotonicity of $\psi(r)/r$; the second is immediate. $\square$

I Beta-mixing Coefficients and Coupling

For sigma-fields $\mathcal{A}$ and $\mathcal{B}$, define

$$\beta(\mathcal{A}, \mathcal{B}) = \frac{1}{2} \sup_{r,s} \sum |\mathbb{P}(A_r \cap B_s) - \mathbb{P}(A_r)\mathbb{P}(B_s)|, \quad (108)$$

where the supremum is over finite measurable partitions $\{A_r\} \subset \mathcal{A}$ and $\{B_s\} \subset \mathcal{B}$. For a process $\{W_i\}_{i \in \mathbb{Z}}$,

$$\beta_W(k) = \sup_t \beta\{\sigma(W_i : i \leq t), \sigma(W_i : i \geq t+k)\}. \quad (109)$$

Lemma 39 (Berbee coupling). *Let $Z_1, \dots, Z_q$ be a finite subsequence whose consecutive elements are separated by at least k in the original beta-mixing process. There are independent variables $Z_1^*, \dots, Z_q^*$ with the same respective marginal laws such that*

$$\mathbb{P}\{(Z_1, \dots, Z_q) \neq (Z_1^*, \dots, Z_q^*)\} \leq (q-1)\beta_W(k).$$

Proof. Couple Z_2 to an independent copy conditional on Z_1 , then couple Z_3 conditional on (Z_1, Z_2) , and continue. At each step, the total-variation coupling error is at most $\beta_W(k)$. A union bound over the $q-1$ steps proves the result. $\square$

The proof of Theorem 6 applies this lemma to the joint cluster $W_i = (X_i, \delta_i, E_i)$. This preserves the contemporaneous dependence among the design, spatial effect, and measurement-error vector. Applying the lemma separately to the coordinate processes would not, in general, preserve their joint law and is therefore not used.

J Tracked Source Comparison for the Supplied Supplement

The corrected proofs in Appendices A–I are the operative supplementary material. For transparency and to preserve the full structure of the supplied 133-page source, the original detailed derivations are reproduced below in gray. Red passages state the implemented replacements. The gray derivations are retained only as a line-by-line comparison and are superseded wherever they conflict with the operative corrected proofs.

J.1 Proof of Theorem [source ref.]

Implemented correction. Appendices [source ref.]–[source ref.] above are the operative corrected supplement. They use $w_{ij} = 1/(nm_i)$, the joint cluster $W_i = (X_i, \delta_i, E_i)$, centered local classes, and the critical radius in [source equation]. The gray material below is retained solely for line-by-line comparison with the supplied source.

J.1.1 Corrected Proof Roadmap and Normalization

Write $w_{ij} = 1/(nm_i)$ and

$$\ell_{ij}(f) = \rho_\tau\{y_{ij} - f(x_{ij})\} - \rho_\tau\{y_{ij} - f^*(x_{ij})\}.$$

Knight’s identity and Assumption [source ref.] imply

$$\bar{Z}(f) := \mathbb{E} \sum_{i,j} w_{ij} \ell_{ij}(f) \geq C_0 \Delta^2(f - f^*), \quad C_0 = \min\{\underline{f}/2, R\underline{f}/4\},$$

where $\Delta^2(h) = \int \min\{|h|, h^2\} dQ$. Since $\|f\|_\infty, \|f^*\|_\infty \leq B$,

$$\|f - f^*\|_2^2 \leq (1 \vee 2B) \Delta^2(f - f^*).$$

For every interlaced block $I = \mathcal{I}_{a,\ell}$, Berbee’s lemma is applied once to the joint random element $W_i = (X_i, \delta_i, E_i)$; separate couplings of X_i , δ_i , and E_i are not used. The centered weighted block process is

$$\mathbb{E} \sup_{h \in \mathcal{H}(r)} \left| \frac{1}{n} \sum_{i \in I} \frac{1}{m_i} \sum_{j=1}^{m_i} \varepsilon_{ij} h(x_{ij}^*) \right|, \quad \mathcal{H}(r) = \{f - f^* : f \in \mathcal{F}, \|f - f^*\|_2 \leq r\}.$$

The conditional spatial-effect remainder is controlled separately by the block average of

$$D_i(x, u) = \mathbb{P}(U_{i1} \leq u \mid x_{i1} = x) - \mathbb{P}(U_{i1} \leq u \mid \delta_i, x_{i1} = x), \quad U_{ij} = \delta_i(x_{ij}) + e_{ij}.$$

The low-dimensional Lipschitz/grid condition gives the term $\Gamma_n = (s + 1)L_n \log(nT_n)/n$. Combining curvature, joint coupling, conditional symmetrization, contraction, and the tail-absorption condition leads to the sub-root inequality

$$\Psi_n(r) = \sum_{\ell=1}^{L_n} \sum_{a \in \{e, o\}} \mathcal{R}_{a,\ell}\{\mathcal{H}(c_0^{-1/2} b_B r)\} + \Gamma_n \leq r^2/64.$$

The critical radius r_n is the smallest r satisfying this inequality, and the basic inequality yields $\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}(b_B^2 r_n^2)$, plus the explicit coupling probability $n\beta_W(L_n)$. This roadmap supersedes any later line that normalizes a block by $\sum_{i \in I} m_i$ without multiplying by the block’s weight in the full empirical objective.

J.1.2 Definitions and Notation

To complete our proofs for the theorems, we provide important definitions.

Definition 40. We introduce the Huber-type Huber (1992) loss. Let Δ_{nm}^2 be a mapping of functions to real numbers, the Huber-type loss function is elucidated as

$$\Delta_{nm}^2(f) = \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \min\{|f(x_{ij})|, (f(x_{ij}))^2\}, \quad (110)$$

which, modulo constants, corresponds to a Huber loss, as delineated in Huber (1992). The Huber loss serves as a natural lower bound for the quantile population loss, which provides the primary motivation for focusing on it. Let $P(x_{ij})$ denote the distribution of x_{ij} . The population Huber loss is then given by

$$\Delta^2(f) = \int \min\{|f(t)|, |f(t)|^2\} dP(t). \quad (111)$$

We derive the rate of convergence by using the Huber-type loss and L_2 loss.

Definition 41. Let K be a function class with Rademacher width well defined. We define its Empirical Rademacher width as

$$R_{mn}(K) = \mathbb{E}_{\xi|\{x_{ij}\}} \left[\sup_{f \in K} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \xi_{ij} f(x_{ij}) \right].$$

Definition 42. We define its Rademacher width as

$$R(K) = \mathbb{E}_x \left[\mathbb{E}_{\xi|\{x_{ij}\}} \left[\sup_{f \in K} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \xi_{ij} f(x_{ij}) \right] \right] = \mathbb{E}_{x, \xi} \left[\sup_{f \in K} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \xi_{ij} f(x_{ij}) \right].$$

Definition 43. We define $\hat{Z}_{ij}(f)$ and the empirical loss function $\hat{Z}(f)$ as

$$\begin{aligned} \hat{Z}_{ij}(f) &= \rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij})) \\ &= \rho_\tau(f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij} - f(x_{ij})) - \rho_\tau(\delta_i(x_{ij}) + e_{ij}). \end{aligned} \quad (112)$$

$$\begin{aligned} \hat{Z}(f) &= \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \{\rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))\} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \{\rho_\tau(f_{ij}^* + \delta_i(x_{ij}) + e_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))\}. \end{aligned} \quad (113)$$

We also define $Z_{ij}(f)$ and $Z(f)$ as

$$\begin{aligned} Z_{ij}(f) &:= \mathbb{E}_{\delta, e|x_{ij}} [\rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))] \\ &= \mathbb{E}_{\delta, e|x_{ij}} [\rho_\tau(f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij} - f(x_{ij})) - \rho_\tau(\delta_i(x_{ij}) + e_{ij})]. \end{aligned} \quad (114)$$

$$\begin{aligned}
 Z(f) &:= \mathbb{E}_{\delta, e | \{x_{ij}\}} \left[\hat{Z}(f) \right] \\
 &= \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{\delta, e | \{x_{ij}\}} [\rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))] \\
 &= \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{\delta, e | \{x_{ij}\}} [\rho_\tau(f_{ij}^* + \delta_i(x_{ij}) + e_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))],
 \end{aligned} \tag{115}$$

where we use $\{x_{ij}\}$ as a shorthand for $\{x_{ij}\}_{i=1, j=1}^{n, m_{ij}}$ and $\{\delta_i\}$ as $\{\delta_i\}_{i=1}^n$.

We then define $\tilde{Z}_{ij}(f)$ and $\tilde{Z}(f)$ as

$$\begin{aligned}
 \tilde{Z}_{ij}(f) &:= \mathbb{E}_{e | \delta_i, x_{ij}} [\hat{Z}_{ij}(f)]. \\
 \tilde{Z}(f) &:= \mathbb{E}_{e | \{\delta_i\}, \{x_{ij}\}} [\hat{Z}(f)].
 \end{aligned} \tag{116}$$

We define the population version $\bar{Z}_{ij}(f)$ and the population loss function as

$$\begin{aligned}
 \bar{Z}_{ij}(f) &:= \mathbb{E}_{x, \delta, e} [\hat{Z}_{ij}(f)]. \\
 \bar{Z}(f) &:= \mathbb{E}_{x, \delta, e} [\hat{Z}(f)].
 \end{aligned} \tag{117}$$

Note that the $\delta_i(x_{i1}), \dots, \delta_i(x_{im_i})$ are not independent for $i \in \{1, \dots, n\}$. Similarly, due to the definitions of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$, $\tilde{Z}(f)$, and $Z(f)$, the terms $\hat{Z}_{i1}(f), \dots, \hat{Z}_{im_i}(f)$ are also not independent. Also note, by definition that $f^*(x_{ij}) = F_{y_{ij}|x_{ij}}^{-1}(\tau)$, we know $f^* \in \arg \min Z(f)$, and $Z(f^*) = 0$.

Definition 44. The constrained quantile trend filtering estimator for temporal-spatial data defined in [source equation] will be

$$\hat{f} = \arg \min_{f \in F_k(V)} \hat{Z}(f), \tag{118}$$

where $F_k(V)$ is defined in [source equation].

J.1.3 Proof Sketches of Theorem [source ref.]

Implemented correction. The proof now works on the event obtained by Berbee coupling of the joint clusters. With interlaced spacing $2L$,

$$\mathbb{P}(\Omega_\beta^c) \leq n\beta_W(2L),$$

and this term is carried through the final probability inequality before the critical-radius argument is applied.

Step 1. To get the convergence rate of $\Delta^2(\hat{f} - f^*)$, we will control the quantity $\mathbb{P}(c_0 \Delta^2(\hat{f} - f^*) > t^2)$, where c_0 satisfies [source equation].

We will first consider the temporal dependence. To control that, we would need to use the β -mixing concept that are constructed in Appendix [source ref.]. We then get the setup of the indices for $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \asymp n/L$ in [source equation]. We let $\mathcal{I} := \bigcup_{l=1}^L (\mathcal{J}_{e,l} \cup \mathcal{J}_{o,l})$. For each

fixed $l \in [L]$, we construct copies $x_{ij}^*, e_{ij}^*, \delta_i^*$ for $i \in \mathcal{J}_{e,l}$ (and similarly for $\mathcal{J}_{o,l}$) via the coupling in Appendix [source ref.]. These copies have the same marginal distribution as x_{ij}, e_{ij}, δ_i , and are mutually independent within each block $\mathcal{J}_{e,l}$ (but not across different blocks). Next, we let

$$X_i := (x_{i1}, \dots, x_{im_i}), \quad E_i := (e_{i1}, \dots, e_{im_i}), \quad \Delta_i := (\delta_{i1}, \dots, \delta_{im_i}),$$

and their coupled copies

$$X_i^* := (x_{i1}^*, \dots, x_{im_i}^*), \quad E_i^* := (e_{i1}^*, \dots, e_{im_i}^*), \quad \Delta_i^* := (\delta_{i1}^*, \dots, \delta_{im_i}^*),$$

and then we define

$$\begin{aligned} \Omega_x &:= \{X_i = X_i^* \text{ for all } i \in \mathcal{I}\}, \\ \Omega_e &:= \{E_i = E_i^* \text{ for all } i \in \mathcal{I}\}, \\ \Omega_\delta &:= \{\Delta_i = \Delta_i^* \text{ for all } i \in \mathcal{I}\}, \\ \Omega_\beta &:= \Omega_x \cap \Omega_e \cap \Omega_\delta. \end{aligned} \tag{119}$$

Thus, we get

$$\mathbb{P}\left(c_0 \Delta^2(\hat{f} - f^*) > t^2\right) \leq \mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right).$$

Step 2. Then in Lemma [source ref.], we show $\mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right)$ is $O(\frac{1}{n})$.

Step 3. It remains to bound the joint probability $\mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right)$. On the coupling event Ω_β (see Remark [source ref.]), the blockwise coupling reduces the original β -mixing sequence to an independent-block structure. Consequently, the arguments in Lemmas [source ref.]–[source ref.] (based on symmetrization and Rademacher complexities) yield an upper bound for $\mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right)$ in terms of the corresponding blockwise Rademacher width terms.

Step 4. In details, with the above two steps, under Assumptions [source ref.], [source ref.], [source ref.], and [source ref.], the following results hold for any $t > 0$,

$$\begin{aligned} &\mathbb{P}\left(c_0 \Delta^2(\hat{f} - f^*) > t^2\right) \\ &\leq \frac{c_5}{t^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) \right) + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n}, \end{aligned}$$

where $s = |S|$ is the intrinsic dimension from Assumption [source ref.] (the number of active coordinates of Π), T_n is from Assumption [source ref.]. Constant c_4 depends on $\tilde{c}_0$ in Assumption [source ref.], C_β in Assumption [source ref.], and c_1, c_2 in [source equation], and c_5 depends on constants C and c in Assumption [source ref.].

Step 5. By using [source ref.], we start by writing

$$\begin{aligned} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{f : c_0 \Delta^2(f - f^*) \leq t^2\}) &\leq R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \left\{f : \|f - f^*\| \leq c_0^{-1/2} \cdot b(B) \cdot t\right\}\right), \text{ and} \\ R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{f : c_0 \Delta^2(f - f^*) \leq t^2\}) &\leq R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \left\{f : \|f - f^*\| \leq c_0^{-1/2} \cdot b(B) \cdot t\right\}\right), \end{aligned}$$

where $b(B) = \max\{B^{1/2}, 1\}$ is a constant that depends on B , with B introduced in [source equation].

Let

$$r_n^2 \asymp \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n \right) \right) + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n \right) \right) \right) + \frac{s \cdot \text{poly}(\log n) \log T_n}{n}.$$

Then we get the convergence rate of

$$\Delta^2(\hat{f} - f^*) = O_p(r_n^2). \quad (120)$$

This follows from Lemma [source ref.] and Corollary [source ref.].

The details of the proofs of the above steps are provided as follows.

J.1.4 Technical Results for Proofs of Theorem [source ref.]

J.1.5 Proof of Step 1 and 2

Implemented correction. The separate coordinate copies have been replaced by one copy $W_i^* = (X_i^*, \delta_i^*, E_i^*)$ for each time cluster. Within each interlaced block the W_i^* are independent, have the correct joint marginal law, and preserve $y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$.

To get the convergence rate of $\Delta^2(\hat{f} - f^*)$, we will control the quantity $\mathbb{P}(c_0 \Delta^2(\hat{f} - f^*) > t^2)$, where c_0 satisfies [source equation], we will first consider the temporal dependence. To control that, we would need to use the β -mixing concept. In the following analysis, we will then deal with this dependence.

Specifically, we rely on the condition that $cm \leq m_i \leq Cm$ for some positive constants c, C . Additionally, we define

$$L = \left\lceil \frac{1}{C_\beta} 2 \log(n) \right\rceil, \quad (121)$$

leading us to the conclusion that

$$\beta_x(L) \leq \frac{1}{n^2}, \quad (122)$$

where β_x is defined in Assumption [source ref.]. This bound is equivalently satisfied by $\beta_\delta(L)$ and $\beta_e(L)$. Subsequently, throughout our discussion, we make frequent references to the copies of the random variables x_{ij} , e_{ij} , and δ_i as they are constructed in Appendix [source ref.]. These copies are denoted as x_{ij}^* , e_{ij}^* , and δ_i^* , respectively. Furthermore, as delineated in Appendix [source ref.], during the formation of these copies the quantities $|\mathcal{J}_{e,l}|$ and $|\mathcal{J}_{o,l}|$ in [source equation], which represent the count of even and odd blocks respectively, satisfy $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \asymp n/L$. Thus, there exist positive constants c_1 and c_2 such that

$$c_1 n/L \leq |\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \leq c_2 n/L, \quad L = \left\lceil \frac{1}{C_\beta} 2 \log(n) \right\rceil. \quad (123)$$

Lemma 45. Let $\hat{f}$, f^* belong to $\mathcal{F}$, and let $\Delta^2(\hat{f} - f^*)$ be the Huber loss. Suppose Assumption [source ref.] holds with the β -mixing condition satisfied in [source equation] and the block indices $\mathcal{J}_{e,l}, \mathcal{J}_{o,l}$ satisfying $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \asymp n/L$ as in [source equation]. For each fixed $l \in [L]$, we construct identically distributed copies $\{x_{ij}^*, e_{ij}^*, \delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ that are mutually independent within the block $\mathcal{J}_{e,l}$ (and similarly for $\mathcal{J}_{o,l}$), via the coupling in Appendix [source ref.]. Let $\Omega_x, \Omega_e, \Omega_\delta, \Omega_\beta$ be defined in [source equation]. Then we have

$$\mathbb{P}(c_0 \Delta^2(\hat{f} - f^*) > t^2) \leq \mathbb{P}\left(\left\{c_0 \Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{\epsilon}{2}.$$

Proof. Using a union bound argument and invoking Lemmas [source ref.], [source ref.], and [source ref.], we note that the following sets, $\Omega_x, \Omega_e, \Omega_\delta$ in [source equation], satisfy the conditions

$$\mathbb{P}(\Omega_x^c) \leq n\beta_x(L) \leq \frac{1}{n}, \quad \mathbb{P}(\Omega_e^c) \leq n\beta_e(L) \leq \frac{1}{n}, \quad \text{and} \quad \mathbb{P}(\Omega_\delta^c) \leq n\beta_\delta(L) \leq \frac{1}{n}.$$

Therefore letting

$$\Omega_\beta = \Omega_x \cap \Omega_e \cap \Omega_\delta \tag{124}$$

we have that

$$\begin{aligned} \mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right) &= \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right) \\ &\leq \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{3}{n} \\ &\leq \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{\varepsilon}{2} \end{aligned}$$

for any $\varepsilon > 0$ as n large enough. $\square$

By previous lemma, in consequence, it is enough to bound the term

$$\mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right). \tag{125}$$

To this end, we will work with the indicator of the coupling event. We write $\mathbf{1}_{\{\Omega_\beta\}}$ for the indicator of Ω_β .

We write $\mathbb{P}(\cdot \cap \Omega_\beta)$ for the probability restricted to Ω_β . We denote by $\mathbb{E}_{x,\delta,e}[\cdot]$ the expectation taken over $(x_{ij}, \delta_i, e_{ij})$. And we denote by $\mathbb{E}_{e|x_{ij}}[\cdot]$ the conditional expectation taken over $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$ respectively.

In particular, for any nonnegative random variable X , we write

$$\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[X \mathbf{1}_{\{\Omega_\beta\}}\right]$$

to denote the expectation of X restricted to the coupling event Ω_β .

Remark 46. Moreover, on Ω_β we work with the coupled sample $\{(x_{ij}^*, y_{ij}^*)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, which are i.i.d. within each fixed block $\mathcal{J}_{e,l}$. We now define the star versions of the loss quantities. The star empirical loss is

$$\hat{Z}_{ij}^*(f) := \rho_\tau(f^*(x_{ij}^*) + \delta_i^*(x_{ij}^*) + e_{ij}^* - f(x_{ij}^*)) - \rho_\tau(\delta_i^*(x_{ij}^*) + e_{ij}^*), \tag{126}$$

and the star conditional expectations are

$$\begin{aligned} Z_{ij}^*(f) &:= \mathbb{E}_{\delta^*, e^* | x_{ij}^*} \left[\hat{Z}_{ij}^*(f) \right], \\ \tilde{Z}_{ij}^*(f) &:= \mathbb{E}_{e^* | \delta_i^*, x_{ij}^*} \left[\hat{Z}_{ij}^*(f) \right], \\ \bar{Z}_{ij}^*(f) &:= \mathbb{E}_{x^*, \delta^*, e^*} \left[\hat{Z}_{ij}^*(f) \right]. \end{aligned}$$

By construction, for each fixed $l \in [L]$, the star quantities $\{\hat{Z}_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are built from the i.i.d. copies $(x_{ij}^*, e_{ij}^*, \delta_i^*)_{i \in \mathcal{J}_{e,l}}$, and hence the indices $i \in \mathcal{J}_{e,l}$ are independent (similarly for $\mathcal{J}_{o,l}$). However, the copies across different blocks l are not mutually independent. On the coupling event Ω_β we have $(x_{ij}, e_{ij}, \delta_i) = (x_{ij}^*, e_{ij}^*, \delta_i^*)$ for all $i \in \mathcal{I}$ and $j \in [m_i]$, so that

$$\hat{Z}_{ij}(f) = \hat{Z}_{ij}^*(f), \quad Z_{ij}(f) = Z_{ij}^*(f), \quad \tilde{Z}_{ij}(f) = \tilde{Z}_{ij}^*(f), \quad \bar{Z}_{ij}(f) = \bar{Z}_{ij}^*(f), \quad \text{on } \Omega_\beta. \tag{127}$$

Results that do not invoke the coupling argument remain valid under the original probability measure and will be indicated explicitly when used.

J.1.6 Proof of Step 3

Implemented correction. Every local process is indexed by the centered class

$$\mathcal{H}_t = \{f - f^* : f \in \mathcal{F}, c_0 \Delta^2(f - f^*) \leq t^2\}.$$

This class contains zero, so the relevant suprema are nonnegative and contraction is applied to maps that vanish at the origin.

Lemma [source ref.] – Corollary [source ref.] are some fundamental lemmas support our convergence analysis.

Lemma 47 (Basic Inequality). *By Definitions in [source equation] and [source equation], we have*

$$\hat{Z}(\hat{f}) - \hat{Z}(f^*) \leq 0$$

as the basic inequality.

Lemma 48 (Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$). *The quantities $\hat{Z}_{ij}(f)$ (defined in [source equation]), $Z_{ij}(f)$ (defined in [source equation]), and $r_{ij}(f)$ (defined in [source equation]) are 1-Lipschitz in the following sense: for any $f, g \in \mathcal{F}$ and all (i, j) ,*

$$\begin{aligned} |\hat{Z}_{ij}(f) - \hat{Z}_{ij}(g)| &\leq |f(x_{ij}) - g(x_{ij})|, \\ |Z_{ij}(f) - Z_{ij}(g)| &\leq |f(x_{ij}) - g(x_{ij})|, \\ |r_{ij}(f) - r_{ij}(g)| &\leq |f(x_{ij}) - g(x_{ij})|. \end{aligned}$$

Proof. For $\hat{Z}_{ij}(f)$ and $Z_{ij}(f)$, their Lipschitz condition can be seen in the Lemma A.11 of Zhang et al. (2023). For

$$r_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt.$$

We have

$$\begin{aligned} &|r_{ij}(f) - r_{ij}(g)| \\ &= \int_0^{f(x_{ij}) - g(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt. \\ &\leq |f(x_{ij}) - g(x_{ij})|. \end{aligned}$$

□

Lemma 49. *For any $f, f^* \in \mathcal{F}$, let $h = f - f^*$, and let $h(x_{ij}) = (f - f^*)(x_{ij})$. Then we have*

$$Z_{ij}(f) - Z_{ij}(f^*) = \int_0^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz.$$

Proof. Let I be the indicator function. By equation (B.3) in Belloni et al. (2011), we have

$$\rho_\tau(w - v) - \rho_\tau(w) = -v(\tau - I(w \leq 0)) + \int_0^v \{I(w \leq z) - I(w \leq 0)\} dz. \quad (128)$$

Given any f and x_{ij} , let $w = y_{ij} - f^*(x_{ij})$, $v = f(x_{ij}) - f^*(x_{ij})$. Then taking conditional expectations on above equation with respect to y_{ij} conditioned on x_{ij} on both sides, we have

$$\begin{aligned}
 Z_{ij}(f) - Z_{ij}(f^*) &= \mathbb{E}_{\delta, e|x_{ij}} [\rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))] \\
 &= \mathbb{E}_{\delta, e|x_{ij}} [-h(x_{ij})(\tau - I(y_{ij} - f^*(x_{ij}) \leq 0))] \\
 &\quad + \mathbb{E}_{\delta, e|x_{ij}} \left[\int_0^{h(x_{ij})} \{I(y_{ij} - f^*(x_{ij}) \leq z) - I(y_{ij} - f^*(x_{ij})) \leq 0\} dz \right] \quad (129) \\
 &= 0 + \int_0^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz \\
 &= \int_0^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz,
 \end{aligned}$$

for all $i = 1, \dots, n$. □

Lemma 50. *Under Assumption [source ref.], with R and $\underline{f}$ from Assumption [source ref.], let*

$$C_0 := \min \left\{ \frac{\underline{f}}{2}, \frac{R\underline{f}}{4} \right\}. \quad (130)$$

Then for all $f, f^* \in \mathcal{F}$, we have

$$Z(f) \geq C_0 \Delta_{nm}^2 (f - f^*).$$

Proof. Let $h(x_{ij}) := f(x_{ij}) - f^*(x_{ij})$. By Lemma [source ref.], we have

$$Z_{ij}(f) - Z_{ij}(f^*) = \int_0^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz.$$

First, suppose $h(x_{ij}) \in [0, R]$. By Assumption [source ref.], for any z with $|z| \leq R$, we have $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \geq \underline{f}z$. Therefore,

$$Z_{ij}(f) - Z_{ij}(f^*) \geq \int_0^{h(x_{ij})} \underline{f}z dz = \frac{\underline{f}h(x_{ij})^2}{2}.$$

Next, suppose $h(x_{ij}) \in [-R, 0]$. For $z \in [h(x_{ij}), 0]$ we have $|z| \leq R$, so by Assumption [source ref.], $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \leq \underline{f}z$ (noting $z \leq 0$). Therefore,

$$\begin{aligned}
 Z_{ij}(f) - Z_{ij}(f^*) &= \int_{h(x_{ij})}^0 \{F_{y_{ij}|x_{ij}}(f^*(x_{ij})) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z)\} dz \\
 &\geq \int_{h(x_{ij})}^0 (-\underline{f}z) dz = \frac{\underline{f}h(x_{ij})^2}{2}.
 \end{aligned}$$

Now suppose $h(x_{ij}) > R$. Since $F_{y_{ij}|x_{ij}}$ is monotone non-decreasing, for $z \geq R/2$ we have $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) \geq F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2)$. By Assumption [source ref.] (applied with $R/2 \leq R$), $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \geq \underline{f}R/2$. Therefore,

$$\begin{aligned}
 Z_{ij}(f) - Z_{ij}(f^*) &\geq \int_{R/2}^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz \\
 &\geq \int_{R/2}^{h(x_{ij})} \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} dz \quad (131) \\
 &= (h(x_{ij}) - R/2) \{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\} \\
 &\geq \frac{h(x_{ij})}{2} \cdot \frac{R\underline{f}}{2} = \frac{R\underline{f}}{4} |h(x_{ij})| \geq C_0 |h(x_{ij})|,
 \end{aligned}$$

where the first inequality holds since the integrand is non-negative on $[0, R/2]$, the second by monotonicity of $F_{y_{ij}|x_{ij}}$, and the penultimate uses $h(x_{ij}) - R/2 \geq h(x_{ij})/2$ (since $h(x_{ij}) > R$).

The case $h(x_{ij}) < -R$ is analogous (by reversing the direction of integration) and yields the same bound $Z_{ij}(f) - Z_{ij}(f^*) \geq C_0 |h(x_{ij})|$.

Combining the four cases, for all (i, j) we have

$$Z_{ij}(f) - Z_{ij}(f^*) \geq C_0 \min \{|h(x_{ij})|, h(x_{ij})^2\}.$$

Summing over all (i, j) and using $Z_{ij}(f^*) = 0$, we obtain

$$Z(f) \geq C_0 \Delta_{nm}^2(f - f^*),$$

where $\Delta_{nm}^2(\cdot)$ is the empirical Huber loss defined in [source equation]. $\square$

Corollary 51. *Assume Assumption [source ref.] holds. Then for C_0 defined in [source equation] and for all $f, f^* \in \mathcal{F}$, we have*

$$\bar{Z}(f) \geq C_0 \Delta^2(f - f^*),$$

where $\bar{Z}(f) := \mathbb{E}_{x, \delta, e} [\hat{Z}(f)]$ is defined in [source equation] and $\Delta^2(\cdot)$ is the population Huber loss defined in [source equation].

Proof. By Lemma [source ref.], for each realization of $\{x_{ij}\}_{i=1, j=1}^{n, m_i}$ we have $Z(f) \geq C_0 \Delta_{nm}^2(f - f^*)$. Taking expectation over $\{x_{ij}\}$ on both sides and using the definitions $\bar{Z}(f) = \mathbb{E}_x [Z(f)]$ **the notation here does not match the statement** from [source equation] and $\Delta^2(f - f^*) = \mathbb{E}_x [\Delta_{nm}^2(f - f^*)]$ from [source equation], we obtain the result. $\square$

Lemma [source ref.]–[source ref.] are used to bound the term [source equation].

Lemma 52. *Let $\hat{f}, f^* \in \mathcal{F}$ and let Ω_β be as defined in [source equation]. Let C_0 be the constant from Corollary [source ref.] (defined in [source equation]), satisfying $\bar{Z}(f) \geq C_0 \Delta^2(f - f^*)$. Let c_0 be any constant satisfying*

$$0 < c_0 < C_0. \quad (132)$$

Then for any $t > 0$,

$$\mathbb{P}\left(\{c_0 \Delta^2(\hat{f} - f^*) > t^2\} \cap \Omega_\beta\right) \leq \mathbb{P}\left(\left\{\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} (\bar{Z}(f) - \hat{Z}(f) - \ell(f)) > t^2\right\} \cap \Omega_\beta\right),$$

where $\ell(f) := (C_0 - c_0) \Delta^2(f - f^*)$.

Proof. Assume $c_0 \Delta^2(\hat{f} - f^*) > t^2$, where c_0 satisfies [source equation]. We know Δ^2 is continuous. Next, define $g : [0, 1] \rightarrow \mathbb{R}$ as $g(\mu) = c_0 \Delta^2((1 - \mu)f^* + \mu\hat{f} - f^*)$. Clearly, g is a continuous function with $g(0) = 0$ and $g(1) = c_0 \Delta^2(\hat{f} - f^*)$. Therefore, there exists $\mu_{\hat{f}} \in [0, 1]$ such that $g(\mu_{\hat{f}}) = t^2$. We then let $\tilde{f} = (1 - \mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}$, so that by construction $c_0 \Delta^2(\tilde{f} - f^*) = t^2$.

Then we have

$$\hat{Z}(\tilde{f}) = \hat{Z}\left\{(1 - \mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}\right\} \stackrel{(a)}{\leq} (1 - \mu_{\hat{f}})\hat{Z}(f^*) + \mu_{\hat{f}}\hat{Z}(\hat{f}) \stackrel{(b)}{\leq} 0. \quad (133)$$

Where [source equation] follows the convexity of $\hat{Z}$. The [source equation] follows by the $\hat{Z}(f^*) = 0$, and the basic inequality. First, let

$$\ell(\tilde{f}) := (C_0 - c_0) \Delta^2(\tilde{f} - f^*), \quad (134)$$

Since $\hat{f}, f^* \in \mathcal{F}$, and $\mathcal{F}$ is a convex set, thus, $\tilde{f} \in \mathcal{F}$, and $c_0 \Delta^2(\tilde{f} - f^*) = t^2$, then we have

$$\begin{aligned}
 t^2 &= c_0 \Delta^2(\tilde{f} - f^*) \\
 &= C_0 \Delta^2(\tilde{f} - f^*) - \underbrace{(C_0 - c_0) \Delta^2(\tilde{f} - f^*)}_{:= \ell(\tilde{f})} \\
 &\leq \bar{Z}(\tilde{f}) - \ell(\tilde{f}) \\
 &= \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) + \hat{Z}(\tilde{f}) - \ell(\tilde{f}) \\
 &\stackrel{(a)}{\leq} \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) \\
 &\leq \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right).
 \end{aligned}$$

The first inequality holds by applying the Corollary [source ref.]. The [source equation] follows by expression in the line [source equation].

Thus, we have

$$\mathbb{P} \left(\left\{ c_0 \Delta^2(\hat{f} - f^*) > t^2 \right\} \cap \Omega_\beta \right) \leq \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) > t^2 \right\} \cap \Omega_\beta \right).$$

□

To control the quantity on the right-hand side, we apply Markov's inequality and then utilize symmetrization to control the empirical process with Rademacher complexity. We also need to account for the dependence among $\delta_i(x_{i1}), \dots, \delta_i(x_{im_i})$ for $i \in \{1, \dots, n\}$. The following theorem introduces the decomposition that handles this dependence.

Theorem 53 (Conversion of Huber loss). *Let $\hat{f}, f^* \in \mathcal{F}$, let Ω_β be defined in [source equation]. Suppose the distribution of $\{(x_{ij}, y_{ij})\}_{i=1, j=1}^{n, m_i}$ obey Assumption [source ref.], where c_0 satisfies [source equation], and $\ell(f)$ is defined in [source equation], the following inequality is true for*

any $t > 0$,

$$\begin{aligned}
 & \mathbb{P} \left(\left\{ c_0 \Delta^2 (\hat{f} - f^*) > t^2 \right\} \cap \Omega_\beta \right) \\
 & \leq \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_1} \\
 & + \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_2} \\
 & + \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_3} \\
 & + \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{B_1} \\
 & + \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{B_2} \\
 & + \underbrace{\sum_{l=1}^L \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{B_3},
 \end{aligned} \tag{135}$$

where L is defined in [source equation].

Proof. By Lemma [source ref.], it is true that

$$\begin{aligned}
 & \mathbb{P} \left(\left\{ c_0 \Delta^2 (\hat{f} - f^*) > t^2 \right\} \cap \Omega_\beta \right) \\
 & \stackrel{(a)}{\leq} \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) > t^2 \right\} \cap \Omega_\beta \right) \\
 & = \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > t^2 \right\} \cap \Omega_\beta \right)
 \end{aligned}$$

$$\begin{aligned}
 & + \frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \Bigg) > t^2 \Bigg\} \cap \Omega_\beta \Bigg) \\
 & \leq \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > \frac{t^2}{2} \right\} \cap \Omega_\beta \right) \\
 & \quad + \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > \frac{t^2}{2} \right\} \cap \Omega_\beta \right) \\
 & \leq \sum_{l=1}^L \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{t^2}{2L} \right\} \cap \Omega_\beta \right) \\
 & \quad + \sum_{l=1}^L \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{t^2}{2L} \right\} \cap \Omega_\beta \right).
 \end{aligned}$$

Now we analyze each of the terms above,

$$\begin{aligned}
 & \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{t^2}{2L} \right\} \cap \Omega_\beta \right) \\
 & \stackrel{(a)}{\leq} \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \stackrel{(b)}{\leq} \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} + \right. \right. \\
 & \quad \left. \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right. \\
 & \quad \left. + \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \Bigg] \\
 & = \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\underbrace{\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}}_{A_1} \right] \\
 & \quad + \frac{2L}{t^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\underbrace{\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}}_{A_2} \right]
 \end{aligned}$$

$$\begin{aligned}
 & + \frac{2L}{t^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \underbrace{\left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_3} \\
 & := \frac{2LA_1}{t^2} + \frac{2LA_2}{t^2} + \frac{2LA_3}{t^2},
 \end{aligned}$$

The inequality [source equation] follows from Markov's inequality, and the third inequality [source equation] follows from the triangle inequality.

Similarly, we have

$$\begin{aligned}
 & \mathbb{P} \left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{z}_{ij}(f) - \hat{z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{t^2}{2L} \right\} \cap \Omega_\beta \right) \\
 & \leq \frac{2LB_1}{t^2} + \frac{2LB_2}{t^2} + \frac{2LB_3}{t^2}.
 \end{aligned}$$

□

Now we proceed to bound A_1 , A_2 , A_3 separately. For B_1 , B_2 , B_3 , we state the resulting bound, as the analysis is similar.

Lemma 54. *Let Ω_β be defined in [source equation]. Let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Then we have*

$$\begin{aligned}
 & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \leq \frac{c_5}{2L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}),
 \end{aligned}$$

Implemented correction. *The malformed intersection has been replaced everywhere by the centered class $\mathcal{H}_t$. Equivalently, the corresponding function class is $\mathcal{F}_\Delta(t) = \{f \in \mathcal{F} : c_0 \Delta^2(f-f^*) \leq t^2\}$.*

where c_5 is a constant depending only on the constants C and c in Assumption [source ref.], and $R_{\mathcal{J}_{e,l}}$ is defined in [source equation].

Proof. We have

$$\begin{aligned}
 & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & = \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \frac{m_{i'}}{nm_i} f(x_{ij}^*) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \tag{136}
 \end{aligned}$$

Next, by Assumption [source ref.], for every $i \in \mathcal{J}_{e,l}$ we have $cm \leq m_i \leq Cm$. Combined with [source equation], which gives $|\mathcal{J}_{e,l}| \leq c_2 n/L$ (where c_2 is from [source equation]), we obtain

$$\frac{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}}{n m_i} \leq \frac{|\mathcal{J}_{e,l}| \cdot Cm}{n \cdot cm} \leq \frac{c_2 C}{cL}. \quad (137)$$

Then by Ledoux Talagrand contraction inequality (see Proposition 5.28 in Wainwright (2019)), we have

$$\begin{aligned} & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \frac{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}}{n m_i} f(x_{ij}^*) \right. \\ & \quad \left. \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq \mathbb{E}_{x^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \frac{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}}{n m_i} f(x_{ij}^*) \right] \\ & \stackrel{(a)}{\leq} \frac{c_2 C}{cL} \mathbb{E}_{x^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \right] \\ & = \frac{c_2 C}{cL} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) = \frac{c_5}{2L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}). \end{aligned}$$

The [source equation] holds since [source equation]. Equivalently, letting $c_5 := 2c_2 C/c$, the prefactor becomes $\frac{c_5}{2L}$. $\square$

Bound of A_2 Implemented correction. Independence is now used at the correct level. The coupled time clusters are independent across i within a block. Conditional on (X_i, δ_i) , the measurement-error contributions are symmetrized over (i, j) ; the shared values $\delta_i(x_{ij})$ are never treated as independent across j .

Lemma 55 (Bound of A_2). *Let $\tilde{Z}(f)$, $\hat{Z}(f)$ be defined in [source equation] and [source equation], and let $\tilde{Z}^*(f)$, $\hat{Z}^*(f)$ be the star versions defined in [source equation]. Let Ω_β be defined in [source equation]. Let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let A_2 be the quantity introduced in [source equation]. Then, for any $t > 0$, we have*

$$\begin{aligned} A_2 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}), \end{aligned}$$

Implemented correction. For every block I , the process retains the full-objective weight:

$$\mathcal{R}_I(\mathcal{H}) = \mathbb{E} \sup_{h \in \mathcal{H}} \left| \sum_{i \in I} \sum_{j=1}^{m_i} \frac{\varepsilon_{ij}}{n m_i} h(x_{ij}^*) \right|.$$

No unweighted or within-block-normalized quantity is substituted without multiplying by the block's share of the full criterion.

where c_5 depends only on the constants C and c in Assumption [source ref.], and $R_{\mathcal{J}_{e,l}}$ is defined in [source equation].

Proof. We pass from A_2 to a star-world quantity A_2^* via the coupling, and then bound A_2^* .

By the construction of the coupling event Ω_β (see [source equation]), on Ω_β we have $(x_{ij}, e_{ij}, \delta_i) = (x_{ij}^*, e_{ij}^*, \delta_i^*)$ for all indices i in the coupled index set and all $j \in [m_i]$. Hence, by [source equation], for every $f \in \mathcal{F}$ and every such (i, j) ,

$$\hat{Z}_{ij}(f) = \hat{Z}_{ij}^*(f), \quad \tilde{Z}_{ij}(f) = \tilde{Z}_{ij}^*(f) \quad \text{on } \Omega_\beta. \quad (138)$$

Therefore,

$$\begin{aligned} & \left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \\ &= \left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}^*(f) - \hat{Z}_{ij}^*(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}. \end{aligned} \quad (139)$$

Since f^* belongs to the constraint set and $\tilde{Z}_{ij}^*(f^*) - \hat{Z}_{ij}^*(f^*) = 0$, the supremum on the right-hand side is nonnegative. Hence $\mathbf{1}_{\{\Omega_\beta\}} \leq 1$ implies

$$\begin{aligned} A_2 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}^*(f) - \hat{Z}_{ij}^*(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \mathbb{E}_{x^*, \delta^*, e^*} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}^*(f) - \hat{Z}_{ij}^*(f) \right\} \right] \\ &=: A_2^*. \end{aligned} \quad (140)$$

It remains to bound A_2^* . By the tower property, conditioning on $\{\delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ and $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, we have

$$\begin{aligned} A_2^* &= \mathbb{E}_{x^*, \delta^*, e^*} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}^*(f) - \hat{Z}_{ij}^*(f) \right\} \right] \\ &= \mathbb{E}_{\delta^*, x^*} \left[\mathbb{E}_{e^* | \delta^*, x^*} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}^*(f) - \hat{Z}_{ij}^*(f) \right\} \right] \right]. \end{aligned}$$

Conditioned on δ^* and x^* , the quantity $\tilde{Z}_{ij}^*(f) = \mathbb{E}_{e^* | \delta_i^*, x_{ij}^*} [\hat{Z}_{ij}^*(f)]$ is deterministic, and $\{\hat{Z}_{ij}^*(f) - \tilde{Z}_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are independent mean-zero random variables (since within block $\mathcal{J}_{e,l}$, the e_{ij}^* are i.i.d. given δ^*, x^*). Applying the symmetrization inequality (Section 2.3.1 in van der Vaart et al. (1996)), we obtain

$$A_2^* \leq 2 \mathbb{E}_{\delta^*, x^*} \left[\mathbb{E}_{e^*, s | \delta^*, x^*} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}^*(f) \right] \right]$$

$$= 2\mathbb{E}_{x^*, \delta^*, e^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}^*(f) \right].$$

We then apply the Lipschitz condition from Lemma [source ref.] and the Ledoux-Talagrand contraction principle (Theorem [source ref.]) to obtain

$$\begin{aligned} A_2^* &\leq 2\mathbb{E}_{x^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right] \\ &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}), \end{aligned}$$

where the last equality follows from the definition of $R_{\mathcal{J}_{e,l}}$ in [source equation]. $\square$

Lemma 56 (Simplify $\tilde{Z}_{ij}(f)$). *For any $f, f^* \in \mathcal{F}$, and $t \in \mathbb{R}$, we have*

$$\begin{aligned} \tilde{Z}_{ij}(f) &:= \mathbb{E}_{e|x_{ij}, \delta_i} [\hat{Z}_{i,j}(f)] = (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij}, \delta_i)) \\ &\quad + \int_0^{f(x_{ij}) - f^*(x_{ij})} \left(\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij}, \delta_i) \right. \\ &\quad \left. - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij}, \delta_i) \right) dt. \end{aligned}$$

Implemented correction. *The conditional quantities are now*

$$\tilde{Z}_{ij}(f) = \mathbb{E}(\ell_{ij,f} | X_i, \delta_i), \quad Z_{ij}(f) = \mathbb{E}(\ell_{ij,f} | x_{ij}), \quad \bar{Z}_{ij}(f) = \mathbb{E}\ell_{ij,f}.$$

The same conditioning convention is used in every subsequent identity.

Proof. Let $I(\cdot)$ denote the indicator function. From equation (B.3) in Belloni et al. (2011), for any scalars w, v :

$$\rho_\tau(w - v) - \rho_\tau(w) = -v(\tau - I(w \leq 0)) + \int_0^v (I(w \leq z) - I(w \leq 0)) dz.$$

Let $w = y_{ij} - f^*(x_{ij})$ and $v = f(x_{ij}) - f^*(x_{ij})$. Substituting these into the equation:

$$\begin{aligned} \hat{Z}_{ij}(f) &= \rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij})) \\ &= (f^*(x_{ij}) - f(x_{ij})) (\tau - I(y_{ij} - f^*(x_{ij}) \leq 0)) \\ &\quad + \int_0^{f(x_{ij}) - f^*(x_{ij})} \{I(y_{ij} - f^*(x_{ij}) \leq t) - I(y_{ij} - f^*(x_{ij}) \leq 0)\} dt. \end{aligned}$$

Take conditional expectation $\mathbb{E}_{e|x_{ij}, \delta_i}[\cdot]$ on both sides:

For linear term,

$$\begin{aligned} &\mathbb{E}_{e|x_{ij}, \delta_i} [(f^*(x_{ij}) - f(x_{ij})) (\tau - I(y_{ij} \leq f^*(x_{ij})))] \\ &= (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) | x_{ij}, \delta_i)). \end{aligned}$$

For integral term, for $w = y_{ij} - f^*(x_{ij})$ and $v = f(x_{ij}) - f^*(x_{ij})$, define it as:

$$\mathbb{E}_{e|x_{ij}, \delta_i} \left[\int_0^v (I(w \leq t) - I(w \leq 0)) dt \right]$$

$$\begin{aligned}
 &= \int_0^v \mathbb{E}_{e|x_{ij}, \delta_i} [I(w \leq t) - I(w \leq 0)] dt \quad (\text{Fubini's theorem}) \\
 &= \int_0^v (\mathbb{P}(w \leq t | x_{ij}, \delta_i) - \mathbb{P}(w \leq 0 | x_{ij}, \delta_i)) dt.
 \end{aligned}$$

Substituting $v = f(x_{ij}) - f^*(x_{ij})$ and $w = y_{ij} - f^*(x_{ij})$, we obtain:

$$\begin{aligned}
 \tilde{Z}_{ij}(f) &= (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) | x_{ij}, \delta_i)) \\
 &\quad + \int_0^{f(x_{ij}) - f^*(x_{ij})} (\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij}, \delta_i) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij}, \delta_i)) dt.
 \end{aligned}$$

□

Lemma 57 (Simplify $Z_{ij}(f)$). *For any $f, f^* \in \mathcal{F}$, and $t \in \mathbb{R}$, we have*

$$Z_{ij}(f) := \mathbb{E} [\hat{Z}_{i,j}(f) | x_{ij}] = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij})\} dt.$$

Proof. By Belloni et al. (2011) (Eq. B.3),

$$\rho_\tau(w - v) - \rho_\tau(w) = -v(\tau - I(w \leq 0)) + \int_0^v (I(w \leq z) - I(w \leq 0)) dz.$$

Let $w = y_{ij} - f^*(x_{ij})$, $v = f(x_{ij}) - f^*(x_{ij})$. Substituting gives

$$\begin{aligned}
 \hat{Z}_{ij}(f) &= \rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij})) \\
 &= (f^*(x_{ij}) - f(x_{ij})) (\tau - I(y_{ij} \leq f^*(x_{ij}))) \\
 &\quad + \int_0^v (I(w \leq t) - I(w \leq 0)) dt.
 \end{aligned}$$

By the definition of f^* as the conditional τ -quantile:

$$\begin{aligned}
 \mathbb{E} [\tau - I(y_{ij} \leq f^*(x_{ij})) | x_{ij}] &= \tau - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) | x_{ij}) \\
 &= 0.
 \end{aligned}$$

For the second term, apply Fubini's theorem and quantile property:

$$\begin{aligned}
 &\mathbb{E} \left[\int_0^v (I(w \leq t) - I(w \leq 0)) dt | x_{ij} \right] \\
 &= \int_0^v \mathbb{E} [I(w \leq t) - I(w \leq 0) | x_{ij}] dt \\
 &= \int_0^v (\mathbb{P}(w \leq t | x_{ij}) - \mathbb{P}(w \leq 0 | x_{ij})) dt \\
 &= \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij})\} dt.
 \end{aligned}$$

The last equality holds for both $v \geq 0$ and $v < 0$ through sign-unification as in Lemma [source ref.]. Thus, in the end, we have

$$Z_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) | x_{ij})\} dt.$$

□

Bound of A_1 **Implemented correction.** Knight's identity yields the exact cancellation

$$Z_{ij}(f) - \tilde{Z}_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} D_i(x_{ij}, u) du,$$

where $D_i(x, u) = \mathbb{P}(U_{i1} \leq u \mid x_{i1} = x) - \mathbb{P}(U_{i1} \leq u \mid \delta_i, x_{i1} = x)$. The two linear terms cancel identically.

Lemma 58 (Decomposition of A_1). *Let $\tilde{Z}(f)$ and $Z(f)$ be defined in [source equation] and [source equation], respectively. Let Ω_β be defined in [source equation]. Let $\ell(f)$ be as in [source equation], and let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher random variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let A_1 be the quantity introduced in [source equation]. Suppose $\tilde{c}_0$ is a small constant from Assumption [source ref.], and let $c_4 > 0$ be any fixed positive constant. Let L be as defined in [source equation], and assume $s < d$ as in Assumption [source ref.], and T_n as in Assumption [source ref.]. Define the remainder term as*

$$r_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + u \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + u \mid \delta_i, x_{ij})\} du,$$

for $u \in \mathbb{R}$. Then for any $t > 0$, we have

$$\begin{aligned} A_1 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Proof. By previous Lemmas [source ref.] and [source ref.],

By Lemma [source ref.] where $Z_{ij}(f) = \mathbb{E}_{\delta, e \mid x_{ij}} [\hat{Z}_{ij}(f)]$, then by Belloni et al. (2011) (Eq. B.3), we have

$$\begin{aligned} Z_{ij}(f) &= (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij})) \\ &\quad + \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij})\} dt \\ &\stackrel{(a)}{=} \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij})\} dt. \end{aligned} \tag{141}$$

The equality [source equation] holds because $f^*(x_{ij})$ is the conditional τ -quantile given x_{ij} , so $\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}) = 0$.

By Lemma [source ref.], since $\tilde{Z}_{ij}(f) = \mathbb{E}_{e \mid \delta_i, x_{ij}} [\hat{Z}_{ij}(f)]$ conditions on both x_{ij} and δ_i ,

$$\begin{aligned} \tilde{Z}_{ij}(f) &= (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}, \delta_i)) \\ &\quad + \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}, \delta_i) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}, \delta_i)\} dt. \end{aligned} \tag{142}$$

Subtracting [source equation] and [source equation], we obtain

$$\begin{aligned}
 & Z_{ij}(f) - \tilde{Z}_{ij}(f) \\
 &= \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt \\
 &+ \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid \delta_i, x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij})\} dt \\
 &- (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}, \delta_i)).
 \end{aligned}$$

Thus, we have

$$\begin{aligned}
 & Z_{ij}(f) - \tilde{Z}_{ij}(f) \\
 &= \mathbb{E}_{\delta_i, e \mid x_{ij}} [\hat{Z}_{ij}(f)] - \mathbb{E}_{e \mid \delta_i, x_{ij}} [\hat{Z}_{ij}(f)] \\
 &= \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt \\
 &+ \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid \delta_i, x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij})\} dt \\
 &- (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}, \delta_i)). \\
 &= - (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} - f^*(x_{ij}) \leq 0 \mid \delta_i, x_{ij})) \\
 &+ \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt \\
 &+ \int_0^{f(x_{ij})-f^*(x_{ij})} \{-\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid x_{ij}) + \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid \delta_i, x_{ij})\} dt \\
 &= - (f^*(x_{ij}) - f(x_{ij})) (\tau - \mathbb{P}(y_{i,j} - f^*(x_{ij}) \leq 0 \mid \delta_i, x_{ij})) \\
 &+ \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt \\
 &+ (f(x_{ij}) - f^*(x_{ij})) (-\tau + \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) \mid \delta_i, x_{ij})) \\
 &= \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt.
 \end{aligned}$$

Implemented correction. The A_1 term is now decomposed using the exact spatial identity above. Its centered fluctuation is controlled by conditional symmetrization and contraction, while its conditional mean is controlled by the uniform grid bound for $|I|^{-1} \sum_{i \in I} D_i(x, u)$.

We define $r_{ij}(f)$ the as the integral expression given by the last line:

$$r_{ij}(f) := \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt. \quad (143)$$

Then, we have

$$\begin{aligned}
 A_1 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \underbrace{\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} r_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_4}. \tag{144}
 \end{aligned}$$

Now we bound A_4 , and we have

$$\begin{aligned}
 A_4 &\leq \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\quad + \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \mathbb{E}_{\delta, \delta^*} \left[\underbrace{\mathbb{E}_{x, e | \delta} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_5} \right] \\
 &\quad + \underbrace{\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{A_6}. \tag{145}
 \end{aligned}$$

where we denote

$$A_5 := \mathbb{E}_{x, e | \{\delta_i\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right], \tag{146}$$

and

$$A_6 := \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e, l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [r_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \tag{147}$$

Then by Lemma [source ref.] and [source ref.], we get the bound for terms A_5 and A_6 , thus, conclude the bound for A_1 . $\square$

Corollary 59 (Bound of A_1). *Let $\tilde{Z}(f)$, $Z(f)$ be defined in [source equation] and [source equation]. Let Ω_β be defined in [source equation]. Let $\ell(f)$ be defined in [source equation], let $\{s_{i,j}\}_{i \in \mathcal{I}_{e, l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$*

and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let A_1 be the quantity introduced in [source equation], then we have

$$\begin{aligned}
 A_1 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
 &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n},
 \end{aligned}$$

with a small constant c_4 that depends on $\tilde{c}_0$ in Assumption [source ref.], C_β in Assumption [source ref.], and c_1, c_2 in [source equation], and c_5 is a constant that depends on constants C and c in Assumption [source ref.], and $R_{\mathcal{J}_{e,l}}$ is defined in [source equation].

Proof. Let

$$\ell(f) = \frac{4\tilde{c}_0 C}{c} \Delta^2(f - f^*),$$

By Lemma [source ref.], the preceding decomposition yields the stated bound after controlling the centered fluctuation and the conditional mean remainder. we get the bound of A_1 .

We have

$$\begin{aligned}
 &\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\stackrel{(a)}{\leq} 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
 &\stackrel{(b)}{\leq} 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
 &\stackrel{(c)}{\leq} \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

Here [source equation] follows from the decomposition in Lemma [source ref.]; [source equation] follows from the Lipschitz condition of $r_{ij}(f)$ in Lemma [source ref.] together with the Ledoux-Talagrand contraction principle in Theorem [source ref.]; and [source equation] follows from Lemma [source ref.].

Implemented correction. The three displayed inequalities are now justified in order by: (i) the exact three-way loss decomposition; (ii) Ledoux–Talagrand contraction for the one-Lipschitz centered map r_{ij} ; and (iii) the weighted block Rademacher inequality with weights $1/(nm_i)$. □

Lemma 60 (Bound of A_5). *Let A_5 be the term defined in [source equation]. Let $\{s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let Ω_β be defined in [source equation]. The remainder term $r_{ij}(f)$ is defined in Lemma [source ref.]. Then the following inequality holds:*

$$A_5 \leq 2\mathbb{E}_{x,e,s|\delta} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij}(f) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right].$$

Proof. By the coupling identity [source equation], on Ω_β we have $(x_{ij}, \delta_i) = (x_{ij}^*, \delta_i^*)$ for all $i \in \mathcal{J}_{e,l}, j \in [m_i]$. Since $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are i.i.d. by construction, and $\{\delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ are i.i.d., the quantities $\{r_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are conditionally independent given $\{\delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ on the event Ω_β .

Let $\{\tilde{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be an independent and identically distributed copy of $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, correspondingly, we let $\{\tilde{y}_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be the copy of y_{ij}^* , where each $\tilde{y}_{i,j}$ and y_{ij}^* share the same $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}, \{e_{i,j}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, but **differ only in** $\{\tilde{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$. Then we define the independent and identically distributed copy of $r_{ij}(f)$ as

$$\tilde{r}_{ij}(f, \delta_i^*, \tilde{x}_{ij}) := \int_0^{f(\tilde{x}_{ij}) - f^*(\tilde{x}_{ij})} \{\mathbb{P}(\tilde{y}_{i,j} \leq f^*(\tilde{x}_{ij}) + t) - \mathbb{P}(\tilde{y}_{i,j} \leq f^*(\tilde{x}_{ij}) + t | \delta_i^*)\} dt.$$

Now we control the $\mathbb{E}_{\tilde{x}, x, e | \{\delta_i^*\}} [r_{ij}(f) - \tilde{r}_{ij}(f)]$, the expectation takes on $\{\tilde{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and conditioned on $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}$. So we have

$$\begin{aligned} & \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] - \mathbb{E}_{\tilde{x}|\delta_i} [\tilde{r}_{ij}(f)]\} \\ & \quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \\ & \stackrel{(a)}{\leq} \mathbb{E}_{\tilde{x}, x, e | \{\delta_i^*\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{r_{ij}(f) - \tilde{r}_{ij}(f)\} \right) \right. \\ & \quad \left. \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \end{aligned} \tag{148}$$

The inequality [source equation] is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned} & \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] - \mathbb{E}_{\tilde{x}|\delta_i} [\tilde{r}_{ij}(f)]\} \\ & \quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \\ & \stackrel{(a)}{=} \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{r_{ij}(f) - \mathbb{E}_{x,e|\delta_i} [r_{ij}(f)]\} \cdot \mathbf{1}_{\{\Omega_\beta\}}. \end{aligned} \tag{149}$$

The expectation $\mathbb{E}_{x,e|\delta_i} [r_{ij}(f)]$ takes on $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ (which equal $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ on Ω_β) and conditioned on $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}$. The equality [source equation] holds since $\mathbb{E}_{x,e|\delta_i^*} [r_{ij}(f)]$ and $\mathbb{E}_{\tilde{x},e|\delta_i^*} [\tilde{r}_{ij}(f)]$ are identically distributed.

Combines [source equation] and [source equation], taking expectation of $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ conditioning on $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}$ we have

$$\begin{aligned}
 A_5 &= \mathbb{E}_{x,e|\{\delta_i\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{r_{ij}(f) - \mathbb{E}_{x,e|\delta_i} [r_{ij}(f)]\} \right) \right. \\
 &\quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \left. \right] \\
 &\leq \mathbb{E}_{\tilde{x},x,e|\{\delta_i\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{r_{ij}(f) - \tilde{r}_{ij}(f)\} \right) \right. \\
 &\quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \left. \right] \\
 &\stackrel{(a)}{\leq} \mathbb{E}_{\tilde{x},x,e,s|\{\delta_i\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \{r_{ij}(f) - \tilde{r}_{ij}(f)\} \right) \right. \\
 &\quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \left. \right] \\
 &\stackrel{(b)}{\leq} 2 \mathbb{E}_{x,e,s|\{\delta_i\}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij}(f) \right) \right. \\
 &\quad \cdot \mathbf{1}_{\{\Omega_\beta\}} \left. \right].
 \end{aligned}$$

The [source equation] holds since on Ω_β we have $(x_{ij}, \delta_i) = (x_{ij}^*, \delta_i^*)$, and $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are i.i.d. by construction. Hence, conditioned on $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}$, the $r_{ij}(f) - \tilde{r}_{ij}(f)$ is independently distributed and symmetric with mean zero, and $\{s_{ij}(r_{ij}(f) - \tilde{r}_{ij}(f))\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{r_{ij}(f) - \tilde{r}_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ have the same distribution. The [source equation] follows because $\{-s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are also independent Rademacher variables.

Taking expectation of $\{(\delta_{i1}^*, \dots, \delta_{im_i}^*)\}_{i \in \mathcal{J}_{e,l}}$ **on both sides**, we obtain the result. $\square$

Lemma 61 (Generalized Bound of A_6). *In the multi-dimensional setting where $x_{ij} \in [0, 1]^d$, let Ω_β be defined in [source equation]. Let A_6 be defined as*

$$A_6 := \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2 (f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left(\mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] - \frac{\ell(f)}{2L} \right) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],$$

$\ell(f)$ be defined in [source equation], and $r_{ij}(f)$ be defined in [source equation] as

$$r_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{ij} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) + t \mid \delta_i, x_{ij})\} dt.$$

Under Assumptions [source ref.] and [source ref.], and if $\ell(f) = 4\tilde{c}_0 c_2 \Delta^2 (f - f^*)$, then the following bound holds:

$$A_6 \leq \frac{c_2}{L} \cdot \frac{c_3^2}{4\tilde{c}_0} \cdot \frac{s \log n \log T_n}{n} = \frac{c_4}{L} \cdot \frac{s \log n \log T_n}{n},$$

for some positive constant c_4 that depends on $\tilde{c}_0$, C_β , c_1 , and c_2 , where c_3 depends on C_β and c_1 , and where s and T_n are as defined in Assumption [source ref.].

Proof. We first show the following bound holds in a multidimensional setting where $x_{ij} \in [0, 1]^d$:

$$\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}.$$

We then obtain the result by taking expectations and setting

$$\ell(f) = 4\tilde{c}_0 c_2 \Delta^2(f - f^*).$$

Given that

$$\begin{aligned} \mathbb{E}(r_{ij}(f)|\delta_i) &= \int_{[0,1]^d} \left(\int_0^{f(x)-f^*(x)} (\mathbb{P}(y_{ij} \leq f^*(x_{ij}) + u \mid x_{ij} = x) \right. \\ &\quad \left. - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) + u \mid \delta_i, x_{ij} = x)) du \right) dx. \end{aligned} \quad (150)$$

Thus, we modify the function $\Pi(x, u)$ in [source equation] as follows:

$$\Pi(x, u) := \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x)]. \quad (151)$$

We then have that

$$\begin{aligned} &\left| \int_{[0,1]^d} \int_0^{f(x)-f^*(x)} \Pi(x, u) du dx \right| \\ &= \left| \int_{[0,1]^d} \left[\int_0^{f(x)-f^*(x)} \Pi(x, u) du \right] \cdot \mathbb{I}_{\{f(x)-f^*(x) \leq 1\}} dx \right. \\ &\quad \left. + \int_{[0,1]^d} \left[\int_0^{f(x)-f^*(x)} \Pi(x, u) du \right] \cdot \mathbb{I}_{\{f(x)-f^*(x) > 1\}} dx \right| \\ &\leq \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-1,1]} |\Pi(x, u)| \cdot \mathbb{I}_{\{f(x)-f^*(x) \leq 1\}} dx \\ &\quad + \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-|f(x)-f^*(x)|, |f(x)-f^*(x)|]} |\Pi(x, u)| \\ &\quad \cdot \mathbb{I}_{\{f(x)-f^*(x) > 1\}} dx \end{aligned} \quad (152)$$

Step 1. Now, for the $\max_{x \in [0,1]^d} \max_{u \in [-|f(x)-f^*(x)|, |f(x)-f^*(x)|]} |\Pi(x, u)|$ term above, we have

$$\begin{aligned} &\max_{x \in [0,1]^d} \max_u \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) \right. \right. \\ &\quad \left. \left. - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x) \right] \right|. \end{aligned}$$

This term represents the maximum discrepancy between the probabilities of y_{i1} not exceeding the threshold defined by $f^*(x_{i1}) + u$, with and without the consideration of δ_i .

Let $T_n > 0$ be a constant that depends on n , then we have

$$\begin{aligned}
 & \max_{u > T_n} \max_{x \in [0,1]^d} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) - \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid x_{i1} = x)] \right| \\
 & \leq \max_{u > T_n} \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq u \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq u \mid x_{i1} = x)] \\
 & \leq \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \\
 & \stackrel{(a)}{\leq} \tilde{c}_0 \text{ with high probability.}
 \end{aligned}$$

The [source equation] holds by Assumption [source ref.]. Since

$$\begin{aligned}
 & \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \\
 & \leq \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \leq \tilde{c}_0
 \end{aligned}$$

Similarly, we have

$$\begin{aligned}
 & \max_{u < -T_n} \max_{x \in [0,1]^d} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) - \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid x_{i1} = x)] \right| \\
 & \leq \max_{u < -T_n} \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid x_{i1} = x)] \\
 & \leq \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid x_{i1} = x)] \\
 & \stackrel{(a)}{\leq} \tilde{c}_0 \text{ with high probability.}
 \end{aligned}$$

The [source equation] holds by Assumption [source ref.]. Since

$$\begin{aligned}
 & \max_{x \in [0,1]^d} \max_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq -T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq -T_n \mid x_{i1} = x)] \\
 & \leq \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq -T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq -T_n \mid x_{i1} = x)] \leq \tilde{c}_0.
 \end{aligned}$$

Step 2. Next, for $u \in [-T_n, T_n]$, we define:

$$\max_{x_S \in [0,1]^s} \max_{u \in [-T_n, T_n]} \left\{ \underbrace{[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x_S) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x_S)]}_{:= \Pi(x_S, u)} \right\}.$$

Consider the space $[0,1]^s$. For the i -th dimension of $[0,1]^s$, let $0 = a_{1,i} < \dots < a_{M,i} \leq a_{M+1,i} = 1$ be an evenly spaced partition. Let $c_{k,i}$ be the midpoint of $[a_{k,i}, a_{k+1,i}]$ for $k = 1, \dots, M$. Define the multi-dimensional grid points in $[0,1]^s$ as $c_k = (c_{k,1}, c_{k,2}, \dots, c_{k,s})$. Similarly, let $-T_n = b_1 < \dots < b_M \leq b_{M+1} = T_n$, partition the interval $[-T_n, T_n]$ into M equal parts with midpoints d_l . For each $k = 1, \dots, M$, given vectors $x_S, c_k \in \mathbb{R}^s$ and an index $i \in \{1, 2, \dots, s\}$, we define a new vector $x_S^{\setminus ki} \in \mathbb{R}^s$ via

$$x_{Sj}^{\setminus ki} := \begin{cases} x_{Sj} & \text{if } j \neq i, \\ c_{k,i} & \text{if } j = i. \end{cases}$$

The $x_S^{\setminus ki}$, $i = 1, \dots, s$ indicates with i -th entry of x_S replaced with c_{ki} .

Given a Lipschitz function $\Pi : [0, 1]^s \times [-T_n, T_n] \rightarrow \mathbb{R}$, we approximate $\Pi(x_S, u)$ as follows:

$$\begin{aligned}
 & \max_{x_S \in [0, 1]^s} \max_{u \in [-T_n, T_n]} |\Pi(x_S, u)| \\
 & \leq \max_{k1=1, \dots, M} \max_{k2=1, \dots, M} \cdots \max_{ks=1, \dots, M} \max_{l=1, \dots, M} |\Pi((c_{k1}, c_{k2}, \dots, c_{ks}), d_l)| \\
 & \quad + \max_{x_S \in [0, 1]^s} \max_{u \in [-T_n, T_n]} \min_{l=1, \dots, M} \min_{k1=1, \dots, M} |\Pi(x_S^{\setminus k1}, d_l) - \Pi(x_S, u)| \\
 & \quad + \cdots \\
 & \quad + \max_{x_S \in [0, 1]^s} \max_{u \in [-T_n, T_n]} \min_{l=1, \dots, M} \min_{ks=1, \dots, M} |\Pi(x_S^{\setminus ks}, d_l) - \Pi(x_S, u)|.
 \end{aligned}$$

For $x_S \in [a_{k,1}, a_{k+1,1}] \times \cdots \times [a_{k,s}, a_{k+1,s}]$ and $u \in [b_l, b_{l+1}]$, we estimate:

$$\begin{aligned}
 & |\Pi(x_S^{\setminus ki}, d_l) - \Pi(x_S, u)| \\
 & \leq |\Pi(x_S^{\setminus ki}, u) - \Pi(x_S^{\setminus ki}, d_l)| + |\Pi(x_S^{\setminus ki}, u) - \Pi(x_S, u)| \\
 & \leq \frac{\alpha \cdot 2T_n}{M} + \frac{\alpha}{M} \lesssim \frac{T_n}{M}.
 \end{aligned}$$

The last inequality follows by [source equation]. By Assumption [source ref.], Π is Lipschitz, leading to the following probability bound:

$$\begin{aligned}
 & \mathbb{P} \left(\max_{x_S \in [0, 1]^s} \max_{u \in [-T_n, T_n]} |\Pi(x_S, u)| - \frac{sT_n}{M} \geq t \right) \\
 & \leq \mathbb{P} \left(\max_{k1=1, \dots, M} \max_{k2=1, \dots, M} \cdots \max_{ks=1, \dots, M} \max_{l=1, \dots, M} |\Pi((c_{k1}, c_{k2}, \dots, c_{ks}), d_l)| \geq t \right) \\
 & \leq M^s e^{-2|\mathcal{J}_{e,l}|t^2} \leq M^s e^{-\frac{2c_1 C_\beta n t^2}{\log n}},
 \end{aligned}$$

where the second last inequality holds since $|\Pi(\cdot)| \leq 1$ uniformly, and then using the Hoeffding's inequality and union bound for the $\Pi(\cdot)$ defined in [source equation]. Take $M = \left(sT_n \cdot \sqrt{\frac{n}{\log n}} \right)$, therefore:

$$\max_{x_S \in [0, 1]^s} \max_{u \in [-T_n, T_n]} |\Pi(x, u)| \leq c_3 \sqrt{\frac{s \log n \log T_n}{n}} = O_p \left(\sqrt{\frac{s \log n \log T_n}{n}} \right). \quad (153)$$

where the c_3 is a constant that depends on c_1 and C_β .

Step 3. Returning to [source equation], using Assumption $|\mathcal{J}_{e,l}| \leq c_2 n/L$ in [source equation], we have:

$$\begin{aligned}
 & \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} |\mathbb{E}[r_{ij}(f)|\delta_i]| \stackrel{(a)}{=} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} |\mathbb{E}[r_{ij}(f)|\delta_i]| \\
 & \stackrel{(b)}{\leq} \frac{c_2}{L} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \int_{[0,1]^d} \int_0^{f(x)-f^*(x)} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) \right] \right.
 \end{aligned}$$

$$\begin{aligned}
 & \left| -\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x) \right] du dx \Big| \\
 & \leq \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-1,1]} |\Pi(x, u)| \cdot \mathbb{1}_{\{f(x) - f^*(x) \leq 1\}} dx \\
 & \quad + \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-|f(x) - f^*(x)|, |f(x) - f^*(x)|]} |\Pi(x, u)| \cdot \mathbb{1}_{\{f(x) - f^*(x) > 1\}} dx \\
 & \leq \tilde{c}_0 \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \mathbb{1}_{\{f(x) - f^*(x) > 1\}} dx \\
 & \quad + \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \mathbb{1}_{\{f(x) - f^*(x) \leq 1\}} dx \cdot c_3 \sqrt{\frac{s \log n \log T_n}{n}} \\
 & \leq \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \Delta(f - f^*) \cdot c_3 \sqrt{\frac{s \log n \log T_n}{n}} \\
 & \stackrel{(c)}{\leq} \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \frac{c_3^2 c_2}{L} \frac{1}{4\tilde{c}_0} \frac{s \log n \log T_n}{n} \\
 & \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n},
 \end{aligned}$$

where the equality [source equation] uses the fact for fixed i , $\mathbb{E}[r_{ij}(f)|\delta_i] = \mathbb{E}[r_{i1}(f)|\delta_i]$ for all $j = 1, \dots, m_i$, where the equality [source equation] follows by $|\mathcal{J}_{e,l}| \leq c_2 n/L$, where the inequality [source equation] uses the bound $ab \leq \frac{pa^2}{2} + \frac{b^2}{2p}$ for any $p > 0$, setting $p = 2\tilde{c}_0$.

Thus, we have

$$\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}.$$

Adding expectation and supremum on both sides, we get

$$\begin{aligned}
 & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i} [r_{ij}(f)] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

□

Below, we provide examples under which Assumptions [source equation] and [source equation] from Assumption [source ref.] are satisfied.

Example 62. Assume the event

$$\Omega_1 = \left\{ \max_{i=1,\dots,n} \|\delta_i\|_\infty \leq R_n \right\}$$

occurs with high probability. Furthermore, define $e_{i1} = g(x_{i1})V_{i1}$ where $\{V_{i1}\}$ is independent from $\{x_{i1}\}$, and consider the event

$$\Omega_2 = \left\{ \max_{i=1,\dots,n} |g(x_{i1})| \leq s_n \right\}$$

also occurring with high probability. We then introduce T_n as

$$T_n = \max_{i=1,\dots,n} \{2R_n, s_n \phi_n\},$$

with ϕ_n being a sequence tending to infinity. Then we have

$$\begin{aligned} & \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) \\ & \quad + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \\ & \stackrel{(a)}{\leq} \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [2\mathbb{P}(\|\delta_i\|_\infty \geq R_n) \\ & \quad + 2\mathbb{P}(\Omega_2^c) \\ & \quad + 2\mathbb{P}\left(|V_{i1}| \geq \max\left\{\frac{R_n}{s_n}, \frac{\phi_n}{2}\right\}\right)] \\ & \leq \tilde{c}_0. \end{aligned}$$

The [source equation] holds since we have

$$\begin{aligned} & \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) \\ & \quad + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \\ & \leq \max_{x \in [0,1]^d} \max_{i=1,\dots,n} \left[\mathbb{P}\left(g(x_{i1})V_{i1} \geq \max\left\{R_n, \frac{s_n \phi_n}{2}\right\} \mid x_{i1} = x\right) \right. \\ & \quad + 2\mathbb{P}\left(\|\delta_i\|_\infty \geq \max\left\{R_n, \frac{s_n \phi_n}{2}\right\}\right) \\ & \quad \left. + \mathbb{P}\left(\left\{g(x_{i1})V_{i1} \geq \max\left\{R_n, \frac{s_n \phi_n}{2}\right\}\right\} \mid x_{i1} = x\right) \right] \\ & \leq \max_{x \in [0,1]^d} \max_{i=1,\dots,n} \left[2\mathbb{P}\left(\|\delta_i\|_\infty \geq \max\left\{R_n, \frac{s_n \phi_n}{2}\right\}\right) \right. \\ & \quad + 2\mathbb{P}(\Omega_2^c \mid x_{i1} = x) \\ & \quad \left. + 2\mathbb{P}\left(g(x_{i1})V_{i1} \geq \max\left\{R_n, \frac{s_n \phi_n}{2}\right\} \mid \Omega_2, x_{i1} = x\right) \right] \\ & \leq \tilde{c}_0. \end{aligned}$$

Provided

1. $\mathbb{P}(\Omega_1^c) \rightarrow 0$.
2. $\mathbb{P}(\Omega_2^c) \rightarrow 0$.
3. $\max_{i=1,\dots,n} \mathbb{P}(|V_{i1}| \geq \phi_n) \rightarrow 0$.
4. $T_n \geq \max\{2R_n, \phi_n s_n\}$.

Similarly, we can get for the multidimensional domain $[0, 1]^d$

$$\begin{aligned} & \max_{x \in [0,1]^d} \max_{i=1,\dots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid \delta_i, x_{i1} = x) \\ & \quad + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid x_{i1} = x)] \end{aligned}$$

$$\leq \tilde{c}_0.$$

So in summary, [source equation] and [source equation] of Assumption [source ref.] hold if the above conditions hold for the multidimensional domain $[0, 1]^d$.

Lemma 63 (Bound of A_3). *Let $\bar{Z}(f)$ and $Z(f)$ be defined in [source equation] and [source equation], respectively. Let Ω_β be defined in [source equation]. Let $\{s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher random variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let A_3 be the quantity introduced in [source equation]. Then, for any $t > 0$, we have*

$$\begin{aligned} A_3 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} (\bar{Z}_{ij}(f) - Z_{ij}(f)) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}), \end{aligned}$$

where c_5 depends only on the constants C and c in Assumption [source ref.], and $R_{\mathcal{J}_{e,l}}$ is defined in [source equation].

Proof. We pass from A_3 to a star-world quantity A_3^* via the coupling, and then bound A_3^* .

By the coupling identity [source equation], on Ω_β we have $(x_{ij}, e_{ij}, \delta_i) = (x_{ij}^*, e_{ij}^*, \delta_i^*)$, so that

$$\bar{Z}_{ij}(f) = \bar{Z}_{ij}^*(f), \quad Z_{ij}(f) = Z_{ij}^*(f) \quad \text{on } \Omega_\beta.$$

Therefore,

$$\begin{aligned} &\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} (\bar{Z}_{ij}(f) - Z_{ij}(f)) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \\ &= \left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} (\bar{Z}_{ij}^*(f) - Z_{ij}^*(f)) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}. \end{aligned} \tag{154}$$

Since f^* belongs to the constraint set and $\bar{Z}_{ij}^*(f^*) - Z_{ij}^*(f^*) = 0$, the supremum is nonnegative. Hence $\mathbf{1}_{\{\Omega_\beta\}} \leq 1$ implies

$$A_3 \leq \mathbb{E}_{x^*, \delta^*, e^*} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} (\bar{Z}_{ij}^*(f) - Z_{ij}^*(f)) \right] =: A_3^*. \tag{155}$$

It remains to bound A_3^* . Let $\{\tilde{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be an independent and identically distributed copy of $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, then we define the independent and identically distributed copy of $Z_{ij}^*(f)$ as

$$\tilde{Z}_{ij}(f) := \mathbb{E}_{\delta_i^*, e_{ij}^* | \tilde{x}_{ij}} [\rho_\tau(f^*(\tilde{x}_{ij}) + \delta_i^*(\tilde{x}_{ij}) + e_{ij}^* - f(\tilde{x}_{ij})) - \rho_\tau(\delta_i^*(\tilde{x}_{ij}) + e_{ij}^*)].$$

Now we control $\mathbb{E}_{\tilde{x}|x^*,\delta^*,e^*} [\check{Z}_{ij}(f) - Z_{ij}^*(f)]$, the expectation takes on $\{\tilde{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$. So we have

$$\begin{aligned} & \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \mathbb{E}_{\tilde{x}} [\check{Z}_{ij}(f)] - \mathbb{E}_{\tilde{x}} [Z_{ij}^*(f)] \} \\ & \stackrel{(a)}{\leq} \mathbb{E}_{\tilde{x}} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \check{Z}_{ij}(f) - Z_{ij}^*(f) \} \right) \right]. \end{aligned} \quad (156)$$

The inequality [source equation] is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned} & \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \mathbb{E}_{\tilde{x}} [\check{Z}_{ij}(f)] - Z_{ij}^*(f) \} \\ & \stackrel{(a)}{=} \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \bar{Z}_{ij}^*(f) - Z_{ij}^*(f) \}. \end{aligned} \quad (157)$$

The equality [source equation] holds since given δ_i^* , $\check{Z}_{ij}(f)$ is independent of x^*, e^* and has the same conditional distribution as $Z_{ij}^*(f)$, so $\mathbb{E}_{\tilde{x}} [\check{Z}_{ij}(f)]$ and $\bar{Z}_{ij}^*(f) = \mathbb{E}_{x^*, \delta^*, e^*} [\hat{Z}_{ij}^*(f)]$ are the same quantity.

Combines [source equation] and [source equation], taking expectation of $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ we have

$$\begin{aligned} A_3^* &= \mathbb{E}_{x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \bar{Z}_{ij}^*(f) - Z_{ij}^*(f) \} \right) \right] \\ &\leq \mathbb{E}_{\tilde{x}, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \check{Z}_{ij}(f) - Z_{ij}^*(f) \} \right) \right] \\ &\stackrel{(a)}{=} \mathbb{E}_{\tilde{x}, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \{ \check{Z}_{ij}(f) - Z_{ij}^*(f) \} \right) \right] \\ &\stackrel{(b)}{\leq} 2 \mathbb{E}_{x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}^*(f) \right) \right]. \end{aligned}$$

The [source equation] holds since $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are i.i.d. by construction, and $\{\tilde{x}_{ij}\}$ is an independent copy with the same distribution. Hence the $\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ is independently distributed and symmetric with mean zero, and $\{s_{ij}(\check{Z}_{ij}(f) - Z_{ij}^*(f))\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ have the same distribution. The [source equation] follows because $\{-s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are also independent Rademacher variables.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma [source ref.], and then apply the Ledoux-Talagrand contraction principle in Theorem [source ref.].

Then we have

$$\begin{aligned}
 A_3 &\leq A_3^* \leq 2\mathbb{E}_{x^*, \delta^*, e^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}^*(f) \right] \\
 &\leq 2\mathbb{E}_{x^*, s} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right] \\
 &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}).
 \end{aligned} \tag{158}$$

Where the second inequality follows from the Lipschitz condition and the Ledoux-Talagrand contraction principle, and the last equality follows from Lemma [source ref.]. $\square$

J.1.7 Proof of Step 4

Implemented correction. All blocks are retained:

$$\sum_{\ell=1}^L \sum_{a \in \{e, o\}} \mathcal{R}_{a, \ell} \{\mathcal{H}_2(c_0^{-1/2} b_B t)\}.$$

The proof sums these terms, adds $\mathbb{P}(\Omega_\beta^c)$, and invokes the critical-radius lemma; it does not replace ℓ by 1 or use Theorem [source ref.] circularly.

Lemma 64 (Bound of A_1 , A_2 , and A_3 by Rademacher width). *Let $\hat{f}, f^* \in \mathcal{F}$. Let Ω_β be defined in [source equation]. Suppose the distribution of $\{(x_{ij}, y_{ij})\}_{i=1, j=1}^{n, m_i}$ satisfies Assumption [source ref.].*

Let c_0 satisfy [source equation]. Let $R(\mathcal{F})$ denote the Rademacher width defined in [source ref.], and let $\tilde{c}_0$ be the small constant given in Assumption [source ref.].

Then, for any $t > 0$, the following inequality holds:

$$\begin{aligned}
 \mathbb{P}\left(c_0 \Delta^2(\hat{f} - f^*) > t^2\right) &\leq \frac{c_5}{t^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}) \right. \\
 &\quad \left. + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}) \right) + \frac{c_4}{t^2} \cdot \frac{s \log n \log T_n}{n},
 \end{aligned} \tag{159}$$

where c_5 depends only on the constants C and c in Assumption [source ref.], c_4 depends on $\tilde{c}_0$ in Assumption [source ref.], C_β in Assumption [source ref.], and c_1, c_2 in [source equation], and $R_{\mathcal{J}_{e,l}}, R_{\mathcal{J}_{o,l}}$ are defined in [source equation].

Proof. We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma [source ref.], and then apply the Ledoux-Talagrand contraction principle in Theorem [source ref.]. We apply the Lemmas [source ref.], [source ref.] and [source ref.]. let A_1 be the quantity introduced in [source equation], By Lemma [source ref.], we have

$$\begin{aligned}
 A_1 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f-f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

Similarly, we have for B_1 in [source equation]

$$B_1 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.$$

Let A_2 be the quantity introduced in [source equation], by Lemma [source ref.], we have

$$\begin{aligned} A_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}). \end{aligned}$$

Similarly, we have

$$B_2 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}).$$

By Lemma [source ref.] we have

$$\begin{aligned} A_3 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f-f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \bar{Z}_{ij}(f) - Z_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}). \end{aligned} \tag{160}$$

Similarly, we have

$$B_3 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}).$$

Thus, by Theorem [source ref.], with Ω_β defined in [source equation], we have

$$\begin{aligned} &\mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > t^2 \right) \\ &\stackrel{(a)}{\leq} \sum_{l=1}^L \left(6c_5 \frac{R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\})}{t^2} \right. \\ &\quad \left. + \frac{2c_4}{Lt^2} \frac{s \log n \log T_n}{n} \right) \\ &= 6c_5 \frac{\sum_{l=1}^L R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\})}{t^2} \\ &\quad + \frac{2c_4}{t^2} \frac{s \log n \log T_n}{n} \\ &= \frac{c_5 \left(\sum_{l=1}^L R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) \right)}{t^2} \\ &\quad + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n} \end{aligned}$$

□

Based on the results in Lemma [source ref.] and Lemma [source ref.], we have on event $\Omega_\beta \cup \Omega_{\beta}^c$,

$$\begin{aligned}
 & \mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > t^2 \right) \\
 &= \frac{c_5}{t^2} \left(\sum_{l=1}^L R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\} \right) \right. \\
 & \quad \left. + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\} \right) \right) + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n} + \frac{3}{nt^2} \quad (161) \\
 &= \frac{c_5}{t^2} \left(\sum_{l=1}^L R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\} \right) \right. \\
 & \quad \left. + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\} \right) \right) + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

In the last line, we combine the c_4 with some other numeric constant into one constant c_4 .

J.1.8 Proof of Step 5

Implemented correction. The pointwise comparison is

$$|f - f^*|^2 \leq (1 \vee 2B) \min\{|f - f^*|, |f - f^*|^2\}.$$

Thus $b_B^2 = 1 \vee 2B$ and the $L_2(Q)$ localization radius is $c_0^{-1/2} b_B t$.

Lemma 65. Let $f, f^* \in \mathcal{F}$. Let $B > 0$ be the constant defined in [source equation], and let the Huber-type loss $\Delta^2(f)$ be defined in [source equation]. Then the following inequality holds:

$$\|f - f^*\|_2^2 \leq \max\{B, 1\} \cdot \Delta^2(f - f^*).$$

Proof. We notice that

$$\begin{aligned}
 \|f - f^*\|_2^2 &= \int (f - f^*)^2(x) dx \\
 &= \int_{\{t: |(f-f^*)(t)| \leq 1\}} |(f - f^*)(x)|^2 dx + \int_{\{t: |(f-f^*)(t)| > 1\}} |(f - f^*)(x)|^2 dx \\
 &\leq \int_{\{t: |(f-f^*)(t)| \leq 1\}} |(f - f^*)(x)|^2 dx + B \cdot \int_{\{t: |(f-f^*)(t)| > 1\}} |(f - f^*)(x)| dx \\
 &\leq \max\{B, 1\} \cdot \Delta^2(f - f^*).
 \end{aligned}$$

□

Definition 66. We provide addition definitions used in the below proofs. Let

$$B_2(t) := \{f : \|f - f^*\|_2 \leq t\},$$

$$B_{mn}(t) := \{f : \|f - f^*\|_{nm} \leq t\}.$$

Lemma 67. For $f, f^* \in \mathcal{F}$, there exists a constant $b(B) := \max\{B^{1/2}, 1\}$ that depends on B , such that

$$R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\} \right) \leq R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \quad (162)$$

Proof. By Lemma [source ref.],

$$\Delta^2(f - f^*) \leq t^2 \implies \|f - f^*\|_2^2 \leq \max\{B, 1\} \cdot \Delta^2(f - f^*) \leq \max\{B, 1\} \cdot t^2.$$

Then we have

$$R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{f : c_0 \Delta^2(f - f^*) \leq t^2\}) \leq R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \left\{f : \|f - f^*\|_2^2 \leq c^{-1} \cdot \max\{B, 1\} \cdot t^2\right\}\right).$$

□

Corollary 68. Let $\kappa > 0$ be a constant and if $\{r_n\}$ is a sequence such that

$$\begin{aligned} \lim_{\kappa \rightarrow \infty} \sup_{n \geq 1} & \left\{ \frac{c_5 \sum_{l=1}^L R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2} \right. \\ & \left. + \frac{c_5 \sum_{l=1}^L R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2} + \frac{\frac{c_4 s \log n \log T_n}{n} + \frac{3}{n}}{\kappa^2 r_n^2} \right\} = 0, \end{aligned} \quad (163)$$

then

$$\Delta^2(\hat{f} - f^*) = O_p(r_n^2). \quad (164)$$

Proof. Let $\varepsilon > 0$ be given. Notice that for any $\kappa > 0$ we have that, from Lemma [source ref.] and Lemma [source ref.],

$$\begin{aligned} & \mathbb{P}\left(c_0 \Delta^2(\hat{f} - f^*) > \kappa^2 r_n^2\right) \\ & \leq \frac{c_5 \sum_{l=1}^L R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2} \\ & \quad + \frac{c_5 \sum_{l=1}^L R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2} + \frac{\frac{c_4 s \log n \log T_n}{n} + \frac{3}{n}}{\kappa^2 r_n^2} \\ & \stackrel{(a)}{<} \varepsilon, \end{aligned} \quad (165)$$

where the inequality [source equation] holds and the last by choosing κ large enough.

Then we have

$$\Delta^2(\hat{f} - f^*) = O_p(r_n^2). \quad (166)$$

And by [source equation] setting that

$$r_n^2 \asymp \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n\right)\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n\right)\right) \right) + \frac{s \cdot \text{poly}(\log n) \log T_n}{n}.$$

□

J.1.9 Proof of Theorem [source ref.]

Implemented correction. The formerly undefined radius has been replaced by

$$\Psi_n(r) = C_* \left[\sum_{a,\ell} \mathcal{R}_{a,\ell} \{ \mathcal{H}_{\mathcal{F}}(c_0^{-1/2} b_B r) \} + b_B^2 \Gamma_n \right], \quad r_n = \inf \{ r > 0 : \Psi_n(r) \leq r^2/64 \}.$$

The final conclusion is $\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}(b_B^2 r_n^2)$.

Finally, we present the proof of Theorem [source ref.]. To proceed in the proof of Theorems [source ref.], by using our Lemma [source ref.], we start by writing

$$R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) \leq R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{f : \|f - f^*\| \leq c_0^{-1/2} \cdot b(B) \cdot t\}),$$

where $b(B) = \max\{B^{1/2}, 1\}$ is a constant that depends on B .

From Lemma [source ref.], we get

$$\begin{aligned} & \mathbb{P}(c_0 \Delta^2(\hat{f} - f^*) > t^2) \\ & \leq \frac{c_5}{t^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) \right) + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n}. \end{aligned}$$

By using our Lemma [source ref.], we have

$$\begin{aligned} & \mathbb{P}(c_0 \Delta^2(\hat{f} - f^*) > t^2) \\ & \leq \frac{c_5}{t^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \right) + \frac{c_4}{t^2} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Then by Corollary [source ref.], we have

$$\Delta^2(\hat{f} - f^*) = O_p(t^2), \tag{167}$$

where

$$\begin{aligned} t^2 & \asymp \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \right) + \frac{s \cdot \text{poly}(\log n) \log T_n}{n}, \end{aligned}$$

for some $\mu > 0$.

J.2 Trend Filtering

Implemented correction. For trend filtering, $M_I \asymp N/L$ and the weighted block complexity is

$$\mathcal{R}_I \lesssim \frac{\sqrt{M_I}}{N} V^{1/(2k)} r^{1-1/(2k)} + \frac{B}{N}.$$

Summing blocks and solving the fixed point gives

$$r^2 \asymp V^{2/(2k+1)} (L/N)^{2k/(2k+1)},$$

plus the spatial term Γ_n .

J.2.1 Proof sketch of Theorem [source ref.]

Recall Definition [source ref.] that $F_k(1)$ denote function space

$$F_k(1) := \{f \in \mathcal{F} : [0, 1] \rightarrow \mathbb{R} : f \text{ is } (k-1) \text{ times weakly differentiable}$$

$$\text{and } J_k(f) = TV(f^{(k-1)}) < 1\}.$$

By Corollary [source ref.] or Theorem [source ref.], we would need to get $R_{\mathcal{J}_{e,l}}(F_k(1) \cap B_2(c^{-\frac{1}{2}} \cdot b(B) \cdot t))$ and $R_{\mathcal{J}_{o,l}}(F_k(1) \cap B_2(c^{-\frac{1}{2}} \cdot b(B) \cdot t))$, to get that, we will use an existing Theorem that bounds local Rademacher complexity using the L_2 set with the empirical set.

$$\begin{aligned} R_{\mathcal{J}_{e,l}}(F_k(1) \cap \{f : \|f - f^*\| \leq b\}) &\leq R_{\mathcal{J}_{e,l}}(F_k(1) \cap \{f : \|f - f^*\|_{nm} \leq \sqrt{2b}\}), \text{ and} \\ R_{\mathcal{J}_{o,l}}(F_k(1) \cap \{f : \|f - f^*\| \leq b\}) &\leq R_{\mathcal{J}_{o,l}}(F_k(1) \cap \{f : \|f - f^*\|_{nm} \leq \sqrt{2b}\}). \end{aligned}$$

where $b > 0$ is any constant, where we denote $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij})^2 \right)^{1/2}$

Lastly, we will apply Dudley's entropy integral (see Dudley (1967)) to get the rate. The Lemma [source ref.] provides the details of the last two steps.

J.2.2 Technical Results for Proof of Theorem [source ref.]

Lemma 69 (Trend Filtering). *Let $J_k(\cdot) = TV(\cdot^{(k-1)})$, the k -th order trend filtering operator, for $F_k(1)$, we have*

$$\Delta^2(\hat{f} - f^*) = O_p \left(\frac{\min \left\{ B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}} \right\}}{(nm)^{\frac{2k}{2k+1}}} + \frac{\log n}{n} \right).$$

Proof. By Corollary [source ref.], we look for the radius to satisfy [source equation], By using [source equation], we first analyze the below quantity,

$$\frac{R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\})}{t^2},$$

Also, let $b = c^{-1/2} \max \{B^{1/2}, 1\} \cdot t$.

For any function class $\mathcal{F} \cap B_2(b)$, we have

$$R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap B_2(b)) = \mathbb{E}_{x^*} \left[\mathbb{E}_s \left[\sup_{g \in \mathcal{F} \cap B_2(b)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}^*) \mid \{x_{ij}^*\} \right] \right]$$

Denote

$$R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(b)) = \mathbb{E}_s \left[\sup_{g \in \mathcal{F} \cap B_2(b)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}^*) \mid \{x_{ij}^*\} \right] \quad (168)$$

Also denote by $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij}^*)^2 \right)^{1/2}$

Corollary 2.2 of Bartlett et al. (2005) gives

$$\mathcal{F} \cap B_2(b) \subseteq \mathcal{F} \cap B_{mn}(\sqrt{2}b),$$

with probability at least $1 - \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \geq 1 - \frac{1}{C_\beta c c_1 n m}$, where c appears in Assumption [source ref.]. Define this good event as $\mathcal{E}$. On $\mathcal{E}$ using Dudley's entropy integral (see Dudley (1967)), Equation [source equation], and the assumption $m_i \asymp m$ specified in the statement of the Lemma, with c defined in Assumption [source ref.], we bound the empirical Rademacher complexity defined in Equation [source equation] as follows

$$\begin{aligned} R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(b)) &\leq R_{\mathcal{J}_{elmn}}\left(\left\{g \in F : \|g\|_{mn}^2 \leq 2\tilde{b}^2\right\}\right) \\ &\leq \frac{C_{\text{Dud}}}{\sqrt{\sum_{i \in \mathcal{J}_{e,1}} m_i}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, \|\cdot\|_{mn}, \mathcal{F})} d\epsilon \\ &\leq \frac{C_{\text{Dud}}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, \|\cdot\|_{mn}, \mathcal{F})} d\epsilon. \end{aligned}$$

Under the event $\mathcal{E}^c$, given that $F \subseteq B_\infty(B)$, we have that $R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(b)) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

Set that $\mathcal{F} \equiv F_k(1)$, we obtain that for some positive constant c_7 , and $A > 0$ from Lemma [source ref.].

The Rademacher complexity defined in Equation [source equation] is bounded by

$$\begin{aligned} R_{\mathcal{J}_{e,1}}(F_k(1) \cap B_2(b)) &= \mathbb{E}_{x^*}[R_{\mathcal{J}_{elmn}}(F_k(1) \cap B_2(b))] \\ &\leq \frac{C_{\text{Dud}}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, \|\cdot\|_{mn}, F_k(1))} d\epsilon + \frac{2B \log(n)}{C_\beta c c_1 n m} \\ &\leq \frac{A C_{\text{Dud}} B^{1/2k}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \epsilon^{-1/(2k)} d\epsilon + \frac{2B \log(n)}{C_\beta c c_1 n m} \\ &= c_6 \frac{B^{1/2k} \sqrt{2 \log(n)}}{\sqrt{C_\beta c c_1 n m}} b^{1-1/(2k)} + \frac{2B \log(n)}{C_\beta c c_1 n m}. \end{aligned}$$

Throughout all the proofs, we will use c_6 to denote some absolute constant that only depends on A , C_{Dud} , C_β , c , c_1 , which may change according to the context.

Without loss of generality assume that $b \geq (nm)^{-1/2}$. Since $k/(2k-1) \geq 1/2$, we have that $b \geq (nm)^{-k/(2k-1)}$ and in consequence $b^{(2k-1)/k} \geq (nm)^{-1}$. Therefore $b^{(2k-1)/k} \geq (nm)^{-1/2}$, which can be written as $b^{1-1/(2k)} \geq (nm)^{-1/2}$, which implies $b^{1-1/(2k)}/\sqrt{nm} \geq (nm)^{-1}$. Thus we have that

$$c_6 \frac{B^{1/2k} \sqrt{2 \log(n)}}{\sqrt{C_\beta c c_1 n m}} b^{1-1/(2k)} + \frac{2B \log(n)}{C_\beta c c_1 \sqrt{nm}} b^{1-1/(2k)} < c_6 B \frac{2 \log(n)}{\sqrt{C_\beta c c_1 n m}} b^{1-1/(2k)}. \quad (169)$$

Here we use Now, using [source equation], we see that

$$\begin{aligned} & \frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\})}{t^2} \\ & \leq \frac{c_6 B \log(n)}{t^2 \sqrt{C_\beta c c_1 n m}} b^{1-1/(2k)} = \frac{c_6 c^{-1} \max\{B, B^2\} \log(n) b^{1-1/(2k)}}{c^{-1} \max\{B, 1\} t^2 \sqrt{C_\beta c c_1 n m}} = \frac{c_6 \max\{B, B^2\} \log(n)}{b^{(2k+1)/(2k)} \sqrt{C_\beta c c_1 n m}}. \end{aligned} \quad (170)$$

If $B \geq 1$, from Inequality [source equation], we get

$$\frac{c_6 \max\{B, B^2\} \log(n)}{b^{(2k+1)/(2k)} \sqrt{C_\beta c c_1 n m}} = \frac{c_6 B^2 \log(n)}{b^{(2k+1)/(2k)} \sqrt{C_\beta c c_1 n m}}. \quad (171)$$

Then we let

$$b^2 = \frac{B^{\frac{8k}{2k+1}} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

to make the [source equation] be $O\left(\frac{1}{nm}\right)$.

Furthermore, we know $b^2 = c^{-1} \max\{B, 1\} t^2 = c^{-1} B t^2$ when $B > 1$, so we get

$$t^2 = \frac{c B^{\frac{6k-1}{2k+1}} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

If $B < 1$, from Inequality [source equation],

$$\frac{c_6 \max\{B, B^2\} \log(n)}{b^{(2k+1)/(2k)} \sqrt{C_\beta c c_1 n m}} = \frac{c_6 B \log(n)}{b^{(2k+1)/(2k)} \sqrt{C_\beta c c_1 n m}} \quad (172)$$

then we let

$$b^2 = \frac{B^{\frac{4k}{2k+1}} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

The [source equation] is $O\left(\frac{1}{nm}\right)$.

Furthermore, we know $b^2 = c^{-1} \max\{B, 1\} t^2 = c^{-1} t^2$ when $B \leq 1$, so we get

$$t^2 = \frac{c B^{\frac{4k}{2k+1}} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

Overall, we have

$$t^2 = \frac{c \max\left\{B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}.$$

and

$$R_{\mathcal{J}_{e,l}}\left(F_k(1) \cap B_2\left(c^{-\frac{1}{2}} \cdot b(L) \cdot t\right)\right) \leq t^2 = \frac{c^2 \max\left\{L^{\frac{4k}{2k+1}}, L^{\frac{2k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}. \quad (173)$$

Similarly, we have

$$R_{\mathcal{J}_{o,l}} \left(F_k(1) \cap B_2 \left(c^{-\frac{1}{2}} \cdot b(L) \cdot t \right) \right) \leq \frac{c^2 \max \left\{ L^{\frac{4k}{2k+1}}, L^{\frac{2k-1}{2k+1}} \right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}.$$

Then by Theorem [source ref.] and [source equation], and the choice of $L = \left\lceil \frac{1}{C_\beta} 2 \log(n) \right\rceil$ in [source equation], we have

$$\Delta^2(\hat{f} - f^*) = O_p(r_n^2),$$

by setting that

$$r_n^2 \asymp \frac{\max \left\{ B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}} \right\} \log(n)^{\frac{6k+1}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{s \log^{1+\mu}(n) \log(T_n)}{n},$$

for some $\mu > 0$. □

Lemma 70. Let $\{x_{ij}\}_{i=1,j=1}^{n,m_j}$ be a collection of random variables in $[0, 1]$. Denote

$$\|f\|_{mn}^2 = \frac{1}{n} \sum_{i=1}^n \frac{1}{m_i} \sum_{j=1}^{m_i} f^2(x_{ij}).$$

Then, there exists an absolute constant A such that for $k \geq 1$,

$$\log N(\varepsilon, \|\cdot\|_{mn}, F_k(1)) \leq AB^{1/k} \varepsilon^{-1/k}.$$

Proof. This follows directly from Sadhanala et al. (2016), see page 3050 of that reference. □

J.2.3 Proof of Theorem [source ref.]

Proof. By Lemma [source ref.], we have

$$\Delta^2(\hat{f} - f^*) = O_p \left(\frac{\max \left\{ B^{\frac{6k+1}{2k+1}}, B^{\frac{2k-1}{2k+1}} \right\} \log(n)^{\frac{6k+1}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right),$$

So we get

$$\|\hat{f} - f^*\|_2^2 = O_p \left(\frac{\max \left\{ B^{\frac{6k+1}{2k+1}}, B^{\frac{2k-1}{2k+1}} \right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{\max \{B, 1\} \cdot s \cdot \text{poly}(\log n) \log(T_n)}{n} \right).$$

J.3 High-dimensional Quantile Linear Regression

Implemented correction. For a block I , let X_I^w have row $x_{ij}^\top/m_i$. Then

$$\mathcal{R}_I \leq \frac{2V\sqrt{2\log(2d)}}{n} \mathbb{E} \max_{q \in [d]} \|X_{I,q}^w\|_2.$$

The even and odd blocks use their own matrices. The resulting theorem is a prediction bound; parameter recovery requires an additional design-identifiability condition.

J.3.1 Proof sketch of Theorem [source ref.]

In this case, the constraint set is

$$H(V) = \{f : f(x_{ij}) = x'_{ij}\beta, \|\beta\|_1 \leq V, \beta \in \mathbb{R}^d\}$$

Since this is a compact set, **we apply** Theorem [source ref.] to get an expectation bound by letting $\mathcal{F} \equiv H(V)$. **It therefore remains to bound $R(H(V))$ using the maximal inequality stated below.**

J.3.2 Proof of Theorem [source ref.]

Proof. By Corollary [source ref.],

It suffices to get $R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$ and $R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$ for $l = 1, \dots, n$. Then we have

$$\begin{aligned} R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right) &= \mathbb{E}_{x^*,s} \left[\sup_{\substack{f \in H(V) \\ f \in B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right] \\ &\leq \mathbb{E}_{x^*,s} \left[\sup_{f \in H(V)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right] \\ &= R_{\mathcal{J}_{e,l}}(H(V)) \end{aligned}$$

Let

$$K_V := \{\beta \in \mathbb{R}^d : \|\beta\|_1 \leq V\}.$$

Also, recall $X_{e,l} \in \mathbb{R}^{\sum_{i \in \mathcal{J}_{e,l}} m_i \times d}$ is the matrix with its rows are $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$. Then we have

$$\begin{aligned} \mathbb{E}_{x^*,s} \left[\sup_{f \in H(V)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right] &\leq \mathbb{E} \left[\frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sup_{\beta \in K_1} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} x_{ij}^{*'} \beta \right] \\ &= \frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \mathbb{E} \left[\sup_{v \in XK_1} s'v \right]. \end{aligned} \quad (174)$$

where

$$K_1 = \{v : \|v\|_1 \leq 1\}.$$

By the proof of Theorem 2.4 in Rigollet et al. (2023), there exists a constant $h_2 > 0$ such that for a given fixed design matrix $X_{e,l} \in \mathbb{R}^{\sum_{i \in \mathcal{J}_{e,l}} m_i \times d}$.

$$\mathbb{E} \left[\sup_{v \in XK_1} s'v \right] \leq h_2 (\log d)^{1/2} \max_{j \in [d]} \|X_{e,l,j}\|.$$

Taking expectations on both sides, thus we have

$$R_{\mathcal{J}_{e,l}}(H(V)) \leq \frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \cdot h_2 (\log d)^{1/2} \mathbb{E} \left[\max_{j \in [d]} \|X_{e,l,j}\| \right]$$

$$\begin{aligned}
 &\leq \frac{Lh_2V(\log d)^{1/2} \mathbb{E} \left[\max_{j \in [d]} \|X_{e,l,j}\| \right]}{cc_1nm} \\
 &= \frac{Lh_3V(\log d)^{1/2} \mathbb{E} \left[\max_{j \in [d]} \|X_{e,l,j}\| \right]}{nm}.
 \end{aligned}$$

Similarly, we have

$$\begin{aligned}
 R_{\mathcal{J}_{o,l}}(H(V)) &\leq \frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \cdot h_2(\log d)^{1/2} \mathbb{E} \left[\max_{j \in [d]} \|X_{o,l,j}\| \right] \\
 &\leq \frac{Lh_3V(\log d)^{1/2} \mathbb{E} \left[\max_{j \in [d]} \|X_{o,l,j}\| \right]}{nm}.
 \end{aligned}$$

Thus, we get

$$\begin{aligned}
 c_5 \sum_{l=1}^L (R_{\mathcal{J}_{e,l}}(H(V)) + R_{\mathcal{J}_{o,l}}(H(V))) &+ \frac{c_4 s \log n \log T_n}{n} \\
 &\leq \frac{Lc_5 h_3 V(\log d)^{1/2} \mathbb{E} \left[\sum_{l=1}^L \left(\max_{j \in [d]} \|X_{e,l,j}\| + \max_{j \in [d]} \|X_{o,l,j}\| \right) \right]}{nm} + \frac{c_4 s \log n \log T_n}{n} \\
 &= \frac{h_4 \log n V(\log d)^{1/2} \mathbb{E} \left[\sum_{l=1}^L \left(\max_{j \in [d]} \|X_{e,l,j}\| + \max_{j \in [d]} \|X_{o,l,j}\| \right) \right]}{nm} + \frac{c_4 s \log n \log T_n}{n}
 \end{aligned}$$

The claim of the theorem then follows and we get

$$\Delta^2(\hat{f} - f^*) = O_p \left(\frac{\log n V(\log d)^{1/2} A}{nm} + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right), \text{ and}$$

$$\|\hat{f} - f^*\|_2^2 = O_p \left(\frac{\max\{B, 1\} \log n V(\log d)^{1/2} A}{nm} + \frac{\max\{B, 1\} s \cdot \text{poly}(\log n) \log(T_n)}{n} \right).$$

□

J.4 Quantile K-NN fused lasso

Implemented correction. The graph result now uses the symmetric union K -NN graph and the explicit parameter $\rho(G) = \max_e \|(D_G^\dagger)_e\|_2$. The proved complexity inequality is

$$\sum_I \mathcal{R}_I\{u : \|u\|_{n,m} \leq r, \|D_G u\|_1 \leq 2V_n\} \lesssim r\sqrt{L/N} + \frac{LV_n \rho(G) \sqrt{\log(2|E|)}}{N}.$$

Together with $V_n \lesssim N^{1-1/d} K^{1+1/d}$ and the stated radius, overlap, and boundary-tube events, this gives the in-sample and off-sample bounds in Theorem [source ref.]; no unresolved equation reference remains.

J.4.1 Proof of Theorem [source ref.]

Proof. We will apply the result from Theorem [source ref.], and also incorporate our problem in [source ref.], to get

$$\begin{aligned} R_{\mathcal{J}_{e,l}} \left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{ \theta : \|\theta - \theta^*\| \leq \max \left\{ 1, B^{1/2} \right\} t \right\} \right), \\ R_{\mathcal{J}_{o,l}} \left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{ \theta : \|\theta - \theta^*\| \leq \max \left\{ 1, B^{1/2} \right\} t \right\} \right) \end{aligned}$$

for all $l = 1, \dots, L$, based on the results from Lemma [source ref.].

Then we can show that Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$. The estimator $\hat{\theta}_{V_n}$, defined in [source equation] satisfies that

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) = O_p \left(\frac{\max \{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max \{B, 1\}}{n} + \frac{d \log^{1+\mu}(n) \log(T_n)}{n} \right) \quad (17)$$

$$\begin{aligned} \left\| \hat{\theta}_{V_n} - \theta^* \right\|^2 = O_p \left(\frac{\max \{B^2, B\} \log^{5+\mu}(nm) V_n}{nm} \right. \\ \left. + \frac{\log^2(nm) \max \{B^2, B\}}{n} + \frac{\max \{B, 1\} d \log^{1+\mu}(n) \log(T_n)}{n} \right), \end{aligned}$$

and provided that $V_n \geq \|V_{G^*}\|$.

In addition, based on the result from Lemma [source ref.], $\hat{f}_{V_n}$, defined in [source equation]–[source equation] satisfies that

$$\begin{aligned} \Delta^2(\hat{f}_{V_n} - f^*) = O_p \left(\frac{\max \{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max \{B, 1\}}{n} \right. \\ \left. + \frac{d \log^{1+\mu}(n) \log(T_n)}{n} + \frac{\log^{2+\mu}(nm)}{(nm)^{1/d}} \right) \end{aligned}$$

$$\begin{aligned} \left\| \hat{f}_{V_n} - f^* \right\|_2^2 = O_p \left(\max \{B, B^2\} \left(\frac{\log^{5+2\mu}(nm)}{nm} + \frac{\log^{3+\mu}(nm) V_n}{n} \right) \right. \\ \left. + \max \{B, 1\} \cdot \left(\frac{d \log n}{n} + \frac{\log^{2+\mu/d}(nm)}{(nm)^{1/d}} \right) \right). \end{aligned}$$

□

J.4.2 Additional Definitions for proof of Theorem [source ref.]

We give some definitions that will be encountered in the proof of the lemma.

The lattice graph We introduce the Lattice graph, for more details, we refer readers to Madrid Padilla et al. (2020). Given $N \in \mathbb{N}$, denote the equally spaced lattice graph in $[0, 1]^d$ as $G_{\text{lat}} = (V_{\text{lat}}, E_{\text{lat}})$, where the distance between any two neighbor lattice points is $1/N$.

Therefore, the number of nodes $|V_{\text{lat}}| = N^d$. Without loss of generality, we assume that the nodes of the grid correspond to the points

$$I = P_{\text{lat}}(N) = \left\{ \left(\frac{i_1}{N}, \frac{i_2}{N}, \dots, \frac{i_d}{N} \right) : i_1, \dots, i_d \in \{1, \dots, N\} \right\} \quad (271) \quad (175)$$

Let $\{c_r\}_{r=1}^{N^d}$ be the corresponding centers and define a collection of cells, $\{C(c_r)\}_{r=1}^{N^d}$ as

$$\mathcal{I}_r = C(c_r) = \left\{ z \in [0, 1]^d : c_r = \arg \min_{r' \in P_{\text{lat}}(N)} \|z - r'\|_\infty \right\}. \quad (176)$$

Now, let $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be a collection of signals observed on the sample points $\{x_{ij}\}_{i=1, j=1}^{n, m}$.

Let $\theta_I \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be the modified version of θ according to the grid induced by the lattice graph of size N . Let $\theta^I \in \mathbb{R}^{N^d}$ be the modified version of θ that takes values on the centers of the lattice graph. See section E of Madrid Padilla et al. (2020) for definitions of θ_I and θ^I . For completion, the definitions are included as follows.

Definition 2. Given $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$, let $\{\mathcal{I}_r\}_{r=1}^{N^d}$ be the collection of the cells in the lattice graph G_{lat} and let $\{c_r\}_{r=1}^{N^d}$ be the corresponding centers. Let $\theta_I \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be such that for any $1 \leq \alpha \leq n, 1 \leq \beta \leq m_\alpha$,

$$(\theta_I)_{\alpha, \beta} = \theta_{ij},$$

if $x_{\alpha, \beta}$ and x_{ij} are in the same cell $\mathcal{I}_r$, and that $(i, j) = \arg \min_{1 \leq i' \leq n, 1 \leq j' \leq m} \|x_{i'j'} - c_r\|$. Denote by

$$I_r = \{(i, j) \in [n] \times [m] : P_I(x_{ij}) = c_r\} \quad (177)$$

for $r = 1, \dots, N^d$, where $P_I(x)$ denotes the point in $P_{\text{lat}}(N)$ such that $x \in C(P_I(x))$.

Definition 3. Let $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be a collection of signals observed on the sample points $\{x_{ij}\}_{i=1, j=1}^{n, m_i}$. Let θ^I in $\mathbb{R}^{N^d}$ be such that, for any $r \in [N^d]$

$$\theta_r^I = \begin{cases} \theta_{i_r j_r}, & \text{if } I_r \neq \emptyset \text{ and } (i_r j_r) = \arg \min_{1 \leq i' \leq n, 1 \leq j' \leq m_{i'}} \|x_{i'j'} - c_r\|; \\ 0, & \text{if } I_r = \emptyset. \end{cases}$$

This means that θ_r^I take the value $\theta_{i_r j_r}$, if $x_{i_r j_r}$ is the closest point to the center of c_r among all $\{x_{ij}\}_{i=1, j=1}^{n, m_i}$.

Piecewise Lipschitz functions

Definition 71. For a set $A \subset [0, 1]^d$, we write $B_\varepsilon(A) = \{a : \exists a' \in A, \text{ with } \|a - a'\|_2 \leq \varepsilon\}$. We let ∂A denote the boundary of the set A .

Let $\Omega_\varepsilon := [0, 1]^d \setminus B_\varepsilon(\partial[0, 1]^d)$. We say that a bounded function $f : [0, 1]^d \rightarrow \mathbb{R}$ is piecewise Lipschitz if there exists a set $S \subseteq (0, 1)^d$ that has the following properties:

- The set S has Lebesgue measure zero.
- For some constants $C_S, \varepsilon_0 > 0$, we have that $\text{Vol}(B_\varepsilon(S) \cup ([0, 1]^d \setminus \Omega_\varepsilon)) \leq C_S \varepsilon$ for all $0 < \varepsilon < \varepsilon_0$.
- There exists a positive constant L_f such that if x and x' belong to the same connected component of $\Omega_\varepsilon \setminus B_\varepsilon(S)$, then $|g(x) - g(x')| \leq L_f \|x - x'\|_2$.

J.4.3 Technical Results for proof of Theorem [source ref.]

The remaining proof steps leading to

$$\begin{aligned} \Delta^2(\hat{\theta}_{V_n} - \theta^*) &= O_p \left(\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\|\nabla G_K \theta_{e,l}\|_1 \leq V_n \cap \left\{ \theta : \|\theta_{e,l} - \theta_{e,l}^*\| \leq c^{-\frac{1}{2}} \cdot b(B) \cdot t \right\} \right) \right. \right. \\ &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\|\nabla G_K \theta_{o,l}\|_1 \leq V_n \cap \left\{ \theta : \|\theta_{o,l} - \theta_{o,l}^*\| \leq c^{-\frac{1}{2}} \cdot b(B) \cdot t \right\} \right) \right. \right. \\ &\quad \left. \left. + \frac{s \log n \log T_n}{n} \right) \right), \text{ and} \end{aligned}$$

follows the similar proofs in Corollary [source ref.] to Lemma [source ref.].

Lemma 72. *The estimator $\hat{\theta}_{V_n}$, defined in [source equation], Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$ satisfies that*

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) = O_p(r_n^2),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B, 1\}}{n} + \frac{d \log^{1+\mu}(n) \log(T_n)}{n}.$$

Proof. By definition, we have

$$R_{\mathcal{J}_{e,l}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) = \mathbb{E}_{x^*,s} \left[\sup_{\substack{\|\theta - \theta^*\| \leq b \\ \|\nabla G_K \theta\|_1 \leq V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \theta_{ij}^* \right].$$

Define:

$$(\tilde{s})_{i,j} = \begin{cases} (s)_{i,j}, & \text{for } i \in \mathcal{J}_{e,l}, \text{ and } j \in m_i, \\ 0, & \text{for } i \notin \mathcal{J}_{e,l}, \text{ and } j \notin m_i. \end{cases} \quad (178)$$

One can show that $\{\tilde{s}_{ij}\}_{i=1, j=1}^{n, m_i}$ are independent SubGaussian random variables with SubGaussian parameter 1.

Now by triangle inequality, and let $\Lambda = \theta - \theta^*$, we have

$$\begin{aligned} &R_{\mathcal{J}_{e,1}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) \\ &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{\|\theta - \theta^*\| \leq b \\ \|\nabla G_K \theta\|_1 \leq V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,1}} m_i} \sum_{i \in \mathcal{J}_{e,1}} \sum_{j=1}^{m_i} s_{ij} (\theta_{ij} - \theta_{ij}^* + \theta_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{\|\theta - \theta^*\| \leq b \\ \|\nabla G_K \theta\|_1 \leq V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,1}} m_i} \sum_{i \in \mathcal{J}_{e,1}} \sum_{j=1}^{m_i} s_{ij} (\theta_{ij} - \theta_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\quad + \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\frac{1}{\sum_{i \in \mathcal{J}_{e,1}} m_i} \sum_{i \in \mathcal{J}_{e,1}} \sum_{j=1}^{m_i} s_{ij} \theta_{ij}^* \right] \end{aligned}$$

$$\begin{aligned}
 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|\theta-\theta^*\| \leq b, \|\nabla G_K(\theta-\theta^*)\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n \\ \|\nabla G_K\theta\|_1 \leq V_n, \|\nabla G_K\theta^*\|_1 \leq V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij}(\theta_{ij} - \theta_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \frac{\sum_{i=1}^n m_i}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\asymp \frac{nm}{|\mathcal{J}_{e,l}|m} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \frac{nL}{c_1 n} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \frac{2 \log(n)}{c_1 C_\beta} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],
 \end{aligned}$$

Where the first equality holds by the fact that $\|\nabla G_K\theta\|_1 \leq V_n$ and $\|\nabla G_K\theta^*\|_1 \leq V_n$ implies $\|\nabla G_K A\|_1 \leq 2V_n$. We use the assumption $m_i \asymp m$ specified in the statement of the Lemma.

In this step the term on the right hand side, we get

$$\begin{aligned}
 &R_{\mathcal{J}_{e,l}}(\|\nabla G_K\theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) \\
 &\leq \frac{2 \log(n)}{c_1 C_\beta} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],
 \end{aligned}$$

For the rest of the proofs we give the sketch, the exact proof is provided in Lemma [source ref.].

From Lemma [source ref.] and the observation after Equation [source equation], with a probability of at least

$$1 - \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right),$$

it is satisfied that

$$\sup_{\|A\|_2 \leq b, \|\nabla G_K A\|_1 \leq 2V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n,$$

where $\tilde{b}_2 \geq 0$ is a constant depending on d , C_3 and v_{max} . Moreover, $\tilde{C}$ is $\frac{c_1 c}{c_2 C}$ with c_1 and c_2 positive constants. Here v_{max} , v_{min} , C and c are from Assumptions [source ref.] and [source ref.]. The positive constants C_3 and C_4 only depend on d . To find the quantity $R_{\mathcal{J}_{e,1}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\})$, we will analyze

$$\frac{R_{\mathcal{J}_{e,1}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\})}{t^2} + \frac{R_{\mathcal{J}_{e,1}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\})}{t^2}.$$

See the equation (13.17) in Wainwright (2019). Hence,

$$\begin{aligned} & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\|A\| \leq b, \|\nabla G_K A\|_1 \leq V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} \tilde{s}_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\ & \quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\ & \quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right). \end{aligned}$$

Also with Lemma [source ref.], we have $b^2 = c^{-1} \max\{B, 1\} \cdot t^2$,

Let denote $\frac{1}{\sum_{i=1}^n m_i}$ as C_6 . Now, we analyze

$$\frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\})}{t^2} \quad (179)$$

We assume $b^2 = O\left(\frac{1}{n} + \frac{1}{nm}\right)$, then we have

$$\begin{aligned} & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K \theta\|_1 + \|\nabla G_K \theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\ & \quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\ & \quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right) \\ & \stackrel{(a)}{\leq} \frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n \\ & \quad + B \cdot \left(\frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\ & \quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) \right) \\ & \stackrel{(b)}{\leq} \frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n \end{aligned}$$

$$+ BC_7 \log(n) nm \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) + \frac{BC_8}{n}.$$

The [source equation] holds since we assume $b^2 = O\left(\frac{1}{n} + \frac{1}{nm}\right)$. The [source equation] holds since we assume $K = \text{poly}(\log(nm))$.

Case 1, if the $B \geq 1$ so $b = c^{-1/2} B^{1/2} \cdot t$, Then we let

$$K = 2 \log(n) \log(nm) \leq \log(nm)^{2+\mu}$$

for some $\mu > 0$, we have

$$\begin{aligned} & \frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n + BC_7 \log(n) nm \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) + \frac{BC_8}{n} \\ & \leq \frac{1}{\sum_{i=1}^n m_i} C_9 B K \log(nm) V_n + \frac{BC_8}{n}. \end{aligned}$$

So we have

$$\begin{aligned} & R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) \\ & \leq \frac{1}{\sum_{i=1}^n m_i} C_9 B K \log(nm) V_n + \frac{BC_8}{n} \\ & \leq \frac{1}{\sum_{i=1}^n m_i} C_9 B \log(nm)^{3+l} V_n + \frac{BC_8}{n}. \end{aligned}$$

Thus, we seek a

$$t^2 = \frac{C_9 B \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) BC_8}{n},$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\})}{t^2} = O\left(\frac{1}{\log(nm)}\right).$$

Case 2, if the $B < 1$ so $b = c^{-1/2} \cdot t$,

So we have

$$\begin{aligned} & R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) \\ & \leq \frac{1}{\sum_{i=1}^n m_i} C_9 B K \log(nm) V_n + \frac{BC_8}{n} \\ & \leq \frac{1}{\sum_{i=1}^n m_i} C_{10} \log(nm)^{3+l} V_n + \frac{C_8}{n}. \end{aligned}$$

Thus, we seek a

$$t^2 = \frac{C_{10} \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) C_8}{n},$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c\Delta^2(f - f^*) \leq t^2\})}{t^2} = O\left(\frac{1}{\log(nm)}\right).$$

Overall, we get

$$t^2 = \frac{C_{11} \max\{B, 1\} \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) \max\{B, 1\} C_8}{n}.$$

Furthermore, by Assumption [source ref.], $\frac{1}{\sum_{i=1}^n m_i} \leq \frac{1}{cmn}$, and Equation [source equation], the result The above results hold uniformly for $R_{\mathcal{J}_{e,l}}$ and $R_{\mathcal{J}_{o,l}}$ for all $l = 1, \dots, L$, then we can show that

$$\begin{aligned} & \frac{1}{t^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{ \theta : \|\theta - \theta^*\| \leq \max\{1, B^{1/2}\} t \right\} \right) \right. \\ & \left. + R_{\mathcal{J}_{o,l}} \left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{ \theta : \|\theta - \theta^*\| \leq \max\{1, B^{1/2}\} t \right\} \right) \right) = O\left(\frac{1}{\log(nm)}\right), \end{aligned}$$

by letting

$$t^2 = \frac{C_{12} \max\{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B, 1\} C_{13}}{n}.$$

Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$.

The estimator $\hat{\theta}_{V_n}$, defined in [source equation] satisfies that

Then by Theorem [source ref.], The estimator $\hat{\theta}_{V_n}$, defined in [source equation] satisfies that

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) = O_p(r_n^2),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B, 1\}}{n} + \frac{d \log^{1+\mu}(n) \log(T_n)}{n}.$$

□

Lemma 73. Let Ω_β be defined in [source equation]. Let $\{s_{ij}\}_{i=1, j=1}^{n, m_i} \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, and use the notation $\Lambda = \theta_1 - \theta_2$, for $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$. For some positive constants C_3 and C_4 we have that

$$\begin{aligned} & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\|A\| \leq b, \|\nabla G_K A\|_1 \leq 2V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\ & \quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)}\right) \right. \\ & \quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)}\right) + \frac{1}{nm} \right). \end{aligned}$$

Here $C_3 \geq 0$ is constant depending on d , $v_{\min}$ and $v_{\max}$, and C_5 is constant depending on d , $v_{\min}$ and $v_{\max}$ and L . Moreover, $\tilde{b}_2 > 0$ is a constant depending on d , $v_{\min}$, and $v_{\max}$. Furthermore, $\tilde{C} = \frac{C_{\beta cc1}}{c_2 C}$ with c_1 and c_2 positive constants, and $\tilde{C}$ is a constant in $(0, 1)$. Here $v_{\max}$, $v_{\min}$, C and c' are from Assumptions [source ref.] and [source ref.].

Proof. Let I_r be given in [source equation], and let N in the construction of $P_{\text{lat}}(N)$ of [source equation] is chosen as

$$N = \frac{\left\lceil 3\sqrt{d(nm)^{1/d}} \right\rceil}{aK^{1/d}}, \quad \text{where} \quad a = \frac{1}{(4C_{2,d}c_2v_{\max})^{1/d}},$$

for a positive constants c_2 , and, $v_{\max}$, and C are from Assumptions [source ref.] and [source ref.]. Moreover, $c_{2,d}$ is a positive constants that depends on d . Then there exist positive constants $\bar{b}_1$ and $\bar{b}_2$ depending on d , $v_{\min}$, and $v_{\max}$ with $v_{\min}$ from Assumptions [source ref.] and [source ref.]. Let set $C(\cdot)$ be defined in [source equation]. We start by introducing the notation $x'_{ij} = P_I(x_{ij})$. Then define the event Ω as

$$\Omega : \quad \text{“If } x_{ij} \in C(x'_{ij}) \text{ and } x_{st} \in C(x'_{rs}) \text{ for } x'_{ij}, x'_{st} \in P_{\text{lat}}(N) \text{ with } \|x'_{ij}, x'_{st}\|_2 \leq N^{-1}, \text{ then } x_{ij} \text{ and } x_{st} \text{ are connected in the } K\text{-NN graph.”} \quad (180)$$

Also, define the following events,

$$E_1 = \left\{ \min_{r=1,\dots,N^d} |I_r| \geq \frac{1}{2}b_2K \right\}, \quad (310)$$

and

$$E_2 = \left\{ \max_{r=1,\dots,N^d} |I_r| \leq \frac{3}{2}b_2K \right\}. \quad (311)$$

hold. Observe that by Lemma [source ref.] event Ω hold with probability at least,

$$1 - \frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) - \frac{1}{n},$$

and events $E_1 \cap E_2$ hold with probability at least

$$1 - \frac{4}{C_\beta} C \log(n) nm^d \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) - \frac{1}{n},$$

respectively, where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$ and $\tilde{C} \in (0, 1)$, with c_1 and c_2 are some constant. Next, by (2.67) in Wainwright (2019) we have that the event

$$E_3 = \left\{ \max_{1 \leq i \leq n, 1 \leq j \leq m_i} |s_{ij}| \leq \sqrt{2 \log \left(\sum_{i=1}^n m_i \right)} \right\}$$

happens with probability at least $1 - \frac{1}{\sum_{i=1}^n m_i}$.

Then let $\tilde{\Pi}$ be the orthogonal projection onto the span of $1_{N^d} \in \mathbb{R}^{N^d}$, and $D^\dagger$ is the pseudoinverse of the incidence matrix D from Lemma [source ref.].

Further, using the proof of Theorem 2 in Hütter et al. (2016) (see page 21 in that paper), if $\tilde{s} \in \mathbb{R}^{N^d}$ is a mean zero vector whose coordinates are Rademacher variables or zero values as introduced in [source equation]. it yields that the following two inequalities hold simultaneously on an event E_4 of probability at least $1 - \frac{1}{nm}$,

$$E_4 = \left\{ \left\| (D^\dagger)^\top \tilde{s} \right\|_\infty \leq \rho \sqrt{2 \log(en^d nm)}, \quad \left\| \tilde{\Pi} \tilde{s} \right\|_2 \leq 2 \sqrt{2 \log(enm)} \right\},$$

where $\rho = \max_{1 \leq v \leq |E_{\text{lat}}|} \|D_{\cdot, v}^\dagger\|_2$. Additionally, using propositions 4 and 6 in Hütter et al. (2016), we have that $\rho \leq \widetilde{C(d)}\sqrt{\log(N)}$, where $\widetilde{C(d)}$ is a positive constant depending on d . Hence,

$$\|(D^\dagger)^\top \tilde{s}\|_\infty \leq C(d)\sqrt{\log\left(\frac{nm}{K}\right)\log\left(\frac{nm}{K}nm\right)} \quad \text{and} \quad \|\tilde{\Pi}\tilde{s}\|_2 \leq 2\sqrt{2\log(enm)},$$

with probability at least $1 - 2 \cdot \frac{1}{nm}$. Here, $C(d) > 0$ is a constant depending on d , C , and $v_{\max}$.

Then denote $\mathcal{E} \equiv \Omega \cap E_1 \cap E_2 \cap E_3 \cap E_4$, on the event $\mathcal{E}$, the following is satisfied.

Let $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$ such that $\|A\|_2 \leq b$ and $\|\nabla G_K A\|_1 \leq 2V_n$. And $\{s_{ij}\}_{i=1, j=1}^{n, m_i} = s \in \mathbb{R}^{\sum_{i=1}^n m_i}$ are independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. First, by Lemma [source ref.] and E_3 ,

$$s^\top(\theta_1 - \theta_2) \leq 4\sqrt{2\log(nm)}V_n + s^\top(\theta_{1I} - \theta_{2I}).$$

Then, by Lemma [source ref.], E_2 , and E_4 , we have that

$$s^\top(\theta_{1I} - \theta_{2I}) \leq \max_{r=1, \dots, N^d} |I_r| \left(\|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1^* - \theta_2^*\|_2 + \|(D^\dagger)^\top\|_\infty (\|\nabla_{G_K} \theta_{1I}\|_1 + \|\nabla_{G_K} \theta_{2I}\|_1) \right), \quad (181)$$

Thus, we have

$$\begin{aligned} \frac{1}{\sum_{i=1}^n m_i} s^\top A &\leq \frac{1}{\sum_{i=1}^n m_i} 4\sqrt{2\log(nm)}V_n \\ &\quad + \max_{r=1, \dots, N^d} |I_r| \frac{1}{\sum_{i=1}^n m_i} \left(\|\tilde{\Pi}\tilde{s}\|_2 b + \|(D^\dagger)^\top \tilde{s}\|_\infty \cdot 4V_n \right) \\ &\leq 4 \frac{\sqrt{2\log(\sum_{i=1}^n m_i)}}{\sum_{i=1}^n m_i} V_n + \frac{3}{2} \tilde{b}_1 K \left(2b \frac{\sqrt{2\log(enm)}}{\sum_{i=1}^n m_i} + C(d) \frac{\log\left(\frac{(nm)^2}{K}\right)}{\sum_{i=1}^n m_i} (4V_n) \right) \\ &\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n, \end{aligned}$$

for some positive constants C_3 and C_4 . Here C_4 depends on d .

Finally, on the event $\mathcal{E}$ the following is satisfied. Thus,

$$\sup_{\|A\|_2 \leq b, \|\nabla G_K A\|_1 \leq 2V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} A_{ij} s_{ij} \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n. \quad (182)$$

Under the event $\mathcal{E}^c$, we have that $R_{\mathcal{J}_{elmn}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

$$\begin{aligned} &\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{\|A\| \leq b \\ \|\nabla G_K A\|_1 \leq \|\nabla G_K \theta\|_1 + \|\nabla G_K \theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} A_{ij} s_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} [R_{\mathcal{J}_{elmn}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\})] \\ &\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \end{aligned}$$

$$\begin{aligned}
 & + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) \right. \\
 & \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp \left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right)
 \end{aligned}$$

□

The following lemma gives the error rate for $\hat{f}_{V_n}$ defined in [source equation].

Lemma 74. *In addition, $\hat{f}_{V_n}$, defined in [source equation] satisfies that*

$$\Delta^2(\hat{f}_{V_n} - f^*) = O_p(\tilde{r}_{nm})$$

for

$$\tilde{r}_{nm} = \frac{\max\{B, 1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B, 1\}}{n} + \frac{d \log^{1+\mu}(n) \log(T_n)}{n} + \frac{\log^{2+\mu}(nm)}{(nm)^{1/d}}.$$

Proof. Through this step, notation from Appendix section [source ref.] is used. We start noticing that,

$$\|f^* - \hat{f}_{V_n}\|_2^2 = \int \left(f^*(x) - \hat{f}_{V_n}(x) \right)^2 v(x) dx.$$

Then, adding and subtracting $\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij})$, it can be written as

$$\begin{aligned}
 & \int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) + \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right. \\
 & \quad \left. - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx \\
 & \leq 2 \int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx \\
 & \quad + 2 \int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx.
 \end{aligned}$$

Now we proceed to bound each of the terms

$$\text{Int}_1 = \int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx, \quad (183)$$

and,

$$\text{Int}_2 = \int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx. \quad (184)$$

For the second term Int_2 in Equation [source equation], we observe that

$$\begin{aligned} & \int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx \\ &= \sum_{r=1}^{N^d} \int_{\mathcal{I}_r} \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx. \end{aligned}$$

Moreover, for $x' \in [0, 1]^d$, there exists $c_r(x') \in P_{\text{lat}}(N)$ that satisfies $x' \in C(c_r(x'))$ and

$$\|x' - P_I(x')\|_2 \leq \frac{1}{2N}.$$

Even more, by Lemma [source ref.], with high probability, there exists $(i(x'), j(x')) \in [n] \times [m_i]$ such that $x_{i(x')j(x')} \in C(c_r(x'))$. Hence,

$$\|x' - x_{i(x')j(x')}\|_2 \leq \frac{1}{N}.$$

Denote by $x_{i_1 j_1}, \dots, x_{i_K j_K}$ $[i_1, \dots, i_K] \in [n]$ and $(j_1, \dots, j_K) \in [m_{i_1}] \times \dots \times [m_{i_K}]$ K -nearest neighbors corresponding to $x_{i'(x')j(x')}$. Observe that

$$\|x' - x_{i_r j_r}\|_2 \leq \frac{1}{N} + R_{K, \max}, \quad r = 1, \dots, K$$

where

$$R_{K, \max} = \max_{1 \leq i \leq n, 1 \leq j \leq m_i} R_K(x_{ij}),$$

with $R_K(x)$ denoting the distance from $x \in [0, 1]^d$ to its K th nearest neighbor in the set $\{x_{11}, \dots, x_{nm_n}\}$. Thus, with high probability, for any $x' \in [0, 1]^d$,

$$N_K(x') \subset \{x_{ij} : \|x' - x_{ij}\|_2 \leq \frac{1}{N} + R_{K, \max}\} \subset \{x_{ij} : \|c_r(x') - x_{ij}\|_2 \leq \left(\frac{1}{2} + \frac{1}{N}\right) + R_{K, \max}\}.$$

Hence, we set

$$\tilde{N}(c_r(x')) = \{x_{ij} : \|c_r(x') - x_{ij}\|_2 \leq \left(\frac{1}{2} + \frac{1}{N}\right) + R_{K, \max}\}.$$

As a result, for $r \in [N]^d$,

$$\int_{C(c_r)} \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx,$$

can be bounded as,

$$\begin{aligned} & \int_{C(c_r)} \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx \\ & \leq \frac{1}{K} \int_{C(c_r)} \left(\sum_{x_{ij} \in \tilde{N}(c_r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \right) v(x) dx \end{aligned}$$

$$= \frac{1}{K} \sum_{x_{ij} \in \tilde{N}(c_r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \int_{C(c_r)} v(x) dx.$$

The above leads to an upper bound for the term

$$\int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx.$$

This upper bound has the form,

$$\sum_{r=1}^{N^d} \left(\frac{1}{K} \sum_{x_{ij} \in \tilde{N}(c_r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \right) \int_{C(c_r)} v(x) dx,$$

which can be bounded by

$$\begin{aligned} & \left[\sum_{r=1}^{N^d} \frac{1}{K} \sum_{x_{ij} \in N(r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \right] \max_{1 \leq r \leq N^d} \int_{C(c_r)} v(x) dx \\ & \leq \sum_{i=1}^n \sum_{j=1}^{m_i} \frac{1}{K} \sum_{\substack{r \in [N^d]: \\ \|x_{ij} - c_r\|_2 \leq (\frac{1}{2} + 1)N + R_{k,\max}}} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \max_{1 \leq r \leq N^d} \int_{C(c_r)} v(x) dx \\ & \leq \left[\sum_{i=1}^n \sum_{j=1}^{m_i} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \right] \left[\frac{1}{K} \max_{1 \leq r \leq N^d} \int_{C(c_r)} v(x) dx \right] \\ & \quad \times \max_{(i,j) \in [n] \times [m_i]} \left\{ k : \|x_{ij} - c_r\|_2 \leq \left(\frac{1}{2} + 1 \right) \frac{1}{N} + R_{k,\max} \right\}. \end{aligned}$$

Next, for a set $A \subset \mathbb{R}^d$ and positive constant $\bar{r}$, we define the packing and external covering numbers as

$$N_{\bar{r}}^{\text{pack}}(A) := \max \{ r \in \mathbb{N} : \exists q_1, \dots, q_r \in A, \text{ such that } \|q_i - q_j\|_2 > \bar{r} \forall i \neq j \},$$

$$N_{\bar{r}}^{\text{ext}}(A) := \min \left\{ r \in \mathbb{N} : \exists q_1, \dots, q_r \in \mathbb{R}^d, \text{ such that } \forall x \in A \text{ there exists } r_x \text{ with } \|x - q_{r_x}\|_2 < \bar{r}_x \right\}.$$

Furthermore, by Lemma [source ref.], there exists a constant $\tilde{c}$ such that $R_{K,\max} \leq \frac{\tilde{c}}{N}$ with high probability. This implies that with high probability, for any $x' \in [0, 1]^d$,

$$\begin{aligned} & \max_{(i,j) \in [n] \times [m_i]} \left| \left\{ r \in [N^d] : \|x_{ij} - c_r\|_2 \leq \left(\frac{1}{2} + \frac{1}{N} \right) + R_{K,\max} \right\} \right| \\ & \leq \max_{(i,j) \in [n] \times [m_i]} \left| \left\{ r \in [N^d] : \|x_{ij} - c_r\|_2 \leq \frac{\tilde{C}}{N} \right\} \right| \leq \max_{(i,j) \in [n] \times [m_i]} N_{\frac{1}{N}}^{\text{pack}} \left(B_{\frac{\tilde{C}}{N}}(x_{ij}) \right) \\ & \leq N_{\frac{1}{N}}^{\text{ext}} \left(B_{\frac{\tilde{C}}{N}}(0) \right) = N_1^{\text{ext}} \left(B_{\tilde{C}}^N(0) \right) < C', \end{aligned}$$

for some positive constant C' , where the first inequality follows from $R_{K,\max} \leq \frac{\tilde{c}}{N}$, the second from the definition of packing number, and the remaining inequalities from basic properties of packing and external covering numbers. Therefore, there exists a constant $C_1 > 0$ such that

$$\begin{aligned}
 \text{Int}_2 &= \int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) \mu(dx) \\
 &\leq \left[\sum_{i=1}^n \sum_{j=1}^{m_i} \frac{1}{K} \left(f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right)^2 \right] \frac{C'}{K} \max_{1 \leq r \leq N^d} \int_{C(c_r)} v(x) dx \\
 &\leq \left[\sum_{i=1}^n \sum_{j=1}^{m_i} \left(f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right)^2 \right] \frac{C' c_2 d v_{\max}}{K} \text{Vol} \left(B_{\frac{\bar{b}_1^{1/d}}{(2c_2, d^{C v_{\max} c_2})^{1/d}} \left(\frac{K}{nm} \right)^{1/d}}(x) \right) \\
 &\leq \left[\frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^{m_i} \left(\theta_{i,j} - (\hat{\theta}_{V_n})_{i,j} \right)^2 \right] \tilde{C}_1 v_{\max},
 \end{aligned}$$

where, the second inequality is followed by Equation [source equation]. Then, making use of [source equation], we conclude that

$$\text{Int}_2 = \int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x) dx = O_p(r_{nm}^2), \quad (185)$$

where

$$r_{nm}^2 = \frac{\max \{B^2, B\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max \{B^2, B\}}{n} + \frac{\max \{B, 1\} d \log^{1+\mu}(n) \log(T_n)}{n}.$$

It remains to analyze the expression in Equation [source equation], this is,

$$\text{Int}_1 = \int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx.$$

To this end, we use the notation from Appendix section [source ref.]. Observe that, if S corresponds to the piecewise definition of f^* in Definition [source equation], the term

$$\int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx,$$

can be written as,

$$\sum_{r \in [N]^d} \int_{C(c_r)} \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx,$$

which by convexity can be bounded by,

$$\sum_{r \in [N]^d} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{1}{K} (f^*(x) - f^*(x_{ij}))^2 v(x) dx$$

$$\begin{aligned}
 &= \sum_{r \in [N]^d : C(c_r) \cap S = \emptyset} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{1}{K} (f^*(x) - f^*(x_{ij}))^2 v(x) dx \\
 &+ \sum_{r \in [N]^d : C(c_r) \cap S \neq \emptyset} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{1}{K} (f^*(x) - f^*(x_{ij}))^2 v(x) dx.
 \end{aligned}$$

Therefore, the term

$$\int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx,$$

is bounded by

$$\begin{aligned}
 &4 \left| \left\{ r \in [N]^d : C(c_r) \cap S \neq \emptyset \right\} \right| \|f_0\|_\infty^2 \max_{1 \leq r \leq N^d} \int_{C(c_r)} v(x) dx \\
 &+ \sum_{r \in [N]^d : C(c_r) \cap S = \emptyset} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{1}{K} (f_0(x) - f_0(x_{ij}))^2 v(x) dx,
 \end{aligned}$$

which by Equation [source equation] is smaller than

$$\begin{aligned}
 &\text{Vol} \left(B \frac{\tilde{b}_1^{1/d}}{(2c_{2,d} C v_{\max} c_2)^{1/d} \left(\frac{K}{nm}\right)^{1/d}}(x) \right) (4v_{\max} \|f_0\|_\infty^2) \left| \left\{ r \in [N]^d : C(c_r) \cap S \neq \emptyset \right\} \right| \\
 &+ \sum_{r \in [N]^d : C(c_r) \cap S = \emptyset} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{1}{K} (f^*(x) - f^*(x_{ij}))^2 v(x) dx.
 \end{aligned}$$

Then, using the bound for volume and the piece-wise Lipschitz condition, it is bounded by

$$\begin{aligned}
 &\left(\frac{4\tilde{b}_1 \|f^*\|_\infty^2}{nm} \right) K \left| \left\{ r \in [N]^d : C(c_r) \cap S \neq \emptyset \right\} \right| \\
 &+ \sum_{r \in [N]^d : C(c_r) \cap S = \emptyset} \int_{C(c_r)} \sum_{x_{ij} \in N_K(x)} \frac{L_0}{K} (\|x - x_{ij}\|_2^2) v(x) dx \\
 &\leq \left(\frac{4\tilde{b}_1^d \|f^*\|_\infty^2}{nm} \right) K \left| \left\{ r \in [N]^d : C(c_r) \cap S \neq \emptyset \right\} \right| \\
 &+ L_0 \left(\sum_{r \in [N]^d : C(c_r) \cap S = \emptyset} \int_{C(c_r)} v(x) dx \right) \left(\frac{1}{N} + R_{K,\max} \right)^2 \\
 &\leq \left(\frac{4\tilde{b}_1^d \|f^*\|_\infty^2}{nm} \right) K \left| \left\{ r \in [N]^d : C(c_r) \cap S \neq \emptyset \right\} \right| + L_0 \left(\frac{1}{N} + R_{K,\max} \right)^2.
 \end{aligned}$$

Finally, by the choice of N , Lemma [source ref.] and following the conclusion on Lemma 15 of Madrid Padilla et al. (2020), we get that

$$\text{Int}_2 = \int \left(f^*(x) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) \right)^2 v(x) dx = O_p \left(\left(\frac{K}{nm} \right)^{\frac{1}{d}} \right). \quad (186)$$

From Equation [source equation] and [source equation], and the fact that

$$\Delta^2(\hat{f}_{V_n} - f^*) \leq \|f^* - \hat{f}_{V_n}\|_2^2,$$

we then conclude the claim. $\square$

Lemma 75. *Let m be a Binomial (M, q) random variable. Then, for all $\delta \in (0, 1)$,*

$$\mathbb{P}(m \leq (1 - \delta)Mq) \leq \exp\left(-\frac{1}{3}\delta^2 Mq\right),$$

$$\mathbb{P}(m \geq (1 + \delta)Mq) \leq \exp\left(-\frac{1}{3}\delta^2 Mq\right)$$

Lemma 76. *(See Lemma 6 in Madrid Padilla et al. (2020)). Under Assumption [source ref.]. Denote by $R_k(x)$ the distance from $x \in [0, 1]^d$ to its K th nearest neighbor in the set $\{x_{ij}\}_{i=1, j=1}^{n, m}$. Setting $R_{k, \max} = \max_{1 \leq i \leq n, 1 \leq j \leq m_i} R_k(x_{ij})$, and $R_{k, \min} = \min_{1 \leq i \leq n, 1 \leq j \leq m_i} R_k(x_{ij})$, we have that,*

$$\begin{aligned} & \mathbb{P}\left(a \left(\frac{K}{nm}\right)^{1/d} \leq R_{k, \min} \leq R_{k, \max} \leq \tilde{a} \left(\frac{K}{nm}\right)^{1/d}\right) \\ & \geq 1 - \frac{4}{C_\beta} \left(\log(n)n \exp\left(-\frac{\tilde{C}K}{24 \log(n)}\right)\right) - \frac{4}{C_\beta} \left(\log(n)n \exp\left(-\frac{C_\beta K}{24 \log(n)}\right)\right) - \frac{2}{n}, \end{aligned}$$

where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$, $\alpha = 1/(4(Cc_2, d c_2 v_{\max})^{1/d})$, and $\tilde{a} = (1/(c_1 c_{1, d} c_1 v_{\min}))^{1/d}$ for some positive constants c_2 , and c_1 , $v_{\max}$, $v_{\min}$, C , C_β , and c from Assumption [source ref.]. Moreover, $c_{1, d}$ and $c_{2, d}$ are positive constants that depends on d .

Proof. First note that for any $x \in [0, 1]^d$, by Assumptions 1 we have

$$\mathbb{P}(x_{ij} \in B_r(x)) = \int_{B_r(x)} v(t) dt \leq c_2 r^d v_{\max} =: \mu_{\max} < 1, \quad (187)$$

(for some small enough r and a positive constant c_2 that depends on d . Let $L = \frac{2}{C_\beta} \log(n)$ and $\{x_{ij}^*\}_{i=1, j=1}^{n, m_i}$ defined as in Appendix J and consider the event $\Omega^c = \bigcap_{1 \leq i \leq n} \{x_{ij} = x_{ij}^* \forall 1 \leq j \leq m_i\}$. It is satisfied that $|\mathcal{J}_{e, l}|, |\mathcal{J}_{o, l}| \geq n/L$, from where, there exist positive constants c_1 and c_2 such that

$$c_1 n/L \leq |\mathcal{J}_{e, l}|, |\mathcal{J}_{o, l}| \leq c_2 n/L. \quad (188)$$

Moreover, by Lemma [source ref.] and Assumption [source ref.], we have that $\mathbb{P}(\Omega^c) \leq \frac{1}{n}$. Then,

$$\mathbb{P}\left(\left\{R_{k, \min} \leq a \left(\frac{K}{nm}\right)^{1/d}\right\}\right) \leq \mathbb{P}\left(\left\{R_{k, \min} \leq a \left(\frac{K}{nm}\right)^{1/d}\right\} \cap \Omega\right) + \frac{1}{n}. \quad (189)$$

To analyze the term $\mathbb{P}\left(\left\{R_{k, \min} \leq a \left(\frac{K}{nm}\right)^{1/d}\right\} \cap \Omega\right)$, we observe that

$$\mathbb{P} \left(\left\{ R_{k,\min} \leq a \left(\frac{K}{nm} \right)^{1/d} \right\} \cap \Omega \right) \leq \mathbb{P} \left(\left\{ (\exists i, j) \in [n] \times [m_i] : R_k(x_{ij}) \leq a(K/mn)^{1/d} \right\} \cap \tilde{\Omega} \right) \quad (190)$$

We also have that $R_k(x) \leq r$ if and only if there are at least K observations among $\{x_{ij}\}$ in $B_r(x)$. Therefore, letting

$$r = a(K/(nm))^{1/d}, \quad (191)$$

we have that

$$\begin{aligned} & \mathbb{P} \left(\left\{ (i, j) \in [n] \times [m_i] : R_k(x_{ij}) \leq a(K/(nm))^{1/d} \right\} \cap \tilde{\Omega} \right) \\ & \leq \sum_{i=1}^n \sum_{j=1}^{m_i} \mathbb{P} \left(\left\{ \sum_{t=1}^n \sum_{s=1}^{m_t} 1_{x_{t,s} \in B_r(x_{ij})} \geq K \right\} \cap \tilde{\Omega} \right) \\ & \leq Cnm \mathbb{P} \left(\sum_{t=1}^n \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} \geq K \right), \end{aligned} \quad (192)$$

where the first inequality is followed by a union bound argument together with the last observation. The second inequality is followed from Assumption [source ref.] coupled with the definition of the event Ω at the beginning of the proof. Then, using a union bound argument

$$\mathbb{P} \left(\sum_{i=1}^n \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} \geq K \right) \quad (193)$$

$$\begin{aligned} & \leq \sum_{l=1}^L \mathbb{P} \left(\sum_{t \in \mathcal{J}_{e,l}} \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} > K/(2L) \right) + \sum_{l=1}^L \mathbb{P} \left(\sum_{t \in \mathcal{J}_{o,l}} \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} > K/(2L) \right) \\ & = I_1 + I_2. \end{aligned} \quad (194)$$

We now bound each of the terms I_1 and I_2 . The analysis employed for the second term follows the same line of arguments as for I_1 and is omitted. Using Lemma [source ref.] for any $1 \leq l \leq L$ we have that

$$\sum_{t \in \mathcal{J}_{e,l}} \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} \sim \text{Binomial} \left(\sum_{t \in \mathcal{J}_{e,l}} m_t, \mathbb{P}(x_{1,1}^* \in B_r(x_{ij})) \right). \quad (195)$$

Let $\text{Bin} \sim \text{Binomial} \left(\sum_{t \in \mathcal{J}_{e,l}} m_t, \mu_{\max} \right)$. By Inequality [source equation],

$$\mathbb{P} \left(\sum_{t \in \mathcal{J}_{e,l}} \sum_{s=1}^{m_t} 1_{x_{t,s} \in B_r(x_{ij})} \geq K/(2L) \right) \leq \mathbb{P}(\text{Bin} \geq K/(2L)). \quad (196)$$

By Inequality [source equation], Equation [source equation], Assumption [source ref.] and the definition of $\mu_{\max}$ in Inequality [source equation], observe that

$$2\mu_{\max} \sum_{t \in \mathcal{J}_{e,l}} m_t \leq \frac{2c_{2,d}v_{\max}KCc_2nm}{4Cc_{2,d}c_2v_{\max}nmL} = \frac{K}{2L}. \quad (197)$$

From Inequality [source equation] for any $1 \leq L \leq L$,

$$\mathbb{P}(\text{Bin} \geq K/(2L)) \leq \mathbb{P}(\text{Bin} \geq 2\mu_{\max} \sum_{t \in \mathcal{J}_{e,l}} m_t). \quad (198)$$

Therefore, by [source ref.], Inequality [source equation] and Inequality [source equation]

$$\begin{aligned} \mathbb{P}\left(\sum_{t \in \mathcal{J}_{e,l}} \sum_{s=1}^{m_t} 1_{x_{t,s}^* \in B_r(x_{ij})} \geq K/(2L)\right) &\leq \exp\left(-\frac{1}{3}\mu_{\max} \sum_{t \in \mathcal{J}_{e,l}} m_t\right) \\ &\leq \exp\left(-\frac{c_1 c}{12c_2 C} \frac{K}{L}\right) = \exp\left(-\frac{c_1 c C_\beta}{24c_2 C} \frac{K}{\log(n)}\right). \end{aligned} \quad (199)$$

where the second inequality is followed by Inequality [source equation], the choice of r in Equation [source equation], Assumption [source ref.], and the choice of $\mu_{\max}$ in Inequality [source equation]. The Equality is attained by the choice of L at the beginning of the proof. Therefore, using Inequality [source equation], [source equation], [source equation] and the choice of L at the beginning of the proof, the term I_1 is upper bounded as follows

$$\begin{aligned} I_1 &\leq \frac{2}{C_\beta} \log(n) \exp\left(-\frac{c_1 c}{24c_2 C} \frac{K}{L}\right) \\ &= \frac{2}{C_\beta} \log(n) \exp\left(-\frac{c_1 c C_\beta}{24c_2 C} \frac{K}{\log(n)}\right). \end{aligned} \quad (200)$$

Similarly we obtain that

$$I_2 \leq \frac{2}{C_\beta} \log(n) \exp\left(-\frac{C_\beta c_1 c}{24C_2} \frac{K}{\log(n)}\right). \quad (201)$$

In consequence, by Inequality [source equation], [source equation], [source equation] and [source equation] we obtain that

$$\mathbb{P}\left(\{(i, j) \in [n] \times [m_i] : R_k(x_{ij}) \leq a(K/(nm))^{1/d}\} \cap \Omega\right) \leq \frac{4}{C_\beta} Cnm \log(n) \exp\left(-\frac{c_1 c C_\beta}{24c_2 C} \frac{K}{\log(n)}\right). \quad (202)$$

From Equation [source equation], [source equation] and [source equation] it follows that

$$\mathbb{P}\left(\{R_{k,\min} \leq a(K/(nm))^{1/d}\}\right) \leq \frac{4}{C_\beta} \log(n) \exp\left(-\frac{c_1 c C_\beta}{24c_2 C} \frac{K}{\log(n)}\right) + \frac{1}{n}. \quad (203)$$

Now we analyze $P(R_{k,\max} \geq \tilde{a}(K/nm)^{1/d})$. To this end, we first observe that,

$$\mathbb{P}(x_{ij} \in B_r(x)) = \int_{B_r(x)} v(t) dt \geq c_{1,d} r^d v_{\min} := \mu_{\min} > 0, \quad (204)$$

for a positive constant $c_{1,d}$ that depends on d . Following a similar argument as above, we arrive at

$$\mathbb{P}(R_{k,\max} \geq \tilde{a}(K/nm)^{1/d}) \leq \frac{4}{C_\beta} \log(n) nm \exp\left(-\frac{C_\beta}{24} \frac{K}{\log(n)}\right) + \frac{1}{n}, \quad (205)$$

concluding the proof.

We provide the following lemma to characterize the minimum and maximum number of observations $\{x_{ij}\}$ that fall within each cell $\mathcal{I}_r$, for $r = 1, \dots, N^d$. Remembering that $I_r = \{(i, j) \in [n] \times [m_i] : P_{I_r}(x_{ij}) = c_r\}$ for $r = 1, \dots, N^d$, where $P_{I_r}(x)$ denotes the point in $P_{\text{lat}}(N)$ such that $x \in C(P_{I_r}(x))$. $\square$

Lemma 77. (see Lemma 7 in Madrid Padilla et al. (2020)). Assume that N in the construction of $P_{lat}(N)$ of [source equation] is chosen as

$$N = \left\lceil \frac{3\sqrt{d(nm)^{1/d}}}{aK^{1/d}} \right\rceil, \quad \text{where} \quad a = \frac{1}{(4Cc_{2,d}c_2v_{\max})^{1/d}}, \quad (206)$$

for a positive constants c_2 , and $v_{\max}$, and C are from Assumptions [source ref.] and [source ref.]. Moreover, $c_{2,d}$ is a positive constant that depends on d . Then there exist positive constants $\tilde{b}_1$ and $\tilde{b}_2$ depending on d , $v_{\min}$, and $v_{\max}$ with $v_{\min}$ from Assumption [source ref.], such that for C_β from Assumption [source ref.],

$$\mathbb{P} \left(\max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right) \leq \frac{4}{C_\beta} \left(\log(n)n \exp \left(-\frac{C_\beta}{12} \delta^2 \tilde{b}_2 \frac{\tilde{b}_2K}{\log(n)} \right) + \frac{1}{n} \right), \quad (207)$$

$$\mathbb{P} \left(\min_{r=1,\dots,N^d} |I_r| \leq (1-\delta)\tilde{b}_1K \right) \leq \frac{4}{C_\beta} \left(\log(n)n \exp \left(-\frac{C_\beta}{12} \delta^2 \tilde{b}_2 \frac{\tilde{b}_2K}{\log(n)} \right) + \frac{1}{n} \right), \quad (208)$$

for all $\delta \in (0, 1)$. Moreover, the symmetric K -NN graph has maximum degree $d_{\max}$ satisfying

$$\begin{aligned} \mathbb{P}(d_{\max} \geq 3Cc_2c_{2,d}v_{\max}\tilde{a}^dK) \\ \leq C_{nm} \left(\frac{4}{C_\beta} \log(n) \exp \left(-\frac{C_\beta}{24} \frac{K}{\log(n)} \right) + \frac{4}{C_\beta} C \log(n)nm \exp \left(-\frac{C_\beta}{24} \frac{K}{\log(n)} \right) + \frac{1}{n} \right), \end{aligned} \quad (209)$$

where $\tilde{a} = \frac{1}{(cc_1c_{1,d}v_{\min})^{1/d}}$, with c_1 is a positive constant. Moreover, $c_{1,d}$ is a positive constants that depends on d . Let $C(\cdot)$ be defined in [source equation]. Define the event Ω as: If $x_{ij} \in C(x'_{i,j})$ and $x_{s,t} \in C(x'_{s,t})$ for $x'_{i,j}, x'_{s,t} \in P_{lat}(N)$ with $\|x'_{i,j} - x'_{s,t}\|_2 \leq N^{-1}$, then x_{ij} and $x_{s,t}$ are connected in the K -NN graph. Then

$$\mathbb{P}(\Omega) \geq 1 - \frac{4}{C_\beta} C \log(n)n \exp \left(-\frac{\tilde{C}K}{24 \log(n)} \right) - \frac{1}{n}. \quad (210)$$

where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$.

Proof. By observation, the following holds,

$$\begin{aligned} \|x_{ij} - x_{s,t}\|_2 &\leq \sqrt{d} \|x_{ij} - x_{s,t}\|_\infty \\ &\leq \sqrt{d} \{ \|x_{ij} - x'_{i,j}\|_\infty + \|x'_{i,j} - x'_{s,t}\|_\infty + \|x_{s,t} - x'_{s,t}\|_\infty \} \\ &< 3\sqrt{d}N^{-1} \leq a \left(\frac{K}{nm} \right)^{1/d}, \end{aligned} \quad (211)$$

with $a = 1/(2Cc_2v_{\max})^{1/d}$ and $v_{\max}$ and C are from Assumptions [source ref.] and [source ref.]. The first inequality is followed by a basic inequality between the l_2 and l_∞ norms. The second is due to the triangle inequality, while the third inequality comes after using that $\|x'_{i,j} - x'_{s,t}\|_\infty \leq N^{-1}$ and $x'_{i,j}, x'_{s,t} \in P_{lat}(N)$.

Proof. Through this proof, we make use of the event $\tilde{\Omega}$ defined as $\tilde{\Omega} = \bigcap_{1 \leq i \leq n} \{x_{ij} = x_{ij}^*\}$, where x_{ij} are defined in Appendix [source ref.]. We use

$$L = \frac{2}{C_\beta} \log(n), \quad (212)$$

and for $l \in [L]$

$$\frac{c_1 n}{L} \leq |\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \leq \frac{c_2 n}{L}, \quad (213)$$

with c_1 and c_2 positive constants. First, we analyze the event Ω . To this end, let $x_{ij} \in C(x'_{i,j})$ and $x_{s,t} \in C(x'_{s,t})$ for $x_{ij}^*, x'_{s,t} \in P_{\text{lat}}(N)$ with

$$\|x_{ij}^* - x'_{s,t}\|_2 \leq N^{-1}. \quad (214)$$

Then the following holds,

$$\begin{aligned} \|x_{ij} - x_{s,t}\|_2 &\leq \sqrt{d} \|x_{ij} - x_{s,t}\|_\infty \\ &\leq \sqrt{d} (\|x_{ij} - x_{ij}^*\|_\infty + \|x_{ij}^* - x'_{s,t}\|_\infty + \|x'_{s,t} - x_{s,t}\|_\infty) \\ &< 3\sqrt{d} N^{-1} \\ &\leq a \left(\frac{K}{nm} \right)^{1/d}, \end{aligned}$$

where $a = 1/(2Cc_2v_{\max})^{1/d}$ and $v_{\max}$ and C are from Assumption [source ref.]. The first inequality is followed by a basic inequality between the ℓ_2 and ℓ_∞ norms. The second is due to the triangle inequality, while the third inequality comes after using that $\|x_{ij}^* - x'_{s,t}\|_2 \leq N^{-1}$ and $x_{ij}^*, x'_{s,t} \in P_{\text{lat}}(N)$. Using the definition of N in the statement of the Lemma provides the last inequality. From Inequality [source equation] we observe that $\left\{ \tilde{a} \left(\frac{K}{nm} \right)^{1/d} \leq R_{k,\min} \right\} \subseteq \Omega$. Therefore, using the result from the proof of Lemma [source ref.],

$$\mathbb{P}(\Omega) \geq \mathbb{P} \left(a \left(\frac{K}{nm} \right)^{1/d} \leq R_{k,\min} \right) \geq 1 - \frac{4}{C_\beta} C \log(n) n \exp \left(-\frac{\tilde{C}}{24 \log(n)} \frac{K}{n} \right) - \frac{1}{n}, \quad (215)$$

Next, we proceed to derive an upper bound on the counts $\{I_l\}_{l=1}^{N^d}$. Assume that $x \in P_{\text{lat}}(N)$, and $x' \in C(x)$. Since $x' \in C(x)$, by the definition of $P_{\text{lat}}(N)$ we have that $\|x' - x\|_\infty \leq \frac{1}{2N}$. Therefore, using the basic inequality between the ℓ_2 and ℓ_∞ norms,

Assume that $x \in P_{\text{lat}}(N)$, and $x' \in C(x)$. Since $x' \in C(x)$, by the definition of $P_{\text{lat}}(N)$ we have that $\|x - x'\|_\infty \leq \frac{1}{2N}$. Therefore, using the basic inequality between the ℓ_2 and ℓ_∞ norms,

$$\|x - x'\|_2 \leq \sqrt{d} \|x - x'\|_\infty \leq \frac{\sqrt{d}}{2N} \leq \frac{a}{6} \left(\frac{K}{nm} \right)^{1/d} = \frac{\tilde{b}_1^{1/d}}{(2c_{2,d}v_{\max}2Cc_2)^{1/d}} \left(\frac{K}{nm} \right)^{1/d}, \quad (216)$$

where the third inequality is obtained from the choice of N in the statement of the Lemma, and b_1 is an appropriate constant. Here $c_{2,d} > 0$ is a constant that depends on d . Therefore, from Inequality [source equation].

$$C(x) \subseteq B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}} \left(\frac{K}{nm} \right)^{1/d}(x). \quad (217)$$

On the other hand, let $\tilde{b}_2 > 0$ a constant such that $\left(\frac{(\tilde{b}_2)^{1/d}}{(2c_{1,d}cc_1v_{\min})^{1/d}} \right) \leq \frac{a}{2} \frac{1}{3\sqrt{d}}$, with $c_{1,d} > 0$ a constant that depends on d . Observe that if the following holds

$$\|x - x'\|_2 \leq \left(\frac{\tilde{b}_2}{(2cc_1v_{\min})} \right)^{1/d} \left(\frac{K}{nm} \right)^{1/d}, \quad (218)$$

we have that,

$$\begin{aligned} \|x - x'\|_\infty &\leq \|x - x'\|_2 \leq \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})} \right)^{1/d} \left(\frac{K}{nm} \right)^{1/d} \\ &\leq \frac{2}{3}\sqrt{d} \left(\frac{K}{nm} \right)^{1/d} \leq \frac{1}{2N}. \end{aligned} \quad (219)$$

The third inequality is followed by the choice of $\tilde{b}_2$ and the Equality due to the choice of N in the statement of the Lemma. Therefore,

$$B \frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}} \left(\frac{K}{nm} \right)^{1/d}(x) \subseteq C(x). \quad (220)$$

We are now ready to obtain the bound for $\mathbb{P} \left\{ \max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right\}$. Notice that

$$\mathbb{P} \left\{ \max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right\} \leq \mathbb{P} \left(\left\{ \max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right\} \cap \tilde{\Omega} \right) + \frac{1}{n}, \quad (221)$$

with $\tilde{\Omega}$ defined at the beginning of the proof. Then, by a union bound argument

$$\mathbb{P} \left\{ \max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right\} \quad (222)$$

$$\leq \sum_{r=1}^{N^d} \mathbb{P} \left\{ |I_1| \geq (1+\delta)\tilde{b}_1K \right\} \quad (223)$$

$$\leq \sum_{x \in P_{\text{lat}}(N)} \mathbb{P} \left[\left\{ |C(x)| \geq (1+\delta)\tilde{b}_1K \right\} \cap \tilde{\Omega} \right], \quad (224)$$

where $C(x) = \{(i, j) \in [n] \times [m_i] : P_I(x_{ij}) = x\}$. Moreover, for any $x \in P_{\text{lat}}(N)$ using Equation [source equation] we have that

$$\begin{aligned} \mathbb{P}(x_{1,1} \in C(x)) &\leq \mathbb{P} \left(x_{1,1} \in B \frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}} \left(\frac{K}{nm} \right)^{1/d}(x) \right) \\ &\leq \frac{(\tilde{b}_1)}{(2c_{2,d}v_{\max}C_2)} \left(\frac{K}{nm} \right) c_{2,d}v_{\max}, \end{aligned} \quad (225)$$

where the last inequality follows the same line of arguments as in Inequality [source equation] in the proof of Lemma [source ref.]. Therefore,

$$2Cc_2nm\mathbb{P}(x_{1,1} \in C(x)) \leq 2Cc_2nm \frac{(\tilde{b}_1)}{(2c_{2,d}v_{\max}C_2)} \left(\frac{K}{nm} \right) c_{2,d}v_{\max} = \tilde{b}_1K$$

The above Inequality and Inequality [source equation] lead to

$$\sum_{x \in P_{\text{lat}}(N)} \mathbb{P} \left[\left\{ |C(x)| \geq (1+\delta)\tilde{b}_1K \right\} \cap \tilde{\Omega} \right] \leq \sum_{x \in P_{\text{lat}}(N)} \mathbb{P} \left[\left\{ |C(x)| \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x)) \right\} \right]. \quad (226)$$

Then, using the definition of the event Ω at the beginning of the proof and a union bound argument,

$$\begin{aligned} & \mathbb{P} \left[\{ |C(x)| \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x)) \} \cap \tilde{\Omega} \right] \\ & \leq \sum_{l=1}^L \mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s} \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/(2L) \right) \\ & \quad + \sum_{l=1}^L \mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s} \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/(2L) \right). \end{aligned} \quad (227)$$

Now we analyze each of the terms

$$\mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/L \right).$$

By Inequality [source equation] and Assumption [source ref.]

$$\mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/L \right) \quad (228)$$

$$\leq \mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta) \left(\sum_{i \in \mathcal{J}_{e,l}} m_i \right) \mathbb{P}(x_{1,1}^* \in C(x)) \right). \quad (229)$$

By Lemma [source ref.] it follows that

$$\begin{aligned} & \mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta) \left(\sum_{i \in \mathcal{J}_{e,l}} m_i \right) \mathbb{P}(x_{1,1}^* \in C(x)) \right) \\ & \leq \exp \left(-\frac{1}{3} \delta^2 \left(\sum_{i \in \mathcal{J}_{e,l}} m_i \right) \mathbb{P}(x_{1,1}^* \in C(x)) \right). \end{aligned} \quad (230)$$

Now Equation [source equation] implies that

$$\mathbb{P}(x_{1,1}^* \in C(x)) \geq \mathbb{P} \left(x_{1,1}^* \in B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}} (\frac{K}{nm})^{1/d}}(x) \right) \geq c_{1,d}v_{\min} \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})} \right) \left(\frac{K}{nm} \right), \quad (231)$$

from where, using Inequality [source equation] and Assumption [source ref.] we get that

$$\left(\sum_{i \in \mathcal{J}_{e,l}} m_i \right) (\mathbb{P}(x_{1,1}^* \in C(x))) \geq cc_1 \frac{n}{L} mc_{1,d}v_{\min} \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})} \right) \left(\frac{K}{nm} \right). \quad (232)$$

Thus, by our choice of L in Equation [source equation] and Inequality [source equation]

$$\mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/L \right) \quad (233)$$

$$\leq \exp \left(-\frac{1}{3} \delta^2 cc_1 \frac{n}{L} mc_{1,d}v_{\min} \frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})} \frac{K}{nm} \right) = \exp \left(-\frac{C_\beta}{12} \delta^2 \tilde{b}_2 \frac{K}{\log(n)} \right). \quad (234)$$

A similar analysis applies to

$$\mathbb{P} \left(\sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} 1_{x_{t,s}^* \in C(x)} \geq (1+\delta)2Cc_2nm\mathbb{P}(x_{1,1}^* \in C(x))/(2L) \right). \quad (235)$$

Thus, by Equation [source equation], [source equation], [source equation], [source equation], and [source equation], we obtain that

$$\mathbb{P} \left(\max_{r=1,\dots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1K \right) \leq \frac{4}{C_\beta} \log(n)N^d \exp \left(-\frac{C_\beta}{12} \delta^2 \tilde{b}_2 \frac{K}{\log(n)} \right) + \frac{1}{n}. \quad (236)$$

On the other hand, with a similar argument, we have that

$$\begin{aligned} \mathbb{P} \left(\min_{r=1,\dots,N^d} |I_r| \leq (1-\delta)\tilde{b}_2K \right) &\leq \sum_{x \in P_{\text{lat}}(N)} \mathbb{P} \left(\left\{ |C(x)| \leq (1-\delta)\tilde{b}_2K \right\} \cap \tilde{\Omega} \right) \\ &\leq \sum_{x \in P_{\text{lat}}(N)} \mathbb{P} \left(\left\{ |C(x)| \leq (1-\delta)2cc_1nm\mathbb{P}(x_{1,1}^* \in C(x)) \right\} \cap \tilde{\Omega} \right) + \frac{1}{N} \\ &\leq \frac{4}{C_\beta} \log(n)N^d \exp \left(-\frac{C_\beta}{12} \delta^2 \tilde{b}_2 \frac{K}{\log(n)} \right) + \frac{1}{n}. \end{aligned} \quad (237)$$

Next we proceed to find an upper bound on the maximum degree of the K-NN graph. To this end, we proceed as in the proof of Lemma [source ref.]. First, define the sets

$$\begin{aligned} B_{i,j}(x) &= \left\{ (i', j') \in [n] \times [m_i] \mid (i', j') \neq (i, j); x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x) \right\} \\ B_{i,j} &= \left\{ (i', j') \in [n] \times [m_i] \mid (i', j') \neq (i, j); x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x_{ij}) \right\}, \end{aligned} \quad (238)$$

for $(i, j) \in [n] \times [m_i]$, and $x \in [0, 1]^{\tilde{\Omega}}$. Observe that

$$\begin{aligned} &\mathbb{P}(|B_{i,j}(a)| > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK) \\ &\leq \mathbb{P} \left(\sum_{(i',j') \in [n] \times [m_i] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \geq \frac{3}{2}c_2Cc_{2,d}v_{\max}\tilde{a}^dK \right). \end{aligned} \quad (239)$$

To bound the term

$$\mathbb{P} \left(\sum_{(i',j') \in [n] \times [m_i] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK \right), \quad (240)$$

we use a union bound arguments to get

$$\mathbb{P} \left(\sum_{(i',j') \in [n] \times [m_i] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK \right) \quad (241)$$

$$\leq \sum_{l=1}^L \mathbb{P} \left(\sum_{(i',j') \in \mathcal{J}_{e,l} \times [m_i] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK/(2L) \right) \quad (242)$$

$$+ \sum_{l=1}^L \mathbb{P} \left(\sum_{(i',j') \in \mathcal{J}_{o,l} \times [m_i] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK/(2L) \right). \quad (243)$$

Further, using Lemma [source ref.] for any $l \in [L]$ we have that

$$\sum_{(i',j') \in [n] \times [m_{i'}] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \sim \text{Binomial} \left(\sum_{i' \in J_l} m_{i'} - 1, \mathbb{P}(x_{i',1} \in B_{\tilde{a}(K/nm)^{1/d}}(x)) \right). \quad (244)$$

By Equation [source equation], Inequality [source equation], Assumption [source ref.] and the Inequality [source equation] in the proof of Lemma [source ref.], we have that

$$\frac{3}{2} \mathbb{P} \left(x_{1,1}^* \in B_{\tilde{a}(K/nm)^{1/d}}(x) \right) \left(\sum_{i' \in J_l} m_{i'} - 1 \right) \leq \frac{3}{2} c_2 \mu_{\max} a^d \frac{K}{nm} C \frac{2nm}{L}. \quad (245)$$

Thus, for any $1 \leq l \leq L$,

$$\begin{aligned} & \mathbb{P} \left(\sum_{(i',j') \in \mathcal{J}_{e,l} \times [m_{i'}] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \geq 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K / (2L) \right) \\ & \leq \mathbb{P} \left(\sum_{(i',j') \in \mathcal{J}_{e,l} \times [m_{i'}] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \right. \\ & \quad \left. \geq 3 \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})} \right)^{1/d} \left(\frac{K}{nm} \right)^{1/d} \times \left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1 \right) \right) \end{aligned} \quad (246)$$

Now, from Equation [source equation], Inequality [source equation], and Assumption [source ref.] we have that

$$\frac{cc_1}{2} c_{1,d} v_{\min} \tilde{a}^d \frac{K}{L} \leq c_{1,d} v_{\min} \tilde{a}^d \frac{K}{nm} - \frac{c_1}{L} \leq \mathbb{P} \left(\{x_{1,1}^* \in B_{\tilde{a}(K/nm)^{1/d}}(x)\} \right) \left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1 \right) \quad (247)$$

Thus, using [source ref.] and the definition of $\tilde{a}$ in the statement of the Lemma,

$$\begin{aligned} & \mathbb{P} \left(\sum_{(i',j') \in \mathcal{J}_{e,l} \times [m_{i'}] \setminus \{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \geq 3 \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})} \right)^{1/d} \left(\frac{K}{nm} \right)^{1/d} \right) \\ & \leq \exp \left(-\frac{1}{12} \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})} \right)^{1/d} \left(\frac{K}{nm} \right)^{1/d} \left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1 \right) \right) \\ & \leq \exp \left(-\frac{C_\beta}{24} \frac{K}{\log(n)} \right) \end{aligned} \quad (248)$$

Analogously, the exact same bound can be achieved when the case of $\mathcal{J}_{o,l}$. Hence, using the above inequality together with Equation [source equation], Inequality [source equation] and Inequality [source equation]

$$\mathbb{P} \left(|B_{i,j}(a)| > 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K \right) \leq \frac{4}{C_\beta} \log(n) \exp \left(-\frac{C_\beta}{24} \frac{K}{\log(n)} \right) \quad (249)$$

Therefore,

$$\begin{aligned}\mathbb{P}\left(|B_{i,j}| \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) &= \int_{[0,1]^d} \mathbb{P}\left(|B_{i,j}(x)| \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) p(x) \mu(dx) \\ &\leq \frac{4}{C_\beta} \log(n) N^d \exp\left(-\frac{C_\beta K}{24 \log(n)}\right).\end{aligned}\quad (250)$$

Finally let $d_{i,j}$ be the degree associated with x_{ij} . Using a union bound argument

$$\begin{aligned}\mathbb{P}\left(d_{\max} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) &\leq \mathbb{P}\left(\{d_{\max} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\} \cap \tilde{\Omega}\right) + \frac{1}{n} \\ &\leq Cn\mathbb{P}\left(d_{i,j} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) \cap \tilde{\Omega} + \frac{1}{n}.\end{aligned}\quad (251)$$

Moreover,

$$\begin{aligned}\mathbb{P}\left(\left\{d_{i,j} \leq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right\} \cap \tilde{\Omega}\right) &\geq \mathbb{P}\left(\left\{(i',j') \in [n] \times [m_{i'}], (i',j') \neq (i,j); \|x_{i,j} - x_{i',j'}\|_2 \leq R_{k,\max}\right\} \leq 3v_{\max}Cc_{2,d}\tilde{a}^dK\right) \cap \tilde{\Omega} \\ &\geq \mathbb{P}\left(\left\{(i',j') \in [n] \times [m_{i'}], (i',j') \neq (i,j); \|x_{i,j} - x_{i',j'}\|_2 \leq R_{k,\max}\right\} \leq \tilde{a}\left(\frac{K}{nm}\right)^{1/d}\right) \cap \tilde{\Omega} \\ &\geq 1 - \mathbb{P}\left(R_{k,\max} > \tilde{a}\left(\frac{K}{nm}\right)^{1/d}\right) \cap \tilde{\Omega} - \mathbb{P}\left(|B_{i,j}| > 3v_{\max}Cc_{2,d}\tilde{a}^dK\right),\end{aligned}\quad (252)$$

and the claim follows from Inequality [source equation] and the proof of Lemma [source ref.].

The [source ref.] demonstrated that the count of observations x_{ij} that fall within each cell I_i is proportional to K , with high probability. We leverage this understanding to derive an **upper bound for the prediction error** in the following result. $\square$

Lemma 78 (see Lemma 8 in Madrid Padilla et al. (2020)). *Consider the notation in [source ref.]. Assume that the event Ω in [source equation] intersected with*

$$\min_{r=1,\dots,N^d} |I_r| \geq \frac{1}{2} \tilde{b}_2 K, \quad (253)$$

holds. Then for all $s \in \mathbb{R}^{\sum_{i=1}^n m_i}$, it holds that

$$|s^\top(\theta - \theta_1)| \leq 2\|s\|_\infty \|\nabla_{G_k} \theta\|_1, \quad \forall \theta \in \mathbb{R}^{\sum_{i=1}^n m_i}. \quad (254)$$

Moreover,

$$\|D\theta^I\| \leq \|\nabla_{G_k} \theta\|_1, \quad \forall \theta \in \mathbb{R}^{\sum_{i=1}^n m_i}, \quad (255)$$

where D is the incidence matrix of a d -dimensional grid graph $G_{\text{grid}} = (V_{\text{grid}}, E_{\text{grid}})$ with $V_{\text{grid}} = [N^d]$, where $(r, r') \in E_{\text{grid}}$ if and only if

$$\|P_I(x_{i_r j_r}) - P_I(x_{i_{r'} j_{r'}})\|_2 = \|c_r - c_{r'}\|_2 = \frac{1}{N},$$

where $x_{i_r j_r} \in C(c_r)$ and $x_{i_{r'} j_{r'}} \in C(c_{r'})$.

Proof. We start by introducing the notation $x'_{ij} = P_I(x_{ij})$. To prove [source equation] we proceed in cases. *Case 1.* If $(\theta_I)_{11} = \theta_{11}$. Then clearly $|s_{(11)}| |\theta_{11} - (\theta_I)_{11}| = 0$. *Case 2.* If $(\theta_I)_{11} = \theta_{ij}$ for $(i, j) \neq (1, 1)$. Then

$$\|x'_{11} - x_{ij}\|_\infty \leq \|x'_{11} - x_{11}\|_\infty \leq \frac{1}{2N}.$$

Thus, $x'_{11} = x'_{ij}$, and so $((1, 1), (i, j)) \in E_K$ by the assumption that the event Ω holds. Therefore for every $(i, j) \in [n] \times [m_i]$, there exists $(s_{(ij)}, t_{(ij)}) \in [n] \times [m]$ such that $(\theta_I)_{ij} = \theta_{s_{(ij)}, t_{(ij)}}$ and either $(i, j) = (s_{(ij)}, t_{(ij)})$ or $(i, j), (s_{(ij)}, t_{(ij)}) \in E_K$. Hence,

$$|s^\top(\theta - \theta_1)| \leq \sum_{i=1}^n \sum_{j=1}^{m_i} |s_{ij}| |\theta_{i,j} - \theta_{s_{(ij)}, t_{(ij)}}| \leq 2 \|s\|_\infty \|\nabla_{G_k} \theta\|_1.$$

To verify [source equation], we observe that

$$\|D\theta^I\|_1 = \sum_{(r, r') \in E_{\text{grid}}} |\theta_{i_r, j_r} - \theta_{i_{r'}, j_{r'}}|.$$

Now, if $(r, r') \in E_{\text{grid}}$, then x_{i_r, j_r} and $x_{i_{r'}, j_{r'}}$ are neighboring cells in I . This implies that $((i_r, j_r), (i_{r'}, j_{r'}))$ is an edge in the K-NN graph. Thus, every edge in the grid graph G_{grid} corresponds to an edge in the K-NN graph and the mapping is injective. Note that here we have used the fact that [source equation] ensures that every cell has at least one point, provided that K is large enough. The claim is then followed. $\square$

Lemma 79 (see Lemma [source ref.] in Madrid Padilla et al. (2020).). *Consider the notation in [source ref.] and assume that that event Ω in [source equation] happens. We have that for $s, \theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$, it holds that*

$$s^\top(\theta_1 - \theta_2) \leq 2\|s\|_\infty (\|\nabla_{G_k} \theta_1\|_2 + \|\nabla_{G_k} \theta_2\|_2) = s^\top(\theta_{1I} - \theta_{2I}).$$

Proof. We observe that

$$s^\top(\theta_1 - \theta_2) = s^\top(\theta_1 - \theta_{1I}) + s^\top(\theta_{1I} - \theta_{2I}) + s^\top(\theta_{2I} - \theta_2),$$

and the claim is followed by Lemma [source ref.]. $\square$

Lemma 80 (see Lemma 10 in Madrid Padilla et al. (2020).). *Assume that the event Ω from Lemma [source ref.] intersected with*

$$\left\{ \min_{r=1, \dots, N^d} |I_r| > \frac{1}{2} \tilde{b}_2 K \right\}, \quad (256)$$

holds. Let $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$ and $\{s_{i,j}\}_{i=1, j=1}^{n, m_i} = s \in \mathbb{R}^{\sum_{i=1}^n m_i}$ sub-Gaussian with parameter 1. Then, we have that

$$\|s^\top(\theta_1 - \theta_2)\| \leq \max_{r=1, \dots, N^d} |I_r| \left(\|\tilde{\Pi}, \tilde{s}\|_2 \|\theta_1 - \theta_2\|_2 + \|(D^\dagger)^\top \tilde{s}\|_\infty \times (\|(\nabla_{G_K} \theta_{1I})\|_1 + \|(\nabla_{G_K} \theta_{2I})\|_1) \right) \quad (257)$$

where $\tilde{s} \in \mathbb{R}^{N^d}$ is a mean zero vector whose coordinates are sub-Gaussian with the same constants 1. Here, $\tilde{\Pi}$ is the orthogonal projection onto the span of $1_{N^d} \in \mathbb{R}^{N^d}$, and $D^\dagger$ is the pseudoinverse of the incidence matrix D from Lemma [source ref.].

Proof. Using the notation from the proof of Lemma [source ref.], we have that

$$\begin{aligned} s^\top(\theta_{1I} - \theta_{2I}) &= \sum_{r=1}^{N^d} \sum_{(i', j') \in I_r} s_{[i', j']}(\theta_{1[i_r, j_r]} - \theta_{2[i_r, j_r]}) \\ &= \max_{r=1, \dots, N^d} |I_r| \tilde{s}^\top(\theta_1^I - \theta_2^I), \end{aligned} \quad (258)$$

where

$$\tilde{s}_j = \left(\max_{r=1, \dots, N^d} |I_r| \right)^{-1} \sum_{(i', j') \in I_r} s_{[i', j']}. \quad (259)$$

Clearly, the $\tilde{s}_1, \dots, \tilde{s}_{N^d}$ are also sub-Gaussian with the same constants as the original errors $\{s_{i,j}\}_{i=1, j=1}^{n, m_i}$.

Moreover, let $\tilde{\Pi}$ be the orthogonal projection onto the span of $1_{N^d} \in \mathbb{R}^{N^d}$. By Hölder's inequality and by the triangle inequality,

$$\begin{aligned} s^\top(\theta_1 - \theta_2) &\leq \left\{ \max_{r=1, \dots, N^d} |I_r| \right\} \left\{ \|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1^I - \theta_2^I\|_2 \right\} + \|(D^+)^\top \tilde{s}\|_\infty \|D(\theta_1 - \theta_2)\|_1 \\ &\leq \left\{ \max_{r=1, \dots, N^d} |I_r| \right\} \left\{ \|\tilde{\Pi}\tilde{s}\|_2 \|\hat{\theta}^I - \theta^{*,I}\|_2 + \|(D^+)^\top \tilde{s}\|_\infty (\|D\theta_1^I\|_1 + \|D\theta_2^I\|_1) \right\}. \end{aligned} \quad (260)$$

Next we observe that

$$\|\theta_1^I - \theta_2^I\|_2 = \left(\sum_{r=1}^{N^d} (\theta_{1[i_r, j_r]} - \theta_{2[i_r, j_r]})^2 \right)^{1/2} \leq \left(\sum_{i=1}^n \sum_{j=1}^{m_i} (\theta_{1,i,j} - \theta_{2,i,j})^2 \right)^{1/2}. \quad (261)$$

Therefore, combining [source equation], [source equation] and Lemma [source ref.] we arrive at

$$s^\top(\theta_{1I} - \theta_{2I}) \leq \left\{ \max_{r=1, \dots, N^d} |I_r| \right\} \left\{ \|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1 - \theta^*\|_2 \right\} + \|(D^+)^\top \tilde{s}\|_\infty (\|\nabla_{G_k} \theta_1\|_1 + \|\nabla_{G_k} \theta_2\|_1).$$

□

J.5 For $f^* \notin \mathcal{F}$ And $\mathcal{F}$ is not convex

Implemented correction. The nonconvex proof chooses a function-valued near-best approximation $f^\circ \in \mathcal{F}$, sets $\phi_n = \|f^\circ - f^*\|_\infty$, and works on

$$\mathcal{H}^* = \{\mu(f - f^*) : f \in \mathcal{F}, \mu \in [0, 1]\}.$$

A single absolute empirical-process supremum at radius $r^2 + C\phi_n^2$ replaces the duplicated one-sided decomposition and yields $O_{\mathbb{P}}\{b_B^2(r_n^2 + \phi_n^2)\}$.

J.5.1 Technical Results for proof of Theorem [source ref.]

Lemma 81. Assume Assumption [source ref.] holds, then there exists c_0 such that for all $h = f - f^*$, we have

$$Z_{ij}(f) \leq \frac{\bar{f}h^2(x_{ij})}{2}.$$

Proof. Supposing that $|h(x_{ij})| = |f(x_{ij}) - f^*(x_{ij})|$, by Assumption [source ref.] and Lemma [source ref.], we have

$$\begin{aligned} Z_{ij}(f) - Z_{ij}(f^*) &= \int_0^{h(x_{ij})} \left\{ F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \right\} dz \\ &\leq \int_0^{h(x_{ij})} \bar{f}z dz \\ &\leq \frac{\bar{f}h^2(x_{ij})}{2}. \end{aligned}$$

Thus, by the fact $Z_{ij}(f^*) = 0$, we have $Z_{ij}(f) \leq \frac{\bar{f}h^2(x_{ij})}{2}$. $\square$

Corollary 82. On event Ω , let $\hat{f}$, belong to $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, where $\mathcal{F}$ is not necessarily convex, and let $\Delta^2(\hat{f} - f^*)$ be the huber loss. Let f_{Proj} in $\mathcal{F}$ be

$$f_{Proj} := \inf_{f \in \mathcal{F}(L,r)} \|f - f^*\|_\infty, \quad (262)$$

such that

$$\|f_{Proj} - f^*\|_\infty \leq \phi_{nm}.$$

Let

$$\tilde{t}^2 := t^2 + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2, \quad (263)$$

Then we have

$$\begin{aligned} &\mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 \right) \\ &\leq \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) + \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left(\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right) > \tilde{t}^2 \right). \end{aligned}$$

The c_0 satisfies [source equation], and $\ell(f) := (C_0 - c_0)\Delta^2(f - f^*)$.

Proof. Assume

$$c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 = t^2 + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 \quad (264)$$

where c_0 satisfies [source equation], we know Δ^2 is continuous. Moreover let f_{Proj} in $\mathcal{F}$ such that

$$\|f_{Proj} - f^*\|_\infty \leq \phi_{nm}$$

Consider $g : [0, 1] \rightarrow \mathbb{R}$ such that

$$g(\mu) = c_0 \Delta^2((1 - \mu)f^* + \mu\hat{f} - f^*)$$

Clearly, g is continuous with

$$g(0) = 0, \quad g(1) = c_0 \Delta^2(\hat{f} - f^*)$$

Therefore, there is $\mu_{\hat{f}} \in [0, 1]$ such that

$$g(\mu_{\hat{f}}) = \tilde{t}^2.$$

Hence, for any $\tilde{f}$ between $\hat{f}$ and f^* , by letting $\tilde{f} = (1 - \mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}$,
By construction we have

$$c_0 \Delta^2(\tilde{f} - f^*) = \tilde{t}^2.$$

So far we have just use the continuity of Δ^2 .

Furthermore, we have

$$\tilde{f} \in \text{Star}(\mathcal{F}, f^*),$$

note that $\tilde{f}$ may not stay in $\mathcal{F}$ anymore.

Then by the convexity of $\hat{Z}$, we have that

$$\begin{aligned} \hat{Z}(\tilde{f}) &\leq (1 - \mu_{\hat{f}})\hat{Z}(f^*) + \mu_{\hat{f}}\hat{Z}(\hat{f}) \\ &\stackrel{(a)}{=} \mu_{\hat{f}}\hat{Z}(\hat{f}) \\ &\stackrel{(b)}{\leq} \mu_{\hat{f}}\hat{Z}(f_{\text{Proj}}), \end{aligned} \tag{265}$$

Where [source equation] holds since $\hat{Z}(f^*) = 0$, and [source equation] holds since $\hat{f}$ is the minimizer of the $\hat{Z}$ on $\mathcal{F}$ and we have $f, f_{\text{Proj}} \in \mathcal{F}$.

Let

$$\ell(\tilde{f}) := (C_0 - c_0)\Delta^2(\tilde{f} - f^*), \tag{266}$$

observe that

$$\begin{aligned} \tilde{t}^2 &= c_0 \Delta^2(\tilde{f} - f^*) \\ &= C_0 \Delta^2(\tilde{f} - f^*) - (C_0 - c_0)\Delta^2(\tilde{f} - f^*) \\ &\leq \bar{Z}(\tilde{f}) - \ell(\tilde{f}) \end{aligned}$$

by Corollary [source ref.] and definition of $\ell(f)$ in [source equation].

Then, adding and subtracting $\hat{Z}(\tilde{f})$ we get that

$$\tilde{t}^2 \leq \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) + \hat{Z}(\tilde{f}) - \ell(\tilde{f})$$

and by equation [source equation],

$$\tilde{t}^2 \leq \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \hat{Z}(f),$$

implies

$$\tilde{t}^2 \leq \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\hat{Z}(f_{\text{Proj}}).$$

Then, adding and subtracting $\mu_{\hat{f}}\bar{Z}(f_{\text{Proj}})$, we get

$$\begin{aligned}\tilde{t}^2 &\leq \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\hat{Z}(f_{\text{Proj}}) - \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}) \\ &\quad - \mu_{\hat{f}}l(f_{\text{Proj}}) + \mu_{\hat{f}}l(f_{\text{Proj}}).\end{aligned}$$

which implies

$$\begin{aligned}\tilde{t}^2 &\leq \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\left[\hat{Z}(f_{\text{Proj}}) - \bar{Z}(f_{\text{Proj}}) - l(f_{\text{Proj}})\right] \\ &\quad + \mu_{\hat{f}}l(f_{\text{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}). \\ &\stackrel{(a)}{\leq} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\ &\quad + \mu_{\hat{f}} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \\ &\quad + \mu_{\hat{f}}l(f_{\text{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}})\end{aligned}$$

The [source equation] holds since $f_{\text{Proj}} \in \text{Star}(\mathcal{F}, f^*)$ and

$$c_0\Delta^2(f_{\text{Proj}} - f^*) \leq c_0 \|f_{\text{Proj}} - f^*\|_2^2 \leq c_0 \|f_{\text{Proj}} - f^*\|_\infty^2 \leq c_0\phi_{nm}^2 \leq C_0\phi_{nm}^2 \leq \tilde{t}^2,$$

where the last inequality holds by [source equation].

Then, observe that $f^* \in \text{Star}(\mathcal{F}, f^*)$, and $c_0\Delta^2(f^* - f^*) = 0$.

Thus

$$\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \geq \hat{Z}(f^*) - \bar{Z}(f^*) - l(f^*) = 0,$$

and

$$\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \geq \bar{Z}(f^*) - \hat{Z}(f^*) - l(f^*) = 0. \quad (267)$$

Furthermore, $0 \leq \mu_{\hat{f}} \leq 1$, and

$$\begin{aligned}0 &\stackrel{(a)}{\leq} l(f_{\text{Proj}}) = (C_0 - c_0)\Delta^2(f_{\text{Proj}} - f^*) \\ &\leq C_0 \|f_{\text{Proj}} - f^*\|_2^2 \leq C_0\phi_{nm}^2\end{aligned}$$

where [source equation] holds by [source equation].

Therefore,

$$\begin{aligned}\tilde{t}^2 &\leq \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\ &\quad + \mu_{\hat{f}} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f-f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \\ &\quad + C_0\phi_{nm}^2 + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}).\end{aligned}$$

Then, we use Lemma [source ref.], we get that

$$\begin{aligned}\bar{Z}(f_{\text{Proj}}) &\leq \frac{\bar{f}}{2} \|f_{\text{Proj}} - f^*\|_2^2 \\ &\leq \frac{\bar{f}}{2} \|f_{\text{Proj}} - f^*\|_\infty^2 \\ &\leq \frac{\bar{f}}{2} \phi_{nm}^2.\end{aligned}$$

Thus, since $0 \leq \mu_{\hat{f}} \leq 1$, we have

$$\begin{aligned}\tilde{t}^2 &\leq \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right] \\ &\quad + \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right] \\ &\quad + \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2,\end{aligned}$$

where the first inequality is by [source equation] and hence,

$$\begin{aligned}\tilde{t}^2 := t^2 + \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 &\leq \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right] \\ &\quad + \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right].\end{aligned}$$

□

Theorem 83 (Conversion of Huber loss). *Let $\hat{f}, f^* \in \mathcal{F}$, let Ω_β be defined in [source equation]. Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1, j=1}^{n, m_i}$ obey Assumption [source ref.], where c_0 satisfies [source equation], and $\ell(f)$ is defined in [source equation], the following inequality is true for any $t > 0$, let*

$$\begin{aligned}\tilde{A}_1 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ \tilde{A}_2 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ \tilde{A}_3 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],\end{aligned} \tag{268}$$

$$\begin{aligned}
 \tilde{B}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{B}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{B}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]
 \end{aligned} \tag{269}$$

$$\begin{aligned}
 \tilde{E}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{E}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{E}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],
 \end{aligned} \tag{270}$$

$$\begin{aligned}
 \tilde{F}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{F}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 \tilde{F}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right],
 \end{aligned} \tag{271}$$

$$\begin{aligned}
 &\mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 \right) \\
 &\leq \sum_{l=1}^L \frac{4L}{\tilde{t}^2} \left(\tilde{A}_1 + \tilde{A}_2 + \tilde{A}_3 + \tilde{B}_1 + \tilde{B}_2 + \tilde{B}_3 + \tilde{E}_1 + \tilde{E}_2 + \tilde{E}_3 + \tilde{F}_1 + \tilde{F}_2 + \tilde{F}_3 \right).
 \end{aligned} \tag{272}$$

Proof. By Corollary [source ref.], it is true that

$$\mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 \right)$$

$$\begin{aligned}
 & \stackrel{(a)}{\leq} \mathbb{P} \left(\underbrace{\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) + \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right)}_{I_1} > \tilde{t}^2 \right) \\
 & \leq \mathbb{P} \left(\underbrace{\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right)}_{I_1} > \frac{\tilde{t}^2}{2} \right) \\
 & \quad + \mathbb{P} \left(\underbrace{\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right)}_{I_2} > \frac{\tilde{t}^2}{2} \right).
 \end{aligned}$$

Then for the first term I_1 above, we have

$$\begin{aligned}
 & \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) > \frac{\tilde{t}^2}{2} \right) \\
 & = \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right. \right. \\
 & \quad \left. \left. + \frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > \frac{\tilde{t}^2}{2} \right) \\
 & \leq \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > \frac{\tilde{t}^2}{4} \right) \\
 & \quad + \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{l=1}^L \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2} \right) > \frac{\tilde{t}^2}{4} \right) \\
 & \leq \sum_{l=1}^L \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right) \\
 & \quad + \sum_{l=1}^L \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right).
 \end{aligned}$$

Now we analyze each of the terms above,

$$\begin{aligned}
 & \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right) \\
 & \stackrel{(a)}{\leq} \frac{4L}{\tilde{t}^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \stackrel{(b)}{\leq} \frac{4L}{\tilde{t}^2} \mathbb{E} \left[\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} + \right. \\
 & \quad \left. \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right. \\
 & \quad \left. + \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right] \\
 & = \frac{4L}{\tilde{t}^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\underbrace{\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}}_{\tilde{A}_1} \right] \\
 & \quad + \frac{4L}{\tilde{t}^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\underbrace{\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}}_{\tilde{A}_2} \right] \\
 & \quad + \frac{4L}{\tilde{t}^2} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\underbrace{\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}}}_{\tilde{A}_3} \right] \\
 & := \frac{4L}{\tilde{t}^2} (\tilde{A}_1 + \tilde{A}_2 + \tilde{A}_3).
 \end{aligned}$$

Similarly, we have

$$\mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{z}_{ij}(f) - \hat{z}_{ij}(f) - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right)$$

$$\begin{aligned}
 &= \underbrace{\frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{B}_1} \\
 &+ \underbrace{\frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{B}_2} \\
 &+ \frac{4L}{\tilde{t}^2} \\
 &\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{B}_3} \\
 &:= \frac{4L}{\tilde{t}^2} \left(\tilde{B}_1 + \tilde{B}_2 + \tilde{B}_3 \right).
 \end{aligned}$$

where the first inequality [source equation] follows from Lemma [source ref.], The inequality [source equation] follows from Markov's inequality, and the third inequality [source equation] follows from the triangle inequality.

Thus, we have

$$I_1 \leq \sum_{l=1}^L \frac{4L}{\tilde{t}^2} \left(\tilde{A}_1 + \tilde{A}_2 + \tilde{A}_3 + \tilde{B}_1 + \tilde{B}_2 + \tilde{B}_3 \right).$$

Then similarly, we have

$$\begin{aligned}
 I_2 &= \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\hat{Z}(f) - \bar{Z}(f) - \ell(f) \right) > \frac{\tilde{t}^2}{2} \right) \\
 &\leq \sum_{l=1}^L \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \bar{Z}_{ij}(f) \right. \right. \\
 &\quad \left. \left. - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right) \\
 &+ \sum_{l=1}^L \mathbb{P} \left(\sup_{\substack{f \in \text{Star}(\mathcal{F},f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \left(\frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \bar{Z}_{ij}(f) \right. \right. \\
 &\quad \left. \left. - \frac{\ell(f)}{2L} \right) > \frac{\tilde{t}^2}{4L} \right)
 \end{aligned}$$

$$\begin{aligned}
 &\leq \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{E}_1} \\
 &+ \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{E}_2} \\
 &+ \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{E}_3} \\
 &+ \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{F}_1} \\
 &+ \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{F}_2} \\
 &+ \underbrace{\sum_{l=1}^L \frac{4L}{\tilde{t}^2} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{F}_3}.
 \end{aligned}$$

We thus get

$$I_2 \leq \sum_{l=1}^L \frac{4L}{\tilde{t}^2} \left(\tilde{E}_1 + \tilde{E}_2 + \tilde{E}_3 + \tilde{F}_1 + \tilde{F}_2 + \tilde{F}_3 \right).$$

To proceed, we would need to bound $\tilde{A}_1, \tilde{A}_2, \tilde{A}_3, \tilde{B}_1, \tilde{B}_2, \tilde{B}_3$, and $\tilde{E}_1, \tilde{E}_2, \tilde{E}_3, \tilde{F}_1, \tilde{F}_2, \tilde{F}_3$ separately. **It is enough to analyze** the $\tilde{A}_1, \tilde{A}_2$, and $\tilde{A}_3$, as for $\tilde{B}_1, \tilde{B}_2, \tilde{B}_3$, the analysis is similar to $\tilde{A}_1, \tilde{A}_2, \tilde{A}_3$, **we state the resulting bound.**

Similarly, we only need to analyze the $\tilde{E}_1, \tilde{E}_2$, and $\tilde{E}_3$, as for $\tilde{F}_1, \tilde{F}_2, \tilde{F}_3$, the analysis is similar to $\tilde{E}_1, \tilde{E}_2$, and $\tilde{E}_3$, **we state the resulting bound.** $\square$

Corollary 84 (Bound of $\tilde{A}_2, \tilde{B}_2$). *Let $\tilde{Z}(f), \hat{Z}(f)$ been defined in [source equation] and [source equation]. Let Ω_β be defined in [source equation]. Let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{A}_2$ be the quantity introduced in [source equation], on event Ω , then we have*

$$\begin{aligned}
 \tilde{A}_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq 2 \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).
 \end{aligned}$$

Similarly, for $\tilde{B}_2$ in [source equation] we have

$$\tilde{B}_2 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).$$

Proof. The proof is a direct application of Corollary [source ref.] by replacing $\mathcal{F}$ with $\text{Star}(\mathcal{F}, f^*)$. $\square$

Corollary 85 (Bound of $\tilde{A}_1, \tilde{B}_1$). *Let Ω_β be defined in [source equation]. Let $\tilde{Z}(f), Z(f)$ been defined in [source equation] and [source equation], let $\ell(f)$ be defined in [source equation], let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{A}_1$ be the quantity introduced in [source equation], then we have*

$$\begin{aligned}
 \tilde{A}_1 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq 2 \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} u_{ij} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
 &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n},
 \end{aligned}$$

with a small constant c_4 that depends on $\tilde{c}_0$ is given in Assumption [source ref.], and c_1, c_2, C_β . Similarly, for $\tilde{B}_1$ in [source equation], we have

$$\tilde{B}_1 \leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n},$$

Proof. The proof is a direct application of Corollary [source ref.] by replacing $\mathcal{F}$ with $\text{Star}(\mathcal{F}, f^*)$. $\square$

Corollary 86 (Bound of $\tilde{A}_3$ and $\tilde{B}_3$). *Let Ω_β be defined in [source equation]. Let $\bar{Z}(f), Z(f)$ been defined in [source equation] and [source equation], let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{A}_3$ be the quantity introduced in expression [source equation], then we have*

$$\tilde{A}_3 = \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \bar{Z}_{ij}(f) - Z_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]$$

$$\begin{aligned}
 &\leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).
 \end{aligned}$$

Similarly, for $\tilde{B}_3$ in [source equation] we have

$$\begin{aligned}
 \tilde{B}_3 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\bar{Z}_{ij}(f) - Z_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).
 \end{aligned}$$

Proof. The proof is a direct application of Corollary [source ref.] by replacing $\mathcal{F}$ with $\text{Star}(\mathcal{F}, f^*)$. $\square$

Corollary 87 (Bound of $\tilde{E}_2, \tilde{F}_2$). *Let $\tilde{Z}(f), \hat{Z}(f)$ been defined in [source equation] and [source equation]. Let Ω_β be defined in [source equation]. Let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{E}_2$ be the quantity introduced in [source equation], on event Ω , then we have*

$$\begin{aligned}
 \tilde{E}_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &\leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).
 \end{aligned}$$

Similarly, for $\tilde{F}_2$ in [source equation] we have

$$\tilde{F}_2 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}).$$

Proof. In the proof, we can see that the $\tilde{E}_2$ and $\tilde{A}_2$ have the same form of upper bound. The proof is similar to the proof of Corollary [source ref.].

First, by Tower property, we have

$$\begin{aligned}
 A_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 &= \mathbb{E} \left[\left(\mathbb{E} \left[\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\hat{Z}_{ij}(f) - \mathbb{E}[\hat{Z}_{ij}(f) \mid \{\delta_i\}]\} \mid \{\delta_i\} \right] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right].
 \end{aligned}$$

To facilitate our analysis, we define more terms. Let $\{\check{e}_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, $\{\check{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be an independent and identically distributed copy of $\{e_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, then we define $\check{Z}_{ij}(f)$ as

$$\check{Z}_{ij}(f) := \rho_\tau(f^*(\check{x}_{ij}) + \delta_i(\check{x}_{ij}) + \check{e}_{ij} - f(\check{x}_{ij})) - \rho_\tau(\delta_i(\check{x}_{ij}) + \check{e}_{ij}).$$

Now we control the $\mathbb{E} \left[\hat{Z}_{ij}(f) - \check{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}, \{x_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]} \right]$, the expectation takes on $\{\check{e}_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, $\{\check{x}_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, and conditioned on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}, \{e_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$.

So we have

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \mathbb{E} \left[\hat{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right] \right. \\ & \quad \left. - \mathbb{E} \left[\check{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right] \right\} \\ & \stackrel{(a)}{\leq} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \hat{Z}_{ij}(f) - \check{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right]. \end{aligned} \quad (273)$$

The equality [source equation] is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \mathbb{E} \left[\hat{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right] \right. \\ & \quad \left. - \mathbb{E} \left[\check{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right] \right\} \\ & \stackrel{(a)}{=} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \hat{Z}_{ij}(f) - \mathbb{E} \left[\hat{Z}_{ij}(f) \mid \{\delta_i\} \right] \right\}. \end{aligned} \quad (274)$$

The second expectation takes on $\{e_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, and conditioned on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$, then the equality [source equation] holds since $\mathbb{E} \left[\check{Z}_{ij}(f) \mid \{\delta_i\}, \{e_{i,j}\}, \{x_{ij}\} \right]$ and $\mathbb{E} \left[\hat{Z}_{ij}(f) \mid \{\delta_i\} \right]$ are the same quantity.

Combines [source equation] and [source equation], taking expectation of $\{e_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ conditioning on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$ we have

$$\begin{aligned} & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \hat{Z}_{ij}(f) - \mathbb{E} \left[\hat{Z}_{ij}(f) \mid \{\delta_i\} \right] \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq \mathbb{E} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \hat{Z}_{ij}(f) \right\} - \check{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \mid \{\delta_i\} \right] \\ & \stackrel{(a)}{=} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \left\{ \hat{Z}_{ij}(f) - \check{Z}_{ij}(f) \right\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \mid \{\delta_i\} \right] \end{aligned}$$

$$\stackrel{(b)}{\leq} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \mid \{\delta_i\} \right].$$

The [source equation] holds since conditioned on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$, the $\{\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ is independently distributed and symmetric with mean zero and $\{s_{ij}(\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f))\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ have the same distribution. The [source equation] follows because $\{-s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are also independent Rademacher variables. Taking expectation of $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$ **on both sides**, we obtain the result.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $u_{ij}(f)$ in Lemma [source ref.], and then apply the Ledoux-Talagrand contraction principle in Theorem [source ref.].

$$\begin{aligned} & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \hat{Z}_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \stackrel{(a)}{\leq} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & = \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}). \end{aligned}$$

□

Lemma 88 (Bound of $\tilde{E}_1$ and $\tilde{F}_1$). *Let Ω_β be defined in [source equation]. Let $\tilde{Z}(f)$, $Z(f)$ be defined in [source equation] and [source equation], and $\ell(f)$ be defined in [source equation], let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{E}_1$ be the quantity introduced in [source equation], with a small constant $\tilde{c}_0$ is given in Assumption [source ref.], then we have*

$$\begin{aligned} \tilde{E}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Similarly, for $\tilde{F}_1$ in [source equation] we have

$$\tilde{F}_1 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f-f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.$$

Proof. Step 1. By previous Lemmas [source ref.] and [source ref.],

$$\tilde{Z}_{ij}(f) - Z_{ij}(f) = \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | \delta_i, x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij})\} dt$$

We define $u_{ij}(f)$ as the integral expression given by the last line:

$$u_{ij}(f) := \int_0^{f(x_{ij}) - f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | \delta_i, x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | x_{ij})\} dt. \quad (275)$$

Then, we have

$$\begin{aligned} \tilde{E}_1 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \underbrace{\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} u_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{\tilde{E}_4}. \end{aligned} \quad (276)$$

Now we bound $\tilde{E}_4$, and we have

$$\begin{aligned} \tilde{E}_4 &\leq \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} u_{ij}(f) - \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} [u_{ij}(f) | \delta_i] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\quad + \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} [u_{ij}(f) | \delta_i] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \underbrace{\mathbb{E}_{\delta, \delta^*} \left[\mathbb{E}_{x, e | \delta} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} u_{ij}(f) - \mathbb{E}_{x, e | \delta_i} [u_{ij}(f)] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \right]}_{E_5(\{\delta_i\}_{i \in \mathcal{J}_{e,l}})} \\ &\quad + \underbrace{\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [u_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]}_{E_6}. \end{aligned}$$

We will bound $E_5(\{\delta_i\}_{i \in \mathcal{J}_{e,l}})$ and E_6 **separately**.

Step 2. First, we will show that

$$E_5(\{\delta_i\}_{i \in \mathcal{J}_{e,l}}) \leq 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} u_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \mid \{\delta_i\} \right].$$

Let $\{\tilde{x}_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$ be an independent and identically distributed copy of $\{x_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$, correspondingly, we let $\{\tilde{y}_{i,j}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$ be the copy of y_{ij} , where each $\tilde{y}_{i,j}$ and y_{ij} share the same $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{I}_{e,l}}$, $\{e_{i,j}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$, but **differ only in** $\{\tilde{x}_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$ and $\{x_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$. Then we define the independent and identically distributed copy of $u_{ij}(f)$ as

$$\tilde{u}_{ij}(f, \delta_i, \tilde{x}_{ij}) := \int_0^{f(\tilde{x}_{ij}) - f^*(\tilde{x}_{ij})} \{\mathbb{P}(\tilde{y}_{i,j} \leq f^*(\tilde{x}_{ij}) + t | \delta_i) - \mathbb{P}(\tilde{y}_{i,j} \leq f^*(\tilde{x}_{ij}) + t)\} dt.$$

Now we control the $\mathbb{E}[u_{ij}(f) - \tilde{u}_{ij}(f) | \{\delta_i\}]$, the expectation takes on $\{\tilde{x}_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$ and conditioned on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{I}_{e,l}}$. So we have

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\mathbb{E}[u_{ij}(f) | \{\delta_i\}] - \mathbb{E}[\tilde{u}_{ij}(f) | \{\delta_i\}]\} \\ & \stackrel{(a)}{\leq} \mathbb{E} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{u_{ij}(f) - \tilde{u}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} | \{\delta_i\} \right]. \end{aligned} \quad (277)$$

The equality [source equation] is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\mathbb{E}[u_{ij}(f) | \{\delta_i\}] - \mathbb{E}[\tilde{u}_{ij}(f) | \{\delta_i\}]\} \\ & \stackrel{(a)}{=} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{u_{ij}(f) - \mathbb{E}[u_{ij}(f) | \{\delta_i\}]\}. \end{aligned} \quad (278)$$

The equality [source equation] holds since $\mathbb{E}[u_{ij}(f) | \{\delta_i\}]$ and $\mathbb{E}[\tilde{u}_{ij}(f) | \{\delta_i\}]$ are identically distributed.

Combines [source equation] and [source equation], taking expectation of $\{x_{ij}\}_{i \in \mathcal{I}_{e,l}, j \in [m_i]}$ conditioning on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{I}_{e,l}}$ we have

$$\begin{aligned} & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{u_{ij}(f) - \mathbb{E}[u_{ij}(f) | \{\delta_i\}]\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq \mathbb{E} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{u_{ij}(f) - \tilde{u}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} | \{\delta_i\} \right] \\ & \stackrel{(a)}{\leq} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \{u_{ij}(f) - \tilde{u}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} | \{\delta_i\} \right] \\ & \stackrel{(b)}{\leq} 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{I}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} u_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} | \{\delta_i\} \right]. \end{aligned}$$

The [source equation] holds since conditioned on $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$, the $u_{ij}(f) - \tilde{u}_{ij}(f)$ is symmetric with mean zero and $\{s_{ij}(u_{ij}(f) - \tilde{u}_{ij}(f))\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and $\{\tilde{u}_{ij}(f) - \hat{u}_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ have the same distribution. The [source equation] follows because $\{-s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are also independent Rademacher variables.

Taking expectation of $\{(\delta_{i1}, \dots, \delta_{im_i})\}_{i \in \mathcal{J}_{e,l}}$ **on both sides**, we obtain the result.

Step 3. Next, we will show that under the Assumption [source ref.], and if $\ell(f) = 4\tilde{c}_0\Delta^2(f - f^*)$, we have

$$E_6 := \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0\Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [u_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \quad (279)$$

We will show that

$$E_6 \leq \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n} = \frac{c_4}{L} \frac{s \log n \log T_n}{n},$$

for a positive constant c_4 , where c_3 is a constant depends on C_β and c_1 .

We first show the following bound holds in a multidimensional setting where $x_{ij} \in [0, 1]^d$:

$$\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E} [u_{ij}(f) | \delta_i] \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}.$$

We then obtain the result by taking expectations and setting

$$\ell(f) = 4\tilde{c}_0 c_2 \Delta^2(f - f^*).$$

Given that

$$\begin{aligned} \mathbb{E}(u_{ij}(f) | \delta_i) &= \int_{[0,1]^d} \left(\int_0^{f(x) - f^*(x)} (\mathbb{P}(y_{ij} \leq f^*(x_{ij}) + u \mid \delta_i, x_{ij} = x) \right. \\ &\quad \left. - \mathbb{P}(y_{ij} \leq f^*(x_{ij}) + u \mid x_{ij} = x)) du \right) dx. \end{aligned} \quad (280)$$

Thus, we modify the function $\Pi(x, u)$ in [source equation] as follows:

$$\Pi(x, u) := \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x)]. \quad (281)$$

Similarly, as [source equation], we then have that

$$\begin{aligned} &\left| \int_{[0,1]^d} \int_0^{f(x) - f^*(x)} \Pi(x, u) du dx \right| \\ &\leq \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-1,1]} |\Pi(x, u)| \cdot \mathbb{I}_{[f(x) - f^*(x) \leq 1]} dx \\ &\quad + \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-|f(x) - f^*(x)|, |f(x) - f^*(x)|]} |\Pi(x, u)| \\ &\quad \cdot \mathbb{I}_{[f(x) - f^*(x) > 1]} dx \end{aligned}$$

Now, for the $\max_{x \in [0,1]^d} \max_{u \in [-|f(x)-f^*(x)|, |f(x)-f^*(x)|]} |\Pi(x, u)|$ term above, we have

$$\max_{x \in [0,1]^d} \max_u \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x) \right] \right|.$$

Let $T_n > 0$ be a constant that depends on n , then we have

$$\begin{aligned} & \max_{u > T_n} \max_{x \in [0,1]^d} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) - \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid x_{i1} = x)] \right| \\ & \stackrel{(a)}{\leq} \tilde{c}_0 \text{ with high probability.} \end{aligned}$$

The [source equation] holds by Assumption [source ref.].

Similarly, we have

$$\begin{aligned} & \max_{u < -T_n} \max_{x \in [0,1]^d} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) - \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid x_{i1} = x)] \right| \\ & \stackrel{(a)}{\leq} \tilde{c}_0 \text{ with high probability.} \end{aligned}$$

The [source equation] holds by Assumption [source ref.].

Next, for $u \in [-T_n, T_n]$, we can follow the proof for [source equation] to get,

$$\max_{x \in [0,1]^d} \max_{u \in [-T_n, T_n]} |\Pi(x, u)| \leq c_3 \sqrt{\frac{s \log n \log T_n}{n}} = O_p \left(\sqrt{\frac{s \log n \log T_n}{n}} \right).$$

where the c_3 is a constant that depends on c_1 and C_β .

Returning to [source equation], we obtain:

$$\begin{aligned} & \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} |\mathbb{E}[u_{ij}(f) | \delta_i]| \\ & \leq \frac{c_2}{L} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \int_{[0,1]^d} \int_0^{f(x)-f^*(x)} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) \right] du dx \right| \\ & = \frac{c_2}{L} \left| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \int_{[0,1]^d} \int_0^{f(x)-f^*(x)} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid x_{i1} = x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x) \right] du dx \right| \\ & \leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Thus, similar to [source equation], we get

$$\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} [u_{ij}(f) | \delta_i] \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]$$

$$\leq \frac{2\tilde{c}_0 c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}.$$

By letting $\ell(f) = 4\tilde{c}_0 \Delta^2(f - f^*)$, we have

$$\begin{aligned} E_6 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{E}_{x, e | \delta_i} [u_{ij}(f)] - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq \frac{c_2}{L} \frac{c_3^2}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Step 4. Back to [source equation], we have

$$\begin{aligned} \tilde{E}_1 &:= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} u_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\ &\stackrel{(a)}{\leq} 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\ &\stackrel{(b)}{\leq} \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}. \end{aligned}$$

Where [source equation] follows the Lipschitz condition of $u_{ij}(f)$ in Lemma [source ref.], and then apply the Ledoux-Talagrand contraction principle in Theorem [source ref.], and apply the Lemma [source ref.]. Finally, we get the conclusion. $\square$

Corollary 89 (Bound of $\tilde{E}_3, \tilde{F}_3$). *Let Ω_β be defined in [source equation]. Let $\bar{Z}(f), Z(f)$ been defined in [source equation] and [source equation], let $\{s_{i,j}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$, let $\tilde{E}_3$ be the quantity introduced in expression [source equation], then we have*

$$\begin{aligned} \tilde{E}_3 &= \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{Z_{ij}(f) - \bar{Z}_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &\leq 2\mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ &= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}). \end{aligned}$$

Similarly, for $\tilde{F}_3$ in [source equation] we have

$$\tilde{F}_3 \leq \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}).$$

Proof. Let $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be an independent and identically distributed copy of $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$, then we define the independent and identically distributed copy of $Z_{ij}(f)$ as

$$Z_{ij}^*(f) := \mathbb{E} [\rho_\tau(f^*(x_{ij}^*) + \delta_i(x_{ij}^*) + e_{ij} - f(x_{ij}^*)) - \rho_\tau(\delta_i(x_{ij}^*) + e_{ij}) \mid x_{ij}^*].$$

Now we control the $\mathbb{E} [Z_{ij}(f) - Z_{ij}^*(f)]$, the expectation takes on x_{ij}^* . So we have

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \mathbb{E} [Z_{ij}(f)] - \mathbb{E} [Z_{ij}^*(f)] \} \\ & \stackrel{(a)}{\leq} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ Z_{ij}(f) - Z_{ij}^*(f) \} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \end{aligned} \quad (282)$$

The equality [source equation] is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned} & \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ \mathbb{E} [Z_{ij}(f)] - \mathbb{E} [Z_{ij}^*(f)] \} \\ & \stackrel{(a)}{=} \sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ Z_{ij}(f) - \bar{Z}_{ij}(f) \}. \end{aligned} \quad (283)$$

The equality [source equation] holds since $\bar{Z}_{ij}(f)$ and $\mathbb{E} [Z_{ij}^*(f)]$ are the same quantity.

Combines [source equation] and [source equation], taking expectation of x_{ij} we have

$$\begin{aligned} & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ Z_{ij}(f) - \bar{Z}_{ij}(f) \} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \leq \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{ Z_{ij}(f) - Z_{ij}^*(f) \} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \stackrel{(a)}{=} \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} \{ Z_{ij}(f) - Z_{ij}^*(f) \} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\ & \stackrel{(b)}{\leq} 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f-f^*) \leq \bar{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right]. \end{aligned}$$

The [source equation] holds since the $\{Z_{ij}(f) - Z_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ is independently distributed and symmetric with mean zero and $\{s_{ij}(Z_{ij}(f) - Z_{ij}^*(f))\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ and

$\{Z_{ij}(f) - Z_{ij}^*(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ have the same distribution. The [source equation] follows because $\{-s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are also independent Rademacher variables.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma [source ref.], and then apply the Ledoux-Talagrand contraction principle in Theorem [source ref.].

Then we have

$$\begin{aligned}
 & \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \{\bar{Z}_{ij}(f) - Z_{ij}(f)\} \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \leq 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} Z_{ij}(f) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & \leq 2 \mathbb{E}_{x, \delta, e, x^*, \delta^*, e^*, s} \left[\left(\sup_{\substack{f \in \text{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \right) \cdot \mathbf{1}_{\{\Omega_\beta\}} \right] \\
 & = \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}).
 \end{aligned} \tag{284}$$

Where the last equality follows from Lemma [source ref.]. $\square$

Lemma 90 (Bound of A_1 , A_2 , and A_3 by Rademacher width). *Let Ω_β be defined in [source equation]. Let $\hat{f}$, belong to $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, where $\mathcal{F}$ is not necessarily convex. Let the star-shaped set $\text{Star}(\mathcal{F}, f^*)$ be defined in [source equation]. Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1, j=1}^{n, m_i}$ obey Assumption [source ref.], where c_0 satisfies [source equation]. Let $R(\mathcal{F})$ be the Rademacher width defined in [source ref.], let $\tilde{c}_0$ be a small constant given in Assumption [source ref.]. Let f_{Proj} of f^* in $\mathcal{F}$ such that*

$$\|f_{\text{Proj}} - f^*\|_\infty \leq \phi_{nm}$$

Let

$$\tilde{t}^2 := t^2 + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2, \quad \tilde{t}^2 = \tilde{t}^2 + \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2.$$

Then we have

$$\begin{aligned}
 & \mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 \right) \\
 & \leq \frac{24c_5 \tilde{t}^2}{\tilde{t}^2} \frac{1}{\tilde{t}^2} \left[\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right. \right. \\
 & \quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right) \right] \\
 & \quad + \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

where $b(B) = \max\{B^{1/2}, 1\}$ is a constant that depends on B , and c is a positive constant that depends on $\mathcal{F}$.

Proof. By [source equation], on event Ω , we have

$$\begin{aligned} & \mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\ & \leq \sum_{l=1}^L \frac{4L}{\tilde{t}^2} \left(\tilde{A}_1 + \tilde{A}_2 + \tilde{A}_3 + \tilde{B}_1 + \tilde{B}_2 + \tilde{B}_3 + \tilde{E}_1 + \tilde{E}_2 + \tilde{E}_3 + \tilde{F}_1 + \tilde{F}_2 + \tilde{F}_3 \right). \end{aligned}$$

Then by Corollary [source ref.], [source ref.], [source ref.], [source ref.], [source ref.], [source ref.], we get the the following result.

$$\begin{aligned} & \mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\ & \leq \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\ & \quad + \frac{16c_4}{\tilde{t}^2} \frac{s \log n \log T_n}{n}. \end{aligned} \tag{285}$$

Based on the results in Lemma [source ref.] and Lemma [source ref.], we have on event $\Omega \cup \Omega^c$,

$$\begin{aligned} & \mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\ & \leq \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\ & \quad + \frac{16c_4}{t^2} \frac{s \log n \log T_n}{n} + \frac{1}{\tilde{t}^2} \frac{3}{n}. \\ & \leq \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\ & \quad + \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n}. \end{aligned} \tag{286}$$

By Lemma [source ref.], we have

$$\begin{aligned} & R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \\ & \leq R_{\mathcal{J}_{e,l}}\left(\text{Star}(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2\right\}\right). \end{aligned}$$

and we have

$$\begin{aligned} & R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\}) \\ & \leq R_{\mathcal{J}_{o,l}}\left(\text{Star}(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2\right\}\right). \end{aligned}$$

Thus, we get

$$\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right)$$

$$\begin{aligned}
 &\leq \frac{24c_5\tilde{t}^2}{\tilde{t}^2} \frac{1}{\tilde{t}^2} \left[\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right. \right. \\
 &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right) \right] \\
 &\quad + \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

□

J.5.2 Proof of Theorem [source ref.]

Proof. From Lemma [source ref.], we get

$$\begin{aligned}
 &\mathbb{P} \left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2 \right) \\
 &\leq \frac{24c_5\tilde{t}^2}{\tilde{t}^2} \frac{1}{\tilde{t}^2} \left[\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right. \right. \\
 &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right) \right] \\
 &\quad + \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n}.
 \end{aligned}$$

Then by Corollary [source ref.], we have

$$\Delta^2(\hat{f} - f^*) = O_p(\tilde{t}^2), \quad (287)$$

where

$$\begin{aligned}
 \tilde{t}^2 &\asymp \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right) \right) \right. \\
 &\quad \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right) \right) \right) \\
 &\quad + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \log^{\mu+1}(n) \log T_n}{n},
 \end{aligned}$$

for some $\mu > 0$.

With such choice of $\tilde{t}^2$, suppose

$$\begin{aligned}
 &\frac{1}{\tilde{t}^2} \left[\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right. \right. \\
 &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right) \right] = O(\gamma_n),
 \end{aligned}$$

with γ_n is a sequence goes to zero with such $\tilde{t}^2$. Then we have

$$\begin{aligned}
 &\frac{24c_5\tilde{t}^2}{\tilde{t}^2} \frac{1}{\tilde{t}^2} \left[\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right. \right. \\
 &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\|^2 \leq c_0^{-1} \cdot \max\{B, 1\} \cdot \tilde{t}^2 \right\} \right) \right) \right]
 \end{aligned}$$

$$+ \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n} = O \left(\gamma_n + \frac{\left(C_0 + \frac{\bar{f}}{2}\right) \phi_{nm}^2}{\tilde{t}^2} \gamma_n + \gamma_n^\mu \right) = O(\gamma_n + \gamma_n^\mu).$$

Thus, it holds that

$$\begin{aligned} \Delta^2(\hat{f} - f^*) &= O_p \left(\sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right\} \right) \right. \right. \\ &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right\} \right) \right. \right. \\ &\quad \left. \left. + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right) \right), \text{ and} \end{aligned}$$

$$\begin{aligned} \|\hat{f} - f^*\|_2^2 &= O_p \left(b^2(B) \cdot \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right\} \right) \right. \right. \\ &\quad \left. \left. + R_{\mathcal{J}_{o,l}} \left(\text{Star}(\mathcal{F}, f^*) \cap \left\{ f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t} \right\} \right) \right. \right. \\ &\quad \left. \left. + b^2(B) 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{b^2(B) \cdot s \cdot \text{poly}(\log n) \log(T_n)}{n} \right) \right). \end{aligned}$$

J.6 Proof of Theorem [source ref.]

Implemented correction. The neural-network theorem is now conditional on explicit approximation, entropy, and architecture-balance conditions for clipped sparse ReLU networks. With $A_N = S_N L_N \log(2S_N)$,

$$\log N(\epsilon) \lesssim A_N \log(CB/\epsilon), \quad A_N L \log N/N \lesssim a_N^2,$$

and $\|f_N^\circ - f^*\|_\infty^2 \lesssim a_N^2$. The proved conclusion is $\|\hat{f} - f^*\|_2^2 = O_{\mathbb{P}}\{b_B^2(a_N^2 + \Gamma_n)\}$; no rate is claimed for arbitrary architectures.

Corollary 91 (Neural Network When $f^* \notin \mathcal{F}$ and $\mathcal{F}$ is not Convex). , Let $F(\mathcal{L}, r)$ defined as in (2) where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ with $\sigma(x) = \max\{x, 0\}$, let $\text{Star}(F(\mathcal{L}, r), f^*)$ follow the definition in [source equation] where f^* not necessarily in $F(\mathcal{L}, r)$, let f_{Proj} in $\mathcal{F}$ such that

$$\|f_{\text{Proj}} - f^*\|_\infty \leq \phi_{nm}$$

Let

$$\tilde{t}^2 := t^2 + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2,$$

then we have

$$\begin{aligned} \Delta^2(\hat{f} - f^*) &= O_p \left(\frac{\max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} \right. \\ &\quad \left. + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right). \end{aligned} \tag{288}$$

Proof. We analyze

$$\frac{R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\})}{\tilde{t}^2},$$

Also, let $\tilde{b} = c^{-1/2} \max\{B^{1/2}, 1\} \cdot \tilde{t}$.

For any function class $\mathcal{F} \cap B_2(\tilde{b})$, we have

$$R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap B_2(\tilde{b})) = \mathbb{E} \left[\mathbb{E} \left[\sup_{g \in \mathcal{F} \cap B_2(\tilde{b})} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}) \mid \{x_{ij}\} \right] \right]$$

Denote

$$R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(\tilde{b})) = \mathbb{E} \left[\sup_{g \in \mathcal{F} \cap B_2(\tilde{b})} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}) \mid \{x_{ij}\} \right] \quad (289)$$

Also denote by $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij})^2 \right)^{1/2}$

Corollary 2.2 of Bartlett et al. (2005) gives

$$\mathcal{F} \cap B_2(\tilde{b}) \subseteq \mathcal{F} \cap B_{mn}(\sqrt{2}\tilde{b}),$$

with probability at least $1 - \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \geq 1 - \frac{1}{C_\beta} \frac{2 \log n}{cc_1 nm}$, where c appears in Assumption [source ref.]. Define this good event as $\mathcal{E}$. On $\mathcal{E}$ using Dudley's entropy integral (see Dudley (1967) or Lemma [source ref.]), Equation [source equation], and the assumption $m_i \asymp m$ specified in the statement of the Lemma, with c defined in Assumption [source ref.], we bound the empirical Rademacher complexity defined in Equation [source equation] as follows

The Rademacher complexity defined in Equation [source equation] is bounded by

$$\begin{aligned} R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(\tilde{b})) &\leq R_{\mathcal{J}_{elmn}}\left(\left\{g \in F : \|g\|_{mn}^2 \leq 2\tilde{b}^2\right\}\right) \\ &\leq 12 \inf_{0 < \alpha < \sqrt{2}\tilde{b}} \left[\alpha + \frac{1}{\sqrt{\sum_{i \in \mathcal{J}_{e,l}} m_i}} \int_{\alpha}^{\delta} \sqrt{\log N(t, \|\cdot\|_{mn}, \mathcal{F})} dt \right] \\ &\leq 12 \inf_{0 < \alpha < \sqrt{2}\tilde{b}} \left[\alpha + \frac{1}{\sqrt{C_\beta cc_1 nm}} \int_{\alpha}^{\delta} \sqrt{\log N(t, \|\cdot\|_{mn}, \mathcal{F})} dt \right]. \end{aligned}$$

Under the event $\mathcal{E}^c$, given that $F \subseteq B_\infty(B)$, we have that $R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(\tilde{b})) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

Set that $\mathcal{F} \equiv \text{Star}(F(\mathcal{L}, r), f^*)$, from Lemma [source ref.], we obtain that.

$$\log N(\epsilon, \|\cdot\|_{nm}, \text{Star}(F(\mathcal{L}, r), f^*)) \leq \log \left(\frac{2B}{\epsilon} \right) + \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1}).$$

Denote $\frac{1}{\sqrt{C_\beta cc_1}}$ as c_8 .

$$R_{\mathcal{J}_{elmn}}(\text{Star}(F(\mathcal{L}, r), f^*) \cap B_{nm}(\sqrt{2}\tilde{b}))$$

$$\begin{aligned}
 &\leq 12 \inf_{0 < \alpha < \sqrt{2\tilde{b}}} \left[\alpha + \frac{c_8}{\sqrt{nm}} \int_{\alpha}^{\sqrt{2\tilde{b}}} \sqrt{\log N(\epsilon, \|\cdot\|_{mn}, \text{Star}(F(\mathcal{L}, r)))} d\epsilon \right] \\
 &\stackrel{(a)}{\leq} 12 \inf_{0 < \alpha < \sqrt{2\tilde{b}}} \left[\alpha + \frac{c_8}{\sqrt{nm}} \int_{\alpha}^{\sqrt{2\tilde{b}}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1})} d\epsilon + \frac{c_8}{\sqrt{nm}} \int_{\alpha}^{\sqrt{2\tilde{b}}} \sqrt{c_1 \log\left(\frac{2B}{\epsilon}\right)} d\epsilon \right] \\
 &\stackrel{(b)}{\leq} \frac{\tilde{b}^2}{mn} + \frac{12c_8}{\sqrt{mn}} \int_{\frac{\tilde{b}^2}{12 \log(nm)}}^{\sqrt{2\tilde{b}}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(\epsilon^{-1})} d\epsilon + \frac{12c_8}{\sqrt{mn}} \int_{\frac{\tilde{b}^2}{12 \log(nm)}}^{\sqrt{2\tilde{b}}} \sqrt{c_1 \log\left(\frac{2B}{\epsilon}\right)} d\epsilon \\
 &\leq \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2\tilde{b}}c_8}{\sqrt{mn}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log\left(\frac{12B \log(mn)}{\tilde{b}^2}\right)} + \frac{12\sqrt{2\tilde{b}}c_8}{\sqrt{mn}} \sqrt{c_1 \log\left(\frac{24B \log(mn)}{\tilde{b}^2}\right)} \\
 &= \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2\tilde{b}}c_8 \mathcal{L} r \sqrt{\log(\mathcal{L} r^2) \log\left(\frac{12B \log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{12\sqrt{2\tilde{b}}c_8 \sqrt{c_1 \log\left(\frac{24B \log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} \\
 &\leq \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2\tilde{b}}c_8 c_9 \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{12B \log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}}.
 \end{aligned}$$

Where inequality [source equation] follows the Corollary [source ref.] and [source equation] holds by choosing $\alpha = \frac{\tilde{b}^2}{12 \log(mn)}$.

The Rademacher complexity defined in Equation [source equation] is bounded by

$$\begin{aligned}
 &R_{\mathcal{J}_{e,l}}(\text{Star}(F(\mathcal{L}, r), f^*) \cap B_2(b)) \\
 &= \mathbb{E}[R_{\mathcal{J}_{e,lmn}}(\text{Star}(F(\mathcal{L}, r), f^*) \cap B_2(b))] \\
 &\leq \frac{\tilde{b}^2}{\log(nm)} + \frac{12\sqrt{2\tilde{b}}c_8 c_9 \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{2Bc_8^2 \log(n)}{nm} \\
 &= \frac{\tilde{b}^2}{\log(nm)} + \frac{c_8 c_9 \tilde{b} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{2Bc_8^2 \log(n)}{nm}
 \end{aligned}$$

Now, we see that

$$\begin{aligned}
 & \frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\})}{\tilde{t}^2} \\
 & \leq \frac{\tilde{b}^2}{\tilde{t}^2 \log(nm)} + \frac{c_8 c_9 \tilde{b} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm} \\
 & = \frac{\tilde{t}^2 c_0^{-1} \max\{B, 1\}}{\tilde{t}^2 \log(nm)} + \frac{\tilde{t}^2 c_8 c_9 c_0^{-1} \max\{B, 1\} \tilde{b} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 c_0^{-1} \max\{B, 1\} \tilde{t}^2 \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm} \\
 & = \frac{\tilde{t}^2 c_0^{-1} \max\{B, 1\}}{\tilde{t}^2 \log(nm)} + \frac{\tilde{t}^2 c_8 c_9 c_0^{-1} \max\{B, 1\} \tilde{b} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 \tilde{b} \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm} \\
 & = \frac{\tilde{t}^2 c_0^{-1} \max\{B, 1\}}{\tilde{t}^2 \log(nm)} + \frac{\tilde{t}^2 c_{10} \max\{B, 1\} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 \tilde{b} \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm} \tag{290}
 \end{aligned}$$

If $B \geq 1$, from Inequality [source equation], we get

$$\begin{aligned}
 & \frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\})}{\tilde{t}^2} \\
 & \leq \frac{\tilde{t}^2 c_0^{-1} \max\{B, 1\}}{\tilde{t}^2 \log(nm)} + \frac{\tilde{t}^2 c_{10} \max\{B, 1\} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 \tilde{b} \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm}. \tag{291}
 \end{aligned}$$

Then we let

$$\tilde{b}^2 = \frac{c_{11} B^2 \log^2(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}$$

The [source equation] is $O\left(\frac{\tilde{t}^2}{\tilde{t}^2 \log(n)} \cdot \frac{1}{\log(n)}\right)$.

Furthermore, we know $\tilde{b}^2 = c_0^{-1} \max\{B, 1\} \tilde{t}^2 = c_0^{-1} \cdot B \cdot \tilde{t}^2$ when $B > 1$, so we get

$$\tilde{t}^2 = \frac{c_{12} B \log^2(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}.$$

If $B < 1$, from Inequality [source equation],

$$\begin{aligned}
 & \frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\})}{\tilde{t}^2} \\
 & \leq \frac{\tilde{t}^2 c_0^{-1} \max\{B, 1\}}{\tilde{t}^2 \log(nm)} + \frac{\tilde{t}^2 c_{10} \max\{B, 1\} \mathcal{L} r (\log(\mathcal{L} r^2))^{1/2} \sqrt{\log\left(\frac{24B \log(nm)}{\tilde{b}^2}\right)}}{\tilde{t}^2 \tilde{b} \sqrt{nm}} + \frac{2B c_8^2 \log(n)}{\tilde{t}^2 nm}. \tag{292}
 \end{aligned}$$

then we let

$$\tilde{b}^2 = \frac{c_{11} \log^2(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}$$

The [source equation] is $O\left(\frac{\tilde{t}^2}{\tilde{t}^2} \frac{1}{\log(n)}\right)$.

Furthermore, we know $\tilde{b}^2 = c_0^{-1} \max\{B, 1\} \tilde{t}^2 = c_0^{-1} \cdot \tilde{t}^2$ when $B \leq 1$, so we get

$$\tilde{t}^2 = \frac{c_{13} \log^2(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}, \text{ when } B \leq 1.$$

So we can let

$$\tilde{t}^2 = \frac{c_{14} \max\{B, 1\} \log^2(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + R_{\mathcal{J}_{o,1}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\})}{\tilde{t}^2} = O\left(\frac{\tilde{t}^2}{\tilde{t}^2} \frac{1}{\log(n)}\right). \quad (293)$$

Then by Theorem [source ref.], we know

$$\begin{aligned} & \mathbb{P}\left(c_0 \Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\ & \leq \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\ & \quad + \frac{c_{12}}{\tilde{t}^2} \frac{s \log n \log T_n}{n}, \end{aligned} \quad (294)$$

and then by [source equation] and the choice of $L = \left\lceil \frac{1}{C_\beta} 2 \log(n) \right\rceil$ in [source equation],

We can let

$$\tilde{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}$$

to make

$$\begin{aligned} & \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\ & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\ & = O\left(\frac{\tilde{t}^2}{\tilde{t}^2} \frac{1}{\log(n)}\right). \end{aligned}$$

Furthermore, we know

$$\tilde{t}^2 = \tilde{t}^2 + \left(C_0 + \frac{\bar{f}}{2}\right) \phi_{nm}^2,$$

so we can let

$$\tilde{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + \left(C_0 + \frac{\bar{f}}{2}\right) \phi_{nm}^2.$$

to make

$$\begin{aligned}
 & \frac{24c_5}{\tilde{t}^2} \sum_{l=1}^L \left(R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right. \\
 & \quad \left. + R_{\mathcal{J}_{o,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) \right) \\
 & = O \left(\frac{1}{\log(n)} + \frac{\left(C_0 + \frac{\bar{f}}{2}\right) \phi_{nm}^2}{\tilde{t}^2} \frac{1}{\log(n)} \right) = O \left(\frac{1}{\log(n)} \right).
 \end{aligned}$$

Overall, we can let

$$\tilde{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \log^{1+\mu}(n) \log(T_n)}{n},$$

for some $\mu > 0$, to have the right hand side of [source equation] to be

$$O \left(\frac{1}{\log(n)} + \frac{1}{\log^\mu(n)} \right).$$

In the end, we have

$$\Delta^2(\hat{f} - f^*) = O_p(r_n^2),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \log^{1+\mu}(n) \log(T_n)}{n}.$$

Thus, we get

$$\begin{aligned}
 \Delta^2(\hat{f} - f^*) &= O_p \left(\frac{\max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} \right. \\
 & \quad \left. + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right), \text{ and}
 \end{aligned}$$

$$\begin{aligned}
 \|\hat{f} - f^*\|_2^2 &= O_p \left(\frac{\max\{B^2, B\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} \right. \\
 & \quad \left. + 2 \max\{B, 1\} \left(C_0 + \frac{\bar{f}}{2} \right) \phi_{nm}^2 + \max\{B, 1\} \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right).
 \end{aligned}$$

Then using the same notations as in Theorem 1 of Kohler et al. (2019), let

$$L = L_{nm} \asymp \log \left(\sum_{i=1}^n m_i \right) \stackrel{(a)}{\asymp} \log(nm), \tag{295}$$

$$r = r_{nm} \asymp \max_{(p,K) \in \mathcal{P}} \left(\sum_{i=1}^n m_i \right)^{\frac{K}{2(2p+K)}} \asymp \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{K}{2(2p+K)}}, \tag{296}$$

where the [source equation] holds by the assumption $m_i \asymp m$, or

$$L = L_{nm} \asymp \log(nm) \left(\max_{(p,K) \in \mathcal{P}} (nm)^{\frac{K}{2(2p+K)}} \right), \quad (297)$$

$$r = r_{nm} \asymp 1. \quad (298)$$

It follows from Theorem 2 of Kohler et al. (2019) that

$$\inf_{f \in \mathcal{F}(L,r)} \|f - f^*\|_\infty \leq \phi_{nm}, \quad (299)$$

where

$$\phi_{nm}^2 = \max_{(p,K) \in \mathcal{P}} \left(\sum_{i=1}^n m_i \right)^{\frac{-2p}{2p+K}} \asymp \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{-2p}{2p+K}}, \quad (300)$$

up to a logarithmic factor depending on n and $\{m_i\}_{i=1}^n$. Then we have

Thus, we get

$$\begin{aligned} \Delta^2(\hat{f} - f^*) &= O_p \left(\frac{\max\{B, 1\} \log^3(n) \cdot \log^2(nm) \cdot \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{2K}{2(2p+K)}}}{nm} \right. \\ &\quad \left. + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{-2p}{2p+K}} + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right) \\ &= O_p \left(\left(\max\{B, 1\} \log^3(n) \cdot \log^2(nm) + 2C_0 + \bar{f} \right) \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{-2p}{2p+K}} + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right), \text{ and} \\ \|\hat{f} - f^*\|_2^2 &= O_p \left(\max\{B^2, B\} \cdot \log^3(n) \cdot \log^2(nm) \cdot \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{-2p}{2p+K}} \right. \\ &\quad \left. + \max\{B, 1\} \cdot (2C_0 + \bar{f}) \cdot \max_{(p,K) \in \mathcal{P}} (nm)^{\frac{-2p}{2p+K}} + \max\{B, 1\} \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right). \end{aligned}$$

□

J.7 Auxiliary Lemmas

Implemented correction. The auxiliary symmetrization and contraction results are stated for centered classes. Cluster-level independence is used after joint coupling; conditional independence over j is invoked only for the measurement-error component given (X_i, δ_i) .

Lemma 92 (Ledoux-Talagrand contraction, see Theorem 7 in Duchi (2009)). *Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be convex and increasing. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\phi(0) = 0$ and be Lipschitz with constant L , i.e., $|\phi(a) - \phi(b)| \leq L|a - b|$. Let σ_i be independent Rademacher random variables. For any $T \subseteq \mathbb{R}^n$,*

$$\mathbb{E} f \left(\frac{1}{2} \sup_{t \in T} \sum_{i=1}^n \sigma_i \phi(t_i) \right) \leq \mathbb{E} f \left(L \cdot \sup_{t \in T} \sum_{i=1}^n \sigma_i t_i \right).$$

Lemma 93 (See Lemma 8 in Liang et al. (2015)). *For any scale $\epsilon > 0$, the covering number of $\mathcal{F} \subset B_2(1)$ and that of $\text{star}(\mathcal{F})$ are bounded in the sense*

$$\log N(\mathcal{F}, \|\cdot\|_{nm}, 2\epsilon) \leq \log N(\text{star}(\mathcal{F}), \|\cdot\|_{nm}, 2\epsilon) \leq \log \frac{2}{\epsilon} + \log N(\mathcal{F}, \|\cdot\|_{nm}, \epsilon).$$

Lemma 94 (Dudley's Chaining, see Lemma 3 in Farrell et al. (2021)). *Let $N(\delta, \mathcal{F}, \|\cdot\|_\infty)$ denote the metric entropy for class $\mathcal{F}$ (with covering radius δ and metric $\|\cdot\|_\infty$), then*

$$\mathbb{E}R_{nm}\{f : f \in \mathcal{F}, \|f\|_{nm} \leq r\} \leq \inf_{0 < \alpha < r} \left\{ 4\alpha + \frac{12}{\sqrt{n}} \int_\alpha^r \sqrt{\log N(\delta, \mathcal{F}, \|\cdot\|_{nm})} d\delta \right\}.$$

Corollary 95. *For any scale $\epsilon > 0$, the covering number of $\mathcal{F} \subset B_\infty(B)$ and then*

$$\log N(\epsilon, \|\cdot\|_{nm}, \text{star}(\mathcal{F})) \leq \log \left(\frac{2B}{\epsilon} \right) + \log N(\epsilon/B, \|\cdot\|_{nm}, \mathcal{F}/B).$$

Proof. We know

$$\mathcal{F}/B \subset B_\infty(1) \subset B_{nm}(1).$$

Then based on Lemma [source ref.], we have

$$\begin{aligned} & \log N(\epsilon, \|\cdot\|_{nm}, \text{star}(\mathcal{F})) \\ & \leq \log N(\epsilon/B, \|\cdot\|_{nm}, \text{star}(\mathcal{F}/B)) \\ & \leq \log \left(\frac{2B}{\epsilon} \right) + \log N(\epsilon/B, \|\cdot\|_{nm}, \mathcal{F}/B). \end{aligned}$$

□

Lemma 96. *Let $F(\mathcal{L}, r)$ defined as in (2) where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ with $\sigma(x) = \max\{x, 0\}$. Let $\mathcal{L}, r \in \mathbb{N}$. Then*

$$\log(N(\epsilon, \|\cdot\|_{nm}, F(\mathcal{L}, r))) \leq \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1}).$$

Proof. It follows from Theorem 6 of Bartlett et al. (2019) (see also the proof of Lemma 19 on page 78 of Kohler et al. (2019)) that the VC-dimension of $F(\mathcal{L}, r)$ satisfies

$$\text{VC}(F(\mathcal{L}, r)) \leq \mathcal{L}^2 r^2 \log(\mathcal{L} r^2).$$

By a slight variant of Lemma 19 of Kohler et al. (2019), where L_1 norm is replaced by L_2 norm (which is implied by Lemma 9.2 and Theorem 9.4 of Györfi et al. (2002)), it follows that

$$\log N(\epsilon, F(\mathcal{L}, r), \|\cdot\|_{nm}) \lesssim \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \{ \log(B^2 \epsilon^{-2}) + \log \log(B^2 \epsilon^{-2}) \} \lesssim \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1}).$$

□

Corollary 97. *Let $F(\mathcal{L}, r)$ defined as in (2) where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ with $\sigma(x) = \max\{x, 0\}$, let $\text{Star}(F(\mathcal{L}, r), f^*)$ follow the definition in [source equation] where f^* not necessarily in $F(\mathcal{L}, r)$, For any scale $\epsilon > 0$, we have*

$$\log N(\epsilon, \|\cdot\|_{nm}, \text{Star}(F(\mathcal{L}, r), f^*)) \leq \log \left(\frac{2B}{\epsilon} \right) + \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1}).$$

Proof. We apply the Lemma [source ref.] and Lemma [source ref.] to obtain the conclusion with the following derivations.

$$\begin{aligned} & \log N(\epsilon, \|\cdot\|_{nm}, \text{Star}(F(\mathcal{L}, r), f^*)) \\ & \leq \log \left(\frac{2B}{\epsilon} \right) + \log N(\epsilon/B, \|\cdot\|_{nm}, F(\mathcal{L}, r)/B) \\ & \leq \log \left(\frac{2B}{\epsilon} \right) + \mathcal{L}^2 r^2 \log(\mathcal{L} r^2) \log(B\epsilon^{-1}). \end{aligned}$$

□

J.8 β -mixing coefficients

Implemented correction. All Berbee couplings are now applied to $W_i = (X_i, \delta_i, E_i)$ on the interlaced sets separated by $2L$. The total mismatch probability is bounded by $n\beta_W(2L)$, preserving the contemporaneous joint law of all cluster components.

In this section, we discuss some results on mixing. For additional proofs and a more comprehensive discussion, the reader is directed to Doukhan (2012). Consider a random variable X in $\mathbb{R}^d$. The σ -algebra generated by X is denoted as $\sigma(X)$. Furthermore, for a collection of random variables $\{X_t\}, t \in I \subset \mathbb{R}$, the associated σ -algebra is denoted by $\sigma(X_t, t \in I)$, which is generated by the random variables $X_t, t \in I$. A sequence of random variables $\{X_t, t \in \mathbb{Z}\} \subset \mathbb{R}^d$ is defined to be β -mixing if

$$\beta_l = \sup_{t \in \mathbb{Z}} \beta(\sigma(X_s, s \leq t), \sigma(X_s, s \geq t+l)) \rightarrow 0, \quad l \rightarrow +\infty,$$

where the β coefficients are defined as,

$$\beta(\sigma(X_s, s \leq t), \sigma(X_s, s \geq t+l)) = \mathbb{E} \sup_{C \in \sigma(X_s, s \geq t+l)} |P(C) - P(C|\sigma(X_s, s \leq t))|.$$

Note, that the right-hand side is the worst-case scenario of how much the past could influence the future. Hence, intuitively, larger β -mixing coefficients indicate a higher degree of dependence, while smaller values suggest more independence.

Theorem 98 (Doukhan (2012) Theorem 1). . *Let E and F be two polish spaces and (X, Y) some $E \times F$ -valued random variables. A random variable Y^* can be defined with the same probability distribution as Y , independent of X and such that*

$$P(Y \neq Y^*) = \beta(\sigma(X), \sigma(Y)).$$

For some measurable function f on $E \times F \times [0, 1]$, and some uniform random variable Δ on the interval $[0, 1]$, Y^ takes the form $Y^* = f(X, Y, \Delta)$.*

Note that Theorem [source ref.] says that Y is originally dependent on X , and Y^* is constructed such that retains Y 's distribution but is independent of X , by introducing the Δ . Let's divide the sequence $[1 \dots n]$ into K groups of size L . For any $L \in \{1, \dots, n\}$ and $K = \lfloor n/L \rfloor$, denote

$$\mathcal{I}_s = \begin{cases} \{(s-1)L, sL\} & \text{when } 1 \leq s \leq K, \\ \{(KL, n)\} & \text{when } s = K+1. \end{cases}$$

So $\mathcal{I}_s$ is a set that contains the start and end indices of the s -th group in a sequence of data points. Specifically, the s -th group starts at the index $(s-1)L$ and ends at index sL . When $s = K+1$, the set $\mathcal{I}_s$ is defined to contain the indices from (KL) to n ,

Denote for $1 \leq s \leq L$,

$$\mathcal{J}_{e,l} = \{i \in \mathcal{I}_s \text{ for } s \% 2 = 0 \text{ and } i \% L = l\} \text{ and } \mathcal{J}_{o,l} = \{i \in \mathcal{I}_s \text{ for } s \% 2 = 1 \text{ and } i \% L = l\}.$$

Note that the $\mathcal{J}_{e,l}$ and $\mathcal{J}_{o,l}$ represent the set of the l indices for K groups. Note that for $i \in \mathcal{I}_s$, we have $i = (s-1)L + l$. Then

$$\frac{K-1}{2} \leq |\mathcal{J}_{e,l}| \leq \frac{K+2}{2} \quad \text{and} \quad \frac{K-1}{2} \leq |\mathcal{J}_{o,l}| \leq \frac{K+2}{2},$$

and that

$$\bigcup_{l=1}^L \mathcal{J}_{e,l} \cup \bigcup_{l=1}^L \mathcal{J}_{o,l} = \{1, \dots, n\}.$$

Lemma 99. *Let $\{x_{ij}\}_{i=1}^{n,m_i} \in [0, 1]^d$ are identically distributed random variables such that $\{\sigma(i)\}_{i=1}^n$ are β -mixing, where $\sigma(i) = \sigma(x_{ij}, j \in [m_i])$. Moreover, suppose that for any $i \in [n]$ fixed, $\{x_{ij}\}_{j=1}^{m_i}$ are independent. For any $l \in [1, L]$, there exists a collection of independent identically distributed random variables $\{x_{ij}^*\}_{j \leq m_i, i \in \mathcal{J}_{e,l}}$, with same distribution as x_{11} , such that for any $i \in \mathcal{J}_{e,l}$,*

$$P(x_{ij}^* \neq x_{ij} \text{ for some } j \in [1, \dots, m_i]) \leq \beta(L).$$

The exact same result is attained for $\mathcal{J}_{o,l}$.
and denote,

$$\mathcal{J}_{o,l,K} = \{l, 2L + l, 4L + l, \dots, (2K)L + l\} \cap [1, n],$$

which implies $\bigcup_{k \geq 0} \mathcal{J}_{o,l,K} = \mathcal{J}_{o,l}$. Therefore, the result is followed by replicating the analysis performed above.

Proof. By Theorem 5, for any $i \in \{1, \dots, n\}$, there exists $\{x_{i,j}^*\}_{j=1}^{m_i}$ such that

$$P(x_{i,j}^* \neq x_{i,j} \text{ for some } j \in \{0, \dots, m_i\}) = \beta(L), \quad (301)$$

where $\{x_{i,j}^*\}_{j=1}^{m_i}$ and $\{x_{i,j}\}_{j=1}^{m_i}$ have the same marginal distribution, and that $\{x_{i,j}^*\}_{j=1}^{m_i}$ is independent of $\sigma(\{x_{t,s}\}_{t=1, s=1}^{n, m_t \setminus \{i,j\}})$. In addition, there exists a uniform random variable Δ_i , independent of σ_z , such that $\{x_{i,j}^*\}_{j=1}^{m_i}$ is measurable with respect to $\sigma(\{x_{i,j}\}_{j=1}^{m_i}, \Delta)$. Note that $\{x_{i,j}\}_{j \in I_r}$ are identically distributed because $\{x_{i,j}^*\}_{j \in I_r}$ have the same marginal density. Therefore it suffices to justify the independence.

Note that $J_{l,1}$ is defined as

$$J_{l,1} = \{L + l, 3L + l, 5L + l, \dots\} \cap [1, n]. \quad (302)$$

Denote

$$J_{l,K} = \{L + l, 3L + l, 5L + l, \dots, (2K + 1)L + l\} \cap [1, n], \quad (303)$$

and so $\bigcup_{k=0}^K J_{l,k} = J_{l,1}$. By induction, suppose $\{x_{i,j}^*\}_{i \in J_{l,K}, 1 \leq j \leq m_i}$ are jointly independent. Let $\{B_{i,j}\}_{i \in J_{l,1}, 1 \leq j \leq m_i}$ be any collection of Borel sets in $[0, 1]^d$. Then

$$\begin{aligned} \mathbb{P}(x_{i,j}^* \in B_{i,j}, i \in J_{l,K+1}, 1 \leq j \leq m_i) &= \mathbb{P}(x_{i,j}^* \in B_{i,j}, i \in J_{l,K}, 1 \leq j \leq m_i) \times \\ &\quad \mathbb{P}(x_{i,j}^* \in B_{i,j}, i \in J_{l,K+1} \setminus J_{l,K}, 1 \leq j \leq m_i), \end{aligned} \quad (304)$$

where the equation follows because $\{x_{i,j}^*\}_{i \in J_{l,K+1} \setminus J_{l,K}, 1 \leq j \leq m_i}$ are independent of $\{x_{i,j}^*\}_{i \in J_{l,K}, 1 \leq j \leq m_i}$. By induction,

$$\mathbb{P}(x_{i,j}^* \in B_{i,j}, i \in J_{l,K}, 1 \leq j \leq m_i) = \prod_{i \in J_{l,K}, 1 \leq j \leq m_i} \mathbb{P}(x_{i,j}^* \in B_{i,j}). \quad (305)$$

Since for any $i \in \{1, \dots, n\}$, $\{x_{i,j}^*\}_{j=1}^{m_i}$ and $\{x_{i,j}\}_{j=1}^{m_i}$ have the same marginal distribution,

$$\mathbb{P} \left(\bigcap_{j=1}^{m_i} \{x_{i,j}^* \in B_{(2k+3)L+j}\} \right) = \prod_{j=1}^{m_i} \mathbb{P}(x_{i,j}^* \in B_{(2k+3)L+j}). \quad (306)$$

So

$$\mathbb{P} \left(\bigcap_{i \in J_{l,K+1}, 1 \leq j \leq m_i} \{x_{i,j}^* \in B_{i,j}\} \right) = \prod_{i \in J_{l,K+1}, 1 \leq j \leq m_i} \mathbb{P}(x_{i,j}^* \in B_{i,j}). \quad (307)$$

This implies that $\{x_{i,j}^*\}_{i \in J_{l,K+1}, 1 \leq j \leq m_i}$ are jointly independent. To analyze the case where the indices are considered in $J_{0,l}$, we follow the same line of arguments, but now we note that

$$J_{0,l} = \{l, 2L+l, 4L+l, \dots\} \cap [1, n], \quad (308)$$

and denote,

$$J_{0,l,K} = \{l, 2L+l, 4L+l, \dots, (2K)L+l\} \cap [1, n], \quad (309)$$

which implies $\bigcup_{k \geq 0} J_{0,l,K} = J_{0,l}$. Therefore, the result is followed by replicating the analysis performed above. $\square$

Lemma 100. *Let $\{\varepsilon_{ij}\}_{i=1}^{n,m_i}$ in $\mathbb{R}$ are identically distributed random variables such that $\{\sigma(i)\}_{i=1}^n$ are β -mixing, where $\sigma(i) = \sigma(\varepsilon_{ij}, j \in [m_i])$. Moreover, suppose that for any $i \in [n]$ fixed, $\{\varepsilon_{ij}\}_{j=1}^{m_i}$ are independent. For any $l \in [1, L]$, there exists a collection of independent identically distributed random variables $\{\varepsilon_{ij}^*\}_{j \leq m_i, i \in \mathcal{J}_{e,l}}$, with same distribution as ε_{11} , such that for any $i \in \mathcal{J}_{e,l}$,*

$$P(\varepsilon_{ij}^* \neq \varepsilon_{ij} \text{ for some } j \in [1, \dots, m_i]) \leq \beta(L).$$

The exact same result is attained for $\mathcal{J}_{o,l}$.

Proof. The arguments used in this proof parallel to those employed in the proof of Lemma [source ref.]. The key distinction lies in our utilization of Borel sets within $\mathbb{R}$ for the present context. $\square$

Lemma 101. *Let $\{\theta_{i,j}\}_{i=1,j=1}^{n,m_i}$ be identically distributed random functions in $[0, 1]^d$, such that $\{\sigma(\theta_{i,j})\}_{j=1}^{m_i}$ are σ -mixing, where $\sigma(\delta_i) = \sigma(\delta)$. For any $l \in [0, \dots, L-1]$, there exists a collection $\{\theta_{i,j}^*\}_{j \in J_l}$ of independent identically distributed random functions in $[0, 1]^d$, with the same distribution as δ_i , such that for any $i \in J_l$,*

$$P(\theta_i^* \neq \theta_i) \leq \beta(L). \quad (310)$$

The exact same result is attained for $J_{0,l}$.

Proof. By Theorem 5, for any $i \in [1, \dots, n]$, there exists θ_i^* such that

$$P(\theta_i^* \neq \theta_i) = \beta(L), \quad (311)$$

where equation [source equation] follows because θ_i^* is independent of $\sigma(\{\theta_t\}_{t \in \mathcal{J}_{e,l}})$. By induction,

$$P(\theta_i^*(t) \in B_i, i \in J_{l,K+1}, 1 \leq t \leq m_i) = \prod_{i \in J_{l,K+1}, 1 \leq t \leq m_i} P(\theta_i^*(t) \in B_i). \quad (312)$$

This implies that $\{\theta_i^*\}_{i \in J_{l,K+1}}$ are jointly independent. To analyze the case where the indices are considered in $J_{0,l}$, we note that

$$J_{0,l} = \{l, 2L + l, 4L + l, \dots\} \cap [1, n], \quad (313)$$

$$J_{0,l,K} = \{l, 2L + l, 4L + l, \dots, (2K)L + l\} \cap [1, n], \quad (314)$$

which implies $\bigcup_{k \geq 0} J_{0,l,K} = J_{0,l}$. Therefore, the result is followed by replicating the analysis performed above. $\square$

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